

A NATIONS OF EMIGRANTS: TRANSATLANTIC EMIGRATION AND ITALIAN NATIONAL IDENTITY *

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Abstract

We study how intergroup contact among immigrants shapes national identity formation within migrant communities and in their places of origin. We assemble a novel dataset on Italian migrants in the US between 1892 and 1924 by linking population census records to Ellis Island ship passenger lists, which contain migrants' municipalities of origin. In the US, exposure to more diverse Italian communities fostered the emergence of a shared national identity, as measured by a novel index of national naming choices and intergroup marriages, while reducing assimilation into American society. In Italy, origin municipalities of migrants exposed to more diverse communities abroad exhibit less cultural particularism, as reflected in naming patterns, street-name designations, and the construction of national monuments. Our findings document the novel nation-building effect of intergroup contact within immigrant communities.

Keywords: National Identity, Migration, Age of Mass Migration

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INTRODUCTION

Nation-states are among the defining political institutions of modern societies. A long-standing view in economics and political science holds that cohesive national identities are primarily created by top-down, elite-driven interventions, such as mass education, centralized bureaucracies, and state propaganda (Mylonas and Tudor, 2021; Rohner and Zhuravskaya, 2024). In this paper, we examine whether a unified national identity can emerge endogenously from bottom-up social interactions, in the absence of coordinated state intervention.

We study Italian migrants in the United States during the Age of Mass Migration (1892–1924). Until unification in 1861, Italy remained little more than a “geographical expression,” as the Austrian Chancellor, Klemens von Metternich, famously described its political and cultural fragmentation. At the turn of the 20th century, the 4.5 million Italians who entered the United States formed a linguistically, culturally, and politically highly heterogeneous group. This setting thus provides an ideal context to examine how inter-group contact shapes the formation of national identity in the absence of coordinated nation-building policies.

We construct a novel dataset linking US census records of Italian immigrants to ship manifests recorded at the Ellis Island immigration station. This linkage allows us to trace the municipality of origin for nearly half a million migrants, yielding an unprecedentedly detailed view of the within-country composition of an immigrant community.

We document two main sets of results. First, within the United States, greater exposure to Italians from other parts of the country increased national identity—as measured by a new indicator of name particularism and by a higher probability of marrying migrants from different municipalities—while slowing assimilation into American society. Second, the national sentiment that emerged within immigrant communities spilled back to their places of origin. Italian municipalities whose migrants were more exposed to diversity in the US subsequently display lower name particularism, more intensive commemoration of national events through street naming patterns, and a higher incidence of monuments dedicated to national causes. Taken together, these findings indicate that social interactions among migrants can contribute to the rise of national identity, thereby reshaping both immigrant communities and their societies of origin, even without centralized nation-building interventions.

Between 1892 and 1924, approximately 4.5 million Italian migrants settled in the United States (Choate, 2008). Italian immigrant communities reflected the deep cultural differences that characterized Italy, which stratified over more than fifteen centuries of political fragmentation (Banti, 2014). Migrants clustered along their local origins, creating “Little Palermos” and “Little Naples” within “Little Italies” (Gabaccia, 2013). Over time, local identities blended within these heterogeneous communities and a sense of shared “national” identity emerged (Vecoli, 1983; Orsi, 1985). While previous

¹The original epigraph reads “*Gli Italiani all'estero alla Madrepatria*” and is engraved on the two ceremonial braziers of the *Vittoriano* monument in Rome, whose perpetual flames symbolize national unity.

literature in economics focused on migrants' cultural assimilation into US societies, these historical and sociological works suggest that migrants developed forms of collective identity distinct from both Americans and their communities of origin. Given that more than half of Italian migrants eventually returned to Italy, this experience plausibly contributed to the formation of national identity in migrants' places of origin (Luconi, 2011; Chiaricati, 2020; Mourlane et al., 2022).

Existing data sources are not well-suited for our analysis because they do not contain information on both migrants' settling location in the US and their places of origin within Italy. To overcome this limitation, we develop an algorithm that links US census records of Italian migrants to ship passenger lists recorded at the Ellis Island immigration station, which report migrants' municipalities of origin. Our approach augments standard record-linkage methods (Abramitzky et al., 2021) with neural-network text embeddings (SBERT) to flexibly match names that were frequently Anglicized after arrival. We match individuals on first and last names, birth year, and immigration year, and, in our preferred specification, we link approximately 470,000 migrants (about 20% of the population). We assess the reliability of the linked dataset in three ways. First, linked migrants constitute a representative sample of the full population along observable characteristics. Second, more than 90% of the established links are classified as highly likely by a large language model (GPT-5). Third, more than 95% of the links are independently validated as highly likely by human reviewers.

We use this novel dataset to test two hypotheses. First, we examine whether exposure to Italians from different parts of the country shaped the formation of national identity in the United States. Second, we study whether inter-group contact among migrants influenced the formation of national identity in their communities of origin.

We conduct an individual-level analysis to estimate how exposure to a more diverse Italian community within an immigrant's neighborhood affects national identity and assimilation. To measure national identity, we exploit the linguistic and cultural divides within Italy documented by De Mauro (1963) to construct a novel index of name particularism, following recent work (e.g., Bazzi et al., 2019; Fouka, 2020). Names that are geographically concentrated within narrowly defined areas reflect stronger local identities, whereas names that are more evenly distributed across the country capture national identity. We complement this measure with information on marriage patterns with individuals from other municipalities, provinces, and regions. We measure assimilation into American society using the name-based indicator developed by Abramitzky et al. (2020).

Estimating the effect of neighborhood diversity on cultural identity raises two distinct identification challenges. First, local diversity may reflect the selective sorting of immigrants from specific municipalities across locations in the United States. Second, local diversity may be correlated with omitted, time-varying local factors that independently influence identity formation and assimilation.

We address this concern using two approaches. First, we construct an instrumental variable that leverages variation in ship composition, following the logic of Dippel and Heblich (2021). Under the hypothesis that interactions during the transatlantic voyage shaped subsequent settlement patterns, quasi-random variation in ship composition generates exogenous differences in immigrants' exposure to diversity upon arrival. Individuals traveling on more diverse ships are more likely to settle

near their shipmates, and therefore to be exposed to more diverse Italian neighborhoods. Empirically, we show that ship-level diversity is a strong predictor of neighborhood diversity and is uncorrelated with observable individual and local characteristics, suggesting that the instrument isolates plausibly exogenous variation in exposure to within-Italian diversity.

Second, we implement a “movers” design in the spirit of Chetty and Hendren (2018). We estimate a difference-in-differences specification that compares individuals who move to more diverse neighborhoods with non-movers before and after the move. This design allows us to assess the parallel trends assumption by showing that movers and non-movers exhibit statistically indistinguishable trends in identity outcomes before relocation.

We find that exposure to more diverse Italian communities decreases name particularism—our measure of national identity—and reduces Americanization. Italians exposed to more diverse immigrant communities are more likely to marry individuals born in other municipalities and provinces, and less likely to marry natives.² Individuals who settle in more versus less diverse neighborhoods are similar upon arrival, but diverge sharply in national and American identity as they spend time in the United States. Quantitatively, ten years after arrival, immigrants exposed to more diverse neighborhoods exhibit a 10% increase in national identity and show no evidence of assimilation into American society. In contrast, immigrants exposed to less diverse neighborhoods do not experience an increase in national identity and give their children 9% more American-sounding names.

These findings suggest that immigrants in more diverse communities developed a shared Italian national identity. A natural implication is that such immigrants would be less likely to return to Italy, where local identities remained more salient. We test this implication using a newly developed estimator of return migration that leverages longitudinally linked census records (Price et al., 2021; Abramitzky et al., 2025). We find that Italian immigrants residing in more diverse communities are approximately 15% more likely to remain in the United States than otherwise comparable immigrants in less diverse areas. Therefore, while roughly half of Italian migrants returned to Italy overall, only about 35% of migrants exposed to highly diverse Italian communities did so. These results suggest that exposure to diversity fostered a distinct form of national identity, distinct from both local particularism in Italy and assimilation into US society.

In the second part of the paper, we examine whether the national identity that formed within immigrant communities spilled over to migrants’ places of origin in Italy. We use three indicators to proxy for the cultural relevance of national identity at the local level: naming patterns, street naming choices, and the construction of national monuments.

Using given names as our primary proxy, we show that Italian municipalities whose migrants were more exposed to diverse Italian communities in the United States experienced a persistent decline in name-based particularism. To address the absence of detailed historical microdata on names, we draw on individual-level casualty records for soldiers who died during World War 1 and on a newly constructed dataset of World War 2 casualties. Using both panel and difference-in-differences

²We implement the supply-side correction proposed by Bazzi et al. (2019) to account for the mechanical increase in out-of-municipality marriage rates in more diverse communities.

specifications, we find that municipalities whose migrants were exposed to higher diversity in the United States shifted from locally concentrated names toward nationally shared names, consistent with the transmission of national identity from migrant communities back to their places of origin.

We corroborate these patterns using street naming as an alternative proxy for collective identity. Municipalities with greater historical exposure to diversity among their US emigrants are significantly more likely to commemorate national figures and events. Conversely, we find no systematic relationship between emigrants' diversity exposure and placebo categories such as religion, art, or literature. This contrast suggests that the results are not driven by a general propensity for commemoration or naming activity, but instead reflect a specific reorientation from local to national identity.

Finally, we examine public monuments as a third, long-run indicator of national sentiment. Focusing on Southern municipalities, we find that exposure to diversity through US migration predicts a higher incidence of monuments commemorating national events, but only in the decades following mass emigration. Exposure to diversity, by contrast, is not associated with national monuments in pre-migration decades, nor with monuments that do not specifically commemorate national events.

Taken together, evidence from names, streets, and monuments consistently indicates that interaction among diverse Italian migrants in the United States contributed to the emergence and diffusion of national identity in Italy.

Benedict Anderson (1983) argues that nations are “imagined communities” formed through shared representations that do not rely on interpersonal contact among their members. This paper documents a complementary perspective: social interactions foster national identity among fragmented groups and, specifically, emigrants contribute to national cohesion in their country of origin. Our findings suggest that today's diasporic communities (Alesina et al., 2003)—often from highly fractionalized societies—can play an important and underappreciated role in fostering nation-building in their countries of origin.

Related Literature. This paper contributes to several strands of literature. First, it relates to a growing body of work on the economics of nation-building (Rohner and Zhuravskaya, 2024).³ A unifying theme in this literature is that national identity emerges in response to top-down interventions, including mass education (Cantoni et al., 2017; Bandiera et al., 2019; Alesina et al., 2021; Blanc and Kubo, 2025), foreign state intervention (Dell and Querubin, 2018; Dehdari and Gehring, 2022), media propaganda (DellaVigna et al., 2014; Esposito et al., 2023), welfare spending (Caprettini and Voth, 2023), and forced population relocation (Bazzi et al., 2019). Our contribution to this literature is to study a setting in which individuals from highly heterogeneous communities were largely insulated from coordinated nation-building policies. In this sense, our findings are related to Depetris-Chauvin et al. (2020), who show that shared experiences contribute to national cohesion. We provide novel evidence that sustained inter-group contact, even in the absence of explicit state intervention, can foster

³The economics of nation-building is a relatively recent field, but it builds on an established literature at the intersection of sociology, political science, and history. “Modernist” theses argue that nation-states emerged in response to industrialization (Gellner, 1983; Anderson, 1983), whereas Smith (1986) maintains that modern nations reflect pre-modern identities. Classic case studies of top-down nation-building processes within this tradition include Weber (1976) on education in France and Hobsbawm and Ranger (1983) on the use of invented tradition as a legitimizing device for national identity.

the formation of national identity.

Second, we contribute to the literature on immigrant assimilation (e.g., Abramitzky et al., 2020; Fouka et al., 2022; Carlitz et al., 2025; Gagliarducci and Tabellini, 2025). A common focus of this work is on how minority groups assimilate into a majority native population. We take a different perspective by studying how exposure to migrants from the same origin country—but from different regions—shapes the dynamics of group identity formation.⁴ Guided by a simple model of identity choice in the spirit of Akerlof and Kranton (2000), we show that greater within-origin diversity among migrants strengthens national identity and slows cultural assimilation into the host society.

Third, our findings relate to the literature on how inter-group contact shapes beliefs, prejudice, and identity. Most studies in this literature focus on majority–minority contact. For example, Bazzi et al. (2019) study the nation-building effects of an extensive relocation program in Indonesia, while Bursztyn et al. (2024) show that proximity to immigrants reduces hostility toward subsequent immigration. Related work examines other settings, such as workplace and educational interaction (Billings et al., 2021; Corno et al., 2022) and ethnic or political cleavages (Dahl et al., 2021; Schindler and Westcott, 2021; Okunogbe, 2024). Building on the contact hypothesis, Lowe (2021) distinguishes between collaborative and adversarial contact and shows that only the former fosters integration. Our work is related to Bagues and Roth (2023), who show that exposure to interregional diversity in Spain fosters national identity formation. We inform this literature by showing that *within*-minority interactions (among migrants) can shape group identity formation and reallocate identity away from majority–minority assimilation.

Fourth, we contribute to the literature on the effects of out-migration on emigration countries (Batista et al., 2025). A growing body of work shows that emigrants can benefit their places of origin through monetary remittances (Khanna et al., 2026), the diffusion of new technologies (Coluccia and Dossi, 2025), increased technology adoption and entrepreneurship (Anelli et al., 2023; Coluccia and Spadavecchia, 2025), and human capital formation (Fernández Sánchez, 2022). Recent studies, in particular, emphasize the political and social consequences of emigration, documenting how migrants diffuse social norms (Levitt, 1998; Levitt and Lamba-Nieves, 2011; Beine et al., 2013; Melki et al., 2024; Godlonton and Theoharides, 2025) and political preferences (Spilimbergo, 2009; Barsbai et al., 2017; Karadja and Prawitz, 2019; Manacorda et al., 2024) to origin communities. We document a novel effect of emigration: its contribution to the formation of national identity in migrants’ countries of origin.

Outline of the Paper. Section I presents a stylized theoretical framework to guide the empirical analysis. Section II provides a concise overview of the historical background. Section III introduces our novel dataset and the additional data used in the analysis. Section IV focuses on the United States and documents that Italian immigrants developed a distinctive group identity, which we refer to as “national identity.” Section V examines how historical out-migration contributed to the rise of national identity in migrants’ communities of origin. Section VI concludes.

⁴In this sense, this work is related to the multidisciplinary field of diaspora studies (among others, see Gilroy, 1993; Edwards, 2003). Hall (2015) emphasizes that cultural identity in diasporic communities is produced in response to displacement and internal heterogeneity. Our findings echo these qualitative narratives, indicating that Italians in the United States developed an identity that is partially distinct from the local identity they acquired in Italy.

I ANALYTICAL FRAMEWORK: A MODEL OF IMMIGRANT GROUP IDENTITY CHOICE

We develop a parsimonious framework to rationalize how exposure to within-group diversity shapes identity formation. We derive two implications which we test in the empirical analysis.

A continuum of individuals resides in a local community. Each individual originates from a municipality m . The community is characterized by a measure of within-Italian diversity $D \in [0, 1]$, which increases with the heterogeneity of origin municipalities.

Agents choose one identity to express, $j \in \{L, N, A\}$, where L , N , and A respectively denote local, Italian, and American identity. Identity choice affects cultural investment.

Following Akerlof and Kranton (2000), identity yields utility if it is socially recognized. The payoff from choosing identity j is $U_j = v_j \cdot s_j(D) - c_j$, where $v_j > 0$ captures the intrinsic utility of identity j , $s_j(D) \in [0, 1]$ denotes the share of individuals in the community for whom identity j is meaningful, and $c_j \geq 0$ is the cost of adopting identity j .

For simplicity, we let $s_L(D) = 1 - D$, $s_N(D) = D$, and $s_A(D) = \bar{s}_A$.⁵ The interpretation is that local identities are recognized only in homogeneous communities, while national identity becomes a useful coordination device as local origins fragment. Exposure to natives is orthogonal to within-Italian diversity, so the salience of American identity is not affected by within-Italian diversity D . This structure is analogous to coordination-based models in which the payoff to identity depends on group size (e.g., Shayo, 2009).

We assume costs satisfy $0 = c_L < c_N < c_A$, reflecting that local identity is inherited, national identity requires modest adaptation, and American identity entails higher linguistic and social effort.

Individuals choose the identity that maximizes their utility $\{U_j\}_{j \in \{L, N, A\}}$. National identity is chosen over local identity whenever $v_N D - c_N \geq v_L(1 - D)$, which defines a threshold level of diversity above which national identity strictly dominates local identity. This yields the following implication.

Proposition I.1. (*Local-to-National Identity*) *Higher within-Italian diversity reduces the prevalence of local identities and increases the prevalence of national Italian identity.*

National identity is chosen over American identity whenever $v_N D - c_N \geq v_A \bar{s}_A - c_A$. Because the left-hand side increases in D while the right-hand side does not, greater within-Italian diversity weakens incentives to adopt American identity, yielding the following testable implication.

Proposition I.2. (*Americanization*) *Higher within-Italian diversity decreases the adoption of American identity.*

The framework highlights identity as a coordination device whose value depends on community composition. High within-group diversity reduces the payoff to narrow local identities while increasing the appeal of broader national identities. Crucially, diversity reallocates identity *within* the immigrant group rather than toward the host society, generating national identity formation.

⁵Results remain unchanged for functional forms of $s_j(\cdot)$ that are increasing and monotone in D .

II HISTORICAL BACKGROUND

Section II.A outlines Italian mass migration to the United States and its regional composition. Section II.B describes Italy's cultural fragmentation during the Age of Mass Migration, while Section II.C documents cultural convergence within Italian immigrant communities in the United States.

II.A The Italian Mass Migration to the United States, 1892–1924

Between 1876 and the early 1920s, approximately 26 million Italians emigrated abroad, constituting the largest episode of voluntary emigration in recorded history (Choate, 2008). The number of emigrants exceeded Italy's population at the time of unification in 1861. By 1913, Italy was the world's second-largest source of emigrants (after India), and its emigration rate was comparable to high-emigration countries such as Ireland and Norway in the nineteenth century (Foerster, 1919).

The United States absorbed approximately 40% of the Italian emigration (Livi-Bacci, 1961). These large flows were facilitated by liberal migration regimes on both sides of the Atlantic (Rosoli, 1998). Until 1917, neither Italy nor the United States imposed binding restrictions on human mobility. Restrictive immigration policies implemented by the US, including a literacy test (1917) and, more importantly, quotas on the number of Italians admitted to enter (1921–1924), abruptly halted Italian mass migration to the United States (Abramitzky and Boustán, 2017).

Italian migrants to the United States were predominantly young men from rural areas of Southern Italy (Foerster, 1919). Because of this geographical selection, migrants were negatively selected at the national level, yet positively selected within their local communities in terms of height, education, and wealth (Spitzer and Zimran, 2018). Anecdotal evidence suggests that migration was often intended as temporary (Cinel, 2002), and quantitative evidence confirms that more than half of Italian migrants eventually returned (Bandiera et al., 2013). Both historical narratives and recent empirical studies emphasize the central role of local networks in shaping migration decisions (Vecoli, 1983; Choate, 2008; Spitzer and Zimran, 2023).

Chain migration and strong localism generated distinctive patterns of geographic concentration both across and within US cities (Vecoli, 1983). Stonecutters from the marble-producing area of Massa clustered in Barre, Vermont; emigrants from Molfetta settled disproportionately in Hoboken, New Jersey; and migrants from Floridia, Sicily, concentrated in Hartford, Connecticut (Corradini, 1923). Residential segregation within cities, particularly in larger urban centers, was also common. As Luigi Villari (1912, p. 216), the Italian consul in New York, observed,

"[...] we can find only Sicilians in a street, only people from Calabria in another street, and immigrants from Abruzzi in a third one. There are even streets where only individuals from a single town live: a colony from Sciacca here, a colony from San Giovanni in Fiore there, a colony from Cosenza somewhere else."

In East Harlem, New York City, even though most Italian migrants originated from the South, segregation by place of origin remained widespread. Neapolitans clustered between 106th and 108th Streets, while migrants from the nearby region of Basilicata settled primarily between 108th and 115th

Streets (Orsi, 1985). In Chicago, segregation was similarly stark, as Italians settled across sixteen distinct neighborhoods according to their origins (Vecoli, 1983). Such patterns of segregation within Italian communities reflected the deep fragmentation that characterized Italy itself. Over time, however, these localized identities gradually gave way to broader forms of shared identity, as Italian communities coalesced into a more unified “Italian American” group. We return to this process in Section II.C.

II.B Cultural Fragmentation in Italy After Unification

Italian national identity emerged largely in response to political unification in 1861 (Riall, 2002). This delay is captured by a famous epigram by Piedmontese statesman Massimo d’Azeglio, who remarked in 1867 that “Italy has been made; Italians remain to be made.” Historians emphasize that national identity was initially articulated as an elite cultural project and only gradually diffused to the broader population (Banti, 2014). Political unification, therefore, preceded social and cultural integration by several decades. Regional divisions, shaped by adverse geography and a long history of political fragmentation, persisted along multiple dimensions, including linguistic particularism and cultural differences, well into the twentieth century, especially along the North-South divide (Dickie, 1999).

Despite the central role attributed to linguistic unity in *Risorgimento* discourse, only a small minority of the population—estimated between 2.5 and 6.5 percent—spoke Italian at the time of unification (De Mauro, 1963).⁶ Four decades later, a public inquiry reported that roughly 50 percent of the population had some knowledge of Italian, yet only 20 percent could use it for everyday communication (Corradini, 1910). The *legge Casati* (1865) introduced compulsory elementary schooling up to age eight in an effort to promote the “national language,” but implementation remained weak. In 1870, between 40 and 65 percent of school-age children did not attend school, and as late as 1906, half of children aged six to ten had received no formal schooling.

Despite state efforts at linguistic homogenization, local dialects remained widespread, even among educated elites. King Vittorio Emanuele II, for instance, conferred with his ministers in his native Piedmontese dialect, while Camillo Benso, Count of Cavour, the architect of unification, preferred French to Italian and thought Sicilians spoke Arabic (Chabod, 1965; Smith, 1985). Dialect use was common even in schools, where, in 1910, 50 percent of teachers reportedly conducted lessons in dialect rather than in Italian, despite formal requirements to the contrary (Corradini, 1910). On the eve of World War I, linguistic fragmentation therefore persisted, particularly in rural areas, from which the majority of Italian migrants to the United States originated.

Linguistic fragmentation was the product of centuries of political and geographic division (De Mauro, 1963). The Apennines, a mountain chain running from the Po Valley in the north to Calabria in the south, created a persistent west-east barrier. In regions such as the Calabrian Sila, neighboring villages were separated by deep chasms and had limited interaction. Before the expansion of the railway network, travel from Rome on the Tyrrhenian coast to Ancona on the Adriatic was easier by ship,

⁶Fewer than 1 percent of the population outside Tuscany and Rome spoke “Italian” as a first language. Italian functioned largely as a written language, accessible primarily to literate individuals. In 1861, the literacy rate averaged around 20 percent nationwide (and was closer to 10 percent in the South), placing a strict upper bound on the share of Italian speakers (Ciccarelli and Weisdorf, 2019).

requiring a voyage of roughly 1,500 kilometers, than by crossing the approximately 210 kilometers through the Apennines (Gilmour, 2011).

The most enduring division within the newly unified Italian state was between the Center–North and the South, the so-called *Questione Meridionale*. Quantitative evidence indicates that this divide widened after unification: the ratio of Southern to Northern GDP per capita declined from roughly 0.8 in 1861 to about 0.5 by the late 1930s, reflecting divergent development trajectories rather than convergence under national institutions (Felice, 2019). Economic divergence reinforced cultural distance between the two areas, as illustrated by the civil conflict that followed unification. This conflict, fought largely by the Southern peasantry and commonly referred to as *brigantaggio*, at its peak required the mobilization of two-fifths of the regular army (Molfese, 1964; Dickie, 1999). Linguistic and educational disparities further deepened these divisions: school attendance remained low in the South, where the use of local dialects persisted well into the postwar period (De Mauro, 1963).

II.C Cultural Convergence in Immigrant Communities in the United States

Historians of the Italian diaspora emphasize that Italy’s cultural fragmentation produced highly localized immigrant communities in the United States (Gabaccia, 2013). Over time, these parochial identities blended into an increasingly homogeneous “Italian American” identity, particularly in response to World War I and the rise of Fascism (Vecoli, 1985; Luconi, 2003). Reflecting this shift, in 1934, the leading Italian-American newspaper, *Il Progresso Italo-Americano*, wrote:

“[...] there cannot be differences between a Lombard and a Roman, between a Venetian and a Sicilian, between a Piedmontese and a Neapolitan. The increasing difficulties in our lives require solidarity, not divisions.”

To underscore the importance of migrant group identity and its influence on their country of origin, Choate (2008) refers to Italy as an “emigrant nation.”

Fragmentation and subsequent homogenization are evident in the evolution of mutual aid societies, grassroots associations that provided financial and social support to immigrants. Initially, these associations were organized along narrow local and regional lines: only migrants from Abruzzo, for instance, could join the *Fratellanza Abruzzese* in Providence, while membership in the *Ateleta Beneficial Association* in Pittsburgh was restricted to migrants from the small village of Ateleta (Luconi, 2003). In East Harlem alone, Covello (1967) counted as many as sixty-four mutual aid societies with origin-based admission criteria. Beginning in the 1910s, however, these organizations increasingly federated across local origins (Gabaccia, 2013). In Milwaukee, for example, all associations merged into a single “Italian Societies of Milwaukee” confederation. The first national Italian association, the “Order of the Sons of Italy,” was established in 1905, and, in the 1920s, its membership increased to 300,000 (Vecoli, 1985).

Language barriers among Italians from different parts of the country were substantial. In 1888, for example, the Irish archbishop of New York sought Italian clergy to minister to newly arrived immigrants, but the priests available to him, largely from Northern Italy, were unable to communicate

with Southern Italians (Choate, 2008). As noted above, most Italian immigrants did not speak standard Italian upon arrival and instead developed a syncretic creole that blended regional dialect—predominantly Neapolitan—with English (Carnevale, 2010). This Italian-American creole became the primary medium of communication across regional lines, and its cultural significance was such that it survived even among returnees in Italy. Linguistic homogenization was further reinforced by the Italian-American press, which predominantly published in standard Italian (Carnevale, 2010). Between 1900 and 1930, the Italian-language daily press ranked second only to German-American newspapers. As Deschamps (1996) argues, these newspapers promoted the use of Italian among migrants more than they did in Italy.⁷

Although virtually all Italian immigrants were Catholic, religious practices varied markedly across regions, and devotion to local saints remained strong. In St. Louis, for instance, Sicilians boycotted the Church of St. Ambrose, which had been erected by Lombards to honor a saint traditionally associated with Milan (Luconi, 2003). Over time, however, the Catholic Church emerged as a force of homogenization. The Missionaries of St. Charles Borromeo, commonly known as the “Scalabrinians,” were founded to minister to Italian migrants abroad and explicitly sought to preserve and cultivate both Catholic practice and a shared Italian cultural identity (Choate, 2008).⁸ Convergence in religious practices emerged within increasingly diverse immigrant communities. In his ethnography of East Harlem, Orsi (1985) documents that Italian immigrants initially venerated a wide array of local saints and patrons brought from their communities of origin. As boundaries between regional groups blurred, however, few survived. Among them, the Madonna of Mount Carmel was the most relevant one. Originally associated with immigrants from parts of Campania and Sicily, devotion to the Madonna gradually became a shared religious experience among community members, irrespective of their origins. This convergence did not result from clerical intervention but instead arose through repeated collective practice in the common urban space, especially during the annual *fešta*.

A final dimension of cultural integration concerns cuisine (Chiaricati, 2020; Soffiati and Grandi, 2024). Although Italian cuisine is a defining component of Italian culture, recent work shows that many contemporary “Italian” foods emerged within immigrant communities in the United States. This process involved two steps: the diffusion of regional foods across immigrant groups and the invention of new culinary practices. Pasta and pizza exemplify the first channel. Originally concentrated in Naples, where it was eaten by hand, pasta spread widely among Italian immigrants to the point that Fascist authorities considered it an American invention. Until the early 1900s, pizza remained a similarly marginal food confined to the slums of Naples and to Neapolitan emigrants. Its modern form emerged in the US, where immigrants introduced toppings and commercialized it

⁷Newspapers were more effective in spreading Italian among migrants than among stayers primarily because migrants gained unprecedented access to basic education upon arriving in the US, thanks to public and denominational (Catholic) schools.

⁸As their founder, Giovanni Battista Scalabrini, put it, “Religion and fatherland [...] become intertwined. [...] May Italy, sincerely reconciled with the Apostolic See, emulate its ancient glories and add a new, undying glory, setting even its faraway children on the shining paths of true civilization and progress” (Scalabrini, 1983, p. 145–161). In doing so, Scalabrini explicitly advocated against the “melting pot” and immigrant assimilation: “Not many years ago, in the United States, there were enormous efforts to Americanize [...] Let us also work so that all Italians abroad possess the same motto, the same steadfastness, the same courage: for Religion and for the Fatherland” (Scalabrini, 1898, p. 39).

through specialized shops. When American troops landed in Sicily in 1943, they were surprised by the absence of pizza shops. The second channel involved invention. Tomato sauce entered Italian cookbooks only in the 1930s after becoming common among emigrants, while dishes such as meatballs and spaghetti Alfredo remain distinctive of Italian-American cuisine.

Did cultural convergence among Italian immigrants in the United States influence Italy? Qualitative historical evidence suggests that it did (Choate, 2008). Partly because of high return migration rates, values and practices acquired abroad plausibly diffused back to migrants’ communities of origin. Return migrants founded mutual aid associations that transcended village boundaries and often evolved into political organizations. In the Apennine village of Montefegatesi, in 1902, returnees financed the construction of a monument to Giuseppe Garibaldi, a symbol of national unity (Wyman, 1993). Cultural feedback also operated through consumption practices: dishes developed within Italian-American communities later entered Italian culinary traditions, although their diasporic origins were frequently obscured (Chiaricati, 2020). More generally, qualitative studies indicate that Italy’s identity as an “emigrant nation” developed partly through interactions among local identities in emigrant communities and the influence that return migrants exerted over their origin communities.

III DATA

This section presents the novel linked dataset of Italian migrants (Section III.A), our methodology to measure national identity (Section III.B), and the additional data we employ in the analysis (Section III.D). We conclude by discussing how we construct final analysis samples (Section III.E).

III.A Linking Ellis Island Ship Passenger Lists to Census Records

III.A.1 Motivation and Overview

Our empirical analysis requires information on the origins and destinations of Italian immigrants in the United States at a fine level of spatial granularity. While the US population census allows us to observe the enumeration district in which individuals resided, it does not report disaggregated information on migrants’ places of origin beyond the country level.⁹

To circumvent this limitation, previous studies have linked US census records to origin-country censuses for the United Kingdom (Coluccia and Dossi, 2025) and Norway (Abramitzky et al., 2012). In this paper, we link detailed microdata from ship rosters recorded at the Ellis Island immigration station, compiled by Coluccia and Spadavecchia (2025), to the US population census. These rosters report individuals’ names, surnames, years of birth, and years of immigration for approximately 4.2 million Italian migrants. Of these, about 2.5 million records are georeferenced to the coordinates of the municipality of origin in Italy and constitute our analysis sample. Coluccia and Spadavecchia (2025) show that migrants with a valid origin municipality are representative of the broader population of Italian immigrants in the United States.

⁹Appendix B provides additional details on the data sources, the implementation of the linking algorithm, and the validation exercises used to assess its reliability.

Relative to prior approaches, this strategy offers two methodological advantages. First, all immigrants should appear in both datasets.¹⁰ Thus, the likelihood of false positive matches is lower than when linking census records across countries, since in that case, not all migrants are observed in both censuses. Second, and more importantly, the data include an additional blocking variable (year of immigration), which substantially reduces the risk of false positives and multiple matches when name, surname, and year of birth do not uniquely identify individuals.

III.A.2 Description of the Linking Algorithm

Linking Ellis Island ship manifests to US census records poses challenges for existing record-linkage algorithms (e.g. Abramitzky et al., 2021). First, census enumerators and Ellis Island officials typically did not speak Italian, leading to frequent transcription errors in both first and last names. Second, many Italian migrants Anglicized their names after arrival (Biavaschi et al., 2017). While spelling errors can be addressed using fuzzy string-matching techniques, name translations are more problematic, as original and translated names are semantically equivalent but orthographically distant.

To address this issue, we develop a linking algorithm that harnesses the flexibility of neural networks to construct embeddings that preserve semantic proximity using bidirectional transformers (SBERT, Reimers and Gurevych, 2019).¹¹ We train a multilingual encoder on a subset of names, which we translate manually and using GPT5. Then, we fine-tune a classifier that uses the encoder’s embeddings to identify name translations. We complement the SBERT-based classifier with a traditional Jaro-Winkler classifier to handle *both* translations and spelling mistakes. In our application, we consider two names a match if either their Jaro-Winkler proximity exceeds .95, a threshold commonly used in the literature (Abramitzky et al., 2021), or their SBERT distance exceeds .6, the fine-tuned optimal distance threshold. Surnames are not subject to translation. Hence, we consider two surnames a match if their Jaro-Winkler similarity exceeds .95.

Armed with this classification rule, we try to link the records in the Ellis Island dataset to the US census. Consider an entry i in the Ellis Island dataset, and let n_i and s_i denote i ’s first and last names. Let b_i and y_i denote i ’s year of birth and year of immigration, respectively. Define $\mathcal{M}_i = \{j \mid b_j \in [b_i - 2, b_i + 2] \wedge y_j \in [y_i - 5, y_i + 5]\}$ as the set of *candidate* matches among US census records. Following (Abramitzky et al., 2021), we allow for measurement error in reported birth and immigration years. Let $\overset{n}{\sim}$ denote the name-equivalence classifier defined above, such that $n_i \overset{n}{\sim} n_j$ if and only if the SBERT similarity between n_i and n_j exceeds 0.6 or their Jaro-Winkler similarity exceeds 0.95. Analogously, $\overset{s}{\sim}$ is the surname-equivalence classifier, such that $s_i \overset{s}{\sim} s_j$ if and only if their Jaro-Winkler similarity exceeds 0.95. The set of *effective matches* is then $\widetilde{\mathcal{M}}_i = \{j \in \mathcal{M}_i \mid n_i \overset{n}{\sim} n_j \wedge s_i \overset{s}{\sim} s_j\}$. The set $\widetilde{\mathcal{M}}_i$ may be empty, in which case i is unmatched; it may contain a single element (a “single” match); or it may contain multiple elements (a “multiple” match). To avoid double-counting, multiple matches are weighted by the inverse of the cardinality of $\widetilde{\mathcal{M}}_i$.

¹⁰ Approximately 95% of Italians who entered the United States between 1895 and 1924 passed through the Ellis Island station (Unrau, 1984).

¹¹ Figure D.1 provides a visual representation of the linking algorithm.

III.A.3 Evaluation of the Linking Approach and Validation

Figure I presents selected visual diagnostics to assess the performance of the linking algorithm.¹² Figure Ia shows the number of immigrants by year of arrival in the full sample (gray bars) and in the linked sample (red dots). The two series exhibit a strong positive correlation, indicating that the algorithm does not systematically over-sample migrants in particular periods. Out of 2,526,024 immigrants in the sample, the baseline algorithm successfully links 473,922 individuals to at least one US census record.

The average matching rate, shown in Figure Ib, is 19%. Matching rates are lower before 1900, primarily because the quality of recorded names and surnames in the Ellis Island data improves substantially between 1895 and 1900. Thereafter, the matching rate remains relatively stable, except during the war years, when overall immigrant inflows contracted markedly. The average matching rate is comparable to that reported in earlier record linkage studies, such as Abramitzky et al. (2021). However, when focusing on non-English individuals, Abramitzky et al. (2021) report substantially lower rates (e.g., 10% for the Germans). More recent approaches, such as Althoff et al. (2025) and Green (2025), achieve higher matching rates. These figures are not directly comparable to ours, as both papers focus on samples composed predominantly of English-speaking individuals. Turning to non-English populations, Postel (2023) constructs a longitudinal linked sample of Chinese immigrants using US census records and attains a 9% matching rate. Our algorithm thus performs on the higher end of match rates for relevant populations.

A high incidence of multiple matches may indicate spurious links. Figure Ic shows that nearly 96% of the matches produced by the algorithm are single, meaning that each Ellis Island record is linked to one census record. The share of records associated with more than two matches is below 1% of the full sample.

Figure Id provides a first assessment of the representativeness of the linked sample. We plot the log number of emigrants by municipality in the full sample on the horizontal axis and in the linked sample on the vertical axis. The solid red line shows the linear fit, while the dashed line represents the 45-degree line. Observations lie below the 45-degree line because the matching rate cannot exceed one. The correlation between the two series is positive and highly statistically significant, indicating that the distribution of municipalities of origin in the linked sample closely mirrors that of the full sample. The estimated slope ($\hat{\beta} = 0.825$) suggests that migrants from smaller municipalities are slightly more likely to be linked, but the magnitude of this difference is modest.

Table C.1 reports selected characteristics of migrants with a match (column 1) and without a match (column 2). Columns 3 and 4 present differences between the two groups. Panel A compares linked and non-linked migrants in the Ellis Island data, while Panel B reports the corresponding comparison using US census records. In the Ellis Island data, linked migrants are younger than non-linked migrants. They are also slightly less likely to originate from Northern Italy and more likely to come from the Islands, primarily Sicily, although these differences are small and not statistically significant. In

¹²Appendix B.IV provides complementary evidence to evaluate the internal reliability of the linking algorithm.

the US census, linked migrants are more likely to be male. This pattern is expected, as matching rates for women are lower due to surname changes upon marriage. Linked migrants are also more likely to be employed. Conditional on labor force participation, however, their industry and occupation distributions closely resemble those of non-linked migrants. The geographic distribution of linked and non-linked migrants is also similar. Overall, the linked sample appears broadly representative of Italian immigrants as observed in the US census.

Validating the linked data against external sources is largely infeasible due to data limitations. In Figure D.4, we therefore implement two alternative procedures to further assess the quality of the links produced by the algorithm.¹³ In particular, to construct “Sample 2,” displayed in red, we provide GPT5 with the same information available to the linking algorithm and ask it to evaluate the reliability of each match. In 91% of cases, the model classifies the matches as correct or very likely to be correct. All results are robust to restricting the analysis to this subset of matches.

III.B Measuring Local, National, and American Identity of Immigrants in the United States

National identity in contemporary settings is typically measured using survey data (Rohner and Zhuravskaya, 2024). Such data are rarely available in historical contexts. To address this limitation, prior work has proxied national identity using first names, focusing on subsets of names with clear ideological or ethnic connotations, such as “pure Turkish” names in the early post-Ottoman period (Assouad, 2025), nationalist names during the Finnish independence movement (Lindfors et al., 2025), or names associated with specific ethnic groups (Bazzi et al., 2019; Fouka, 2020).

In this paper, we propose a novel, disaggregated measure of national identity based on first names. Our approach builds on historical research documenting the extreme geographic fragmentation of naming practices in pre- and post-unification Italy (Delille, 1985; Viazzo, 1989). In this context, first names were tightly linked to local kinship norms, patron-saint cults, and dialectal traditions, resulting in pronounced spatial concentration. We therefore interpret names that were more geographically concentrated as denoting particularistic identities, relative to names that were more evenly distributed across the national territory.

The first measure of particularism is the Local Name Index (LNI). The Local Name Index of name n is

$$LNI_n \equiv \sum_m (s_{mn} - p_m)^2, \quad (1)$$

where $s_{mn} = C_{mn}/N_n$, C_{mn} denotes the total number of individuals named n in municipality m , and $N_n = \sum_m C_{mn}$. Thus, s_{mn} is the share of individuals named n residing in municipality m . Further, let $p_m = B_m/B$, where $B_m = \sum_n C_{mn}$ and $B = \sum_m B_m$, so that p_m represents the population share of municipality m . Intuitively, p_m is the share that name n would exhibit in municipality m if it were evenly distributed across the territory. By construction, the LNI is normalized between zero and one. It is equal to zero if name n is distributed proportionally to population across all municipalities (i.e., $s_{mn} = p_m$ for all m), and it approaches one as the mass of name n becomes increasingly concentrated

¹³Appendix B.V describes these validation procedures in detail.

in a small number of municipalities.

We construct the LNI using municipalities of origin reported in the Ellis Island records. Figure IIa shows the resulting distribution. To illustrate the index, we highlight two names, “Alfio” and “Raimondo,” chosen because they occur with similar overall frequency. Figure IIb plots the spatial distribution of “Alfio,” while Figure IIc reports that of “Raimondo.” The former, which has an LNI of 0.045, is highly concentrated in western Sicily, plausibly reflecting the local cult of Saint Alfio, patron saint of Taormina, near Siracusa. By contrast, “Raimondo” is substantially more evenly distributed across municipalities throughout both Northern and Southern Italy.

Intuitively, the baseline particularism index captures how spatially concentrated a name is relative to a benchmark in which the name is evenly distributed across municipalities. By construction, however, the baseline LNI does not account for spatial proximity between municipalities.¹⁴ To address this limitation, we construct a distance-adjusted local name index (DLNI):

$$DLNI_n \equiv \sum_m \sum_{m'} s_{mn} d_{m,m'} s_{m'n} , \quad (2)$$

where $d_{m,m'}$ denotes the geodesic distance between municipalities m and m' . The index $DLNI_n$ admits a simple interpretation as the expected distance between two randomly drawn individuals sharing name n . Accordingly, higher values of $DLNI_n$ indicate more geographically diffuse, and hence more “national,” names, while lower values indicate stronger local particularism.

To measure assimilation into US society, we follow Abramitzky et al. (2020) and assign to each name its associated Foreign Name Index (FNI). The FNI is normalized to lie between zero and one and takes higher values for names that are relatively more prevalent among immigrants than among natives.¹⁵

The US census data allow us to observe the full set of children born to each Italian immigrant.¹⁶ Depending on the specification, we either take the average name-based indices at the parent level, or construct a yearly parent-level panel in which parents are observed in each year they have a child.

Following Bazzi et al. (2019), we complement the name-based indices with measures of the probability of marrying Italians from other municipalities, provinces, and regions.

III.C Measuring Local and National Identity in Italy

Unlike in the United States, individual-level records from Italian censuses did not survive. To the best of our knowledge, no existing sources allow us to observe the universe of names. Here, we outline one strategy to address this data limitation and present two complementary measures constructed for the Italian case.

¹⁴Consider two names, n_1 and n_2 , each observed in exactly two municipalities. If n_1 is present in two distant municipalities while n_2 is present in two nearby municipalities, it may be desirable for a measure of particularism to classify n_2 as more local than n_1 .

¹⁵Formally, the FNI is defined as $FNI_n \equiv \Pr(\text{Name} = n \mid \text{Immigrant}) / (\Pr(\text{Name} = n \mid \text{Immigrant}) + \Pr(\text{Name} = n \mid \text{Native}))$.

¹⁶See Appendix A.I for additional details on the construction of completed fertility from US census data.

III.C.1 *World War 1 and World War 2 Casualty Data as Individual-Level Repositories of Names*

To circumvent the unavailability of individual census records, we collect names from two alternative sources: the repositories of World War 1 (WW1) deaths and World War 2 (WW2) casualties. Italy suffered approximately 650,000 military deaths in WW1 and 500,000 casualties in WW2.

WWI casualty data have been used in previous economics papers, including Acemoglu et al. (2022) and Coluccia and Spadavecchia (2025).¹⁷ The records report first and last name, father's name, year of birth, municipality of origin, military rank, unit, year, place, and cause of death. We georeference the municipality of origin and disambiguate first and last names. For 570,355 individuals, we observe complete information on name, year of birth, and municipality of birth.

WW2 casualty data have been recently digitized by the Italian Ministry of Defense. To the best of our knowledge, this paper is the first to use these data.¹⁸ We construct a dataset covering the universe of recorded casualties. The records report first name, surname, date and municipality of birth, and date and region of death or disappearance. As with the WWI data, we georeference the municipality of origin and clean first names to merge them with the baseline name indices. The final dataset contains information on name, municipality of origin, and year of birth for 408,940 individuals.

We use these data to construct a yearly series of municipality-level name-based indices. The time span of the resulting series corresponds to the birth cohorts who were eligible for military service during the two World Wars. Universal conscription was in force during World War I for cohorts born between 1874 and 1900 (Isnenghi and Rochat, 2008). Conscription during World War II was not universal, but it expanded substantially over the course of the war, with men born between 1900 and 1925 enrolled in the armed forces (Jowett, 2019). We therefore restrict the analysis to cohorts born between 1875 and 1925. The resulting sample consists almost exclusively of men, since women did not serve in combat roles in the regular army.

III.C.2 *Street Names*

We use street names as a first complementary indicator of national and local identities at the municipality level, building on established practice in urban studies (Azaryahu, 1996; Oto-Peralías, 2018). In Italy, municipal councils have near-complete discretion in naming new streets and, in some cases, the process involves local communities. Street names thus plausibly reflect the preferences of local legislators and community members.

We collect the universe of municipal streets in Italy in 2025 from OpenStreetMap. The resulting dataset contains information on street names, georeferenced locations, and lengths for 849,124 streets.¹⁹ We use a large language model (GPT5) to classify street names that reflect national and local identities. Examples of national identity street names include figures associated with the independence movement, such as Giuseppe Garibaldi and Giuseppe Mazzini, prominent historical figures,

¹⁷The dataset was compiled by the Fascist regime for propaganda purposes and is available at cadutigrandeguerra.it.

¹⁸Individual searches within the dataset are available at difesa.it.

¹⁹We exclude provincial, regional, and national roads, as well as highways, for two reasons. First, their names are not determined by municipal councils. Second, their names are typically administrative codes, as in the case of highways, or reflect only the places where they are located or the locations they connect.

such as Cristoforo Colombo and Giuseppe Verdi, and references to patriotic events, such as Giacomo Matteotti or the Battle of Isonzo. By contrast, local identity street names honor notable individuals from the municipality or the surrounding area.

In addition, we identify street names belonging to three placebo categories, which we use to verify that exposure to immigrant diversity is specifically related to national and local identities rather than to broader shifts in preferences. The placebo categories are arts, literature, and religion. Importantly, the same name may belong to more than one category. For example, Giosuè Carducci, a central figure in Italian literature at the end of the nineteenth century and winner of the Nobel Prize in 1906, was widely regarded as the “poet laureate” of the newly unified Italian state. As such, his name is classified as both national and literary. Moreover, the classification of a given street name generally depends on its location. A street named San Zeno is typically associated with religion, but in Verona, it is also classified as local, since San Zeno is the city’s patron saint. The full prompt used for classification is reported in Appendix A.II.

III.C.3 Monuments

We use monuments as a third indicator of national sentiment (Johnson, 1995). We construct a dataset covering the universe of nationally recognized monuments compiled by the Italian Ministry of Culture.²⁰ The data report the year of designation and the location of each monument. We restrict the sample to monuments designated between 1861 and 2000, since coverage prior to unification depends on the legislation of pre-unitary states. We georeference each monument and assign it to a municipality using GIS boundary data.

In addition, each monument is accompanied by an artistic and a historical description. We apply a large language model (GPT5) to these textual descriptions to distinguish monuments that signify national identity from all other monuments. As with streets, the prompt used for classification is reported in Appendix A.III. Monuments identified as expressing national identity typically commemorate soldiers from the municipality in which the monument is located who died in combat, most often, though not exclusively, during World War I and World War II. The dataset includes information on 32,325 monuments designated between 1861 and 2000, of which 36% (11,774) are classified as reflecting national identity sentiment.

III.D Other Data

III.D.1 United States

We employ the individual-level US census records to construct multiple economic and demographic variables that we include in the regressions as controls and use to check instrument balance (Ruggles et al., 2025). For each geographical unit of analysis—enumeration districts, hexagons, and counties—we assemble data on population, immigration, racial composition, age structure, and occupational and industry structures. All census-based controls are available at the census-decade frequency, i.e., decennially, between 1900 and 1930.

²⁰The dataset is available at dati.cultura.gov.it.

III.D.2 Italy

The Italian Statistical Office (ISTAT) compiled the population of each municipality from population censuses between 1861 and 2000. For each municipality, we compiled data on land area, altitude, and a dummy variable indicating whether the municipality has access to the coastline. In addition, we digitize the literacy rate from historical population censuses (1881, 1901, 1911, 1921, and 1931). Using the data described in Section III.C.1, we compute the number of deaths at the municipality level during WW1 and WW2. Lastly, we employ the railway dataset digitized by Coluccia and Spadavecchia (2025) to measure when each municipality was connected to the Italian railway network.

We complement these with coarser data at the levels of *circondario* and *provincia* compiled by Coluccia and Spadavecchia (2025).²¹ For each *circondario*, we have information on the agricultural and manufacturing employment and urbanization, defined as the share of individuals living in towns with at least 5,000 inhabitants. For each *provincia*, we know employment by manufacturing sectors and have proxies for technology adoption (number and horsepower of engines by power source).

III.E Construction of the Samples

III.E.1 Immigrants in the United States

We construct two samples of Italian immigrants in the United States using census records.

First, we compile a panel of Italian migrants observed between 1900, or the first census in which they appear, and 1930. We link migrants over time using the longitudinal links produced by the Census Linking and Tree projects (Price et al., 2021; Abramitzky et al., 2025; Buckles et al., 2025). In this sample, name-based indices are computed by averaging them across all children born in the ten years preceding each census decade. Migrants' locations correspond to their place of residence in each census year.

Second, we construct a panel in which migrants are observed repeatedly at each child's birth. In this case, a migrant's location is taken from the census record closest to the child's birth year.

In both samples, we measure the diversity of the Italian community at the enumeration district level. Enumeration districts were approximately ten times smaller than counties and, therefore, better suited to capture local immigrant communities, particularly since Italians clustered in urban areas (Eriksson and Ward, 2019). A limitation of enumeration districts is that their boundaries changed across censuses. To control for consistent sub-county characteristics, we construct a hexagonal spatial grid and assign individuals to grid cells using data from Berkes et al. (2023). Because the hexagons are time-invariant, hexagon-fixed effects control for geographic variation at a finer level than counties and remain consistent over time. Panel A in Table I provides descriptive statistics at the neighborhood (Panel A.1) and individual (Panel A.2) levels.

III.E.2 Panel Dataset of Italian Municipalities

The analysis for Italy is conducted at the municipality level. We construct three datasets that differ in their temporal granularity, reflecting the structure of each outcome variable.

²¹In 1926, there were 8,176 municipalities, 290 *circondari*, and 69 *province*.

First, name-based indicators are available at the cohort level. We therefore assemble a yearly panel tracking municipalities between 1870 and 1925. In the empirical analysis, we collapse the data to coarser time intervals to reduce short-term noise and to limit the incidence of empty cells, that is, municipality-year observations with no recorded individuals.

Second, the analysis of street names relies on a cross-sectional dataset in which each municipality is observed once.

Third, for monuments, we observe both their location and year of designation. We aggregate these data at the municipality level by decade and follow municipalities from 1870, corresponding to the 1860 to 1869 decade, through 2000.

In all samples, we use georeferenced coordinates to assign observations to municipalities defined by their 1991 boundaries, as historical GIS data on municipal borders are unavailable. Panel B in Table I lists descriptive statistics for the main outcome and explanatory variables employed in the analysis on national sentiment in Italy.

IV LOCAL, NATIONAL, AND AMERICAN IDENTITIES OF ITALIAN IMMIGRANTS

This section presents the main results we document on the emergence of national identity among Italians in the United States. Section IV.A discusses the empirical approach. In Section IV.B, we present the main results. Section IV.C explores how exposure to diversity influenced return rates of Italians. We conclude by presenting a complementary identification approach in Section IV.D. Section IV.E summarizes the robustness analyses.

IV.A Empirical Strategy

IV.A.1 *Measuring the Diversity of Italian Immigrant Communities*

We measure diversity of the Italian community by municipality of origin at the enumeration district level. Formally, let $d(t)$ be a district at time t . Denote with $n_{d(t)}^m$ the number of immigrants from municipality m located in district $d(t)$, and let $p_{d(t)}^m = n_{d(t)}^m / \sum_m n_{d(t)}^m$ be the share of migrants from that municipality. We measure the diversity of the Italian community as the Shannon entropy:

$$H_{d(t)} = - \sum_m p_{d(t)}^m \log(p_{d(t)}^m). \quad (3)$$

Entropy is weakly positive and increasing in the diversity of the Italian community. Shannon entropy is a widely employed metric in information theory. It reflects the idea that relatively more infrequent encounters are more informative than more frequent encounters (Posch et al., 2026). In our context, this property is appealing because exposure to a small number of immigrants from relatively distant municipalities is plausibly more salient than repeated exposure to many immigrants from the same place of origin. Results, however, are robust to using the fractionalization index, a more standard metric in economics that does not embed this feature.

In Figure D.6, we provide a visual depiction of Italian segregation in the United States. The blue line shows the distribution of the average distance between each immigrant's municipality of origin and that of their neighbors, defined as other Italian immigrants residing in the same enumeration

district. We compare this distribution to counterfactual benchmarks generated by simulating location choices under random sorting.²² The observed distribution exhibits excess mass at shorter distances, indicating that migrants from geographically proximate municipalities were more likely to settle together. However, segregation is far from complete: the average observed distance between neighbors' municipalities is only 14% lower than the counterfactual mean under random sorting, implying that Italian immigrant communities remained considerably diverse.

IV.A.2 Baseline Specification

The baseline analysis employs the individual-level panel dataset described in Section III.E.1. We estimate variations of the following specification:

$$y_{ihdt} = \alpha_{m(i)} + \alpha_h + \alpha_t + \beta \cdot H_d(t) + X_d' \Gamma_1 + X_i' \Gamma_2 + \varepsilon_{ihdt}, \quad (4)$$

where i , h , d , and t denote an individual, the hexagon of residence, the enumeration district, and the census year, respectively. Municipality-of-origin fixed effects ($\alpha_{m(i)}$) control for time-invariant origin characteristics such as average population size, emigration intensity, and geographic location. Area-level fixed effects (α_h) absorb time-invariant local characteristics, including location and proximity to major ports of entry. Time-fixed effects (α_t) capture aggregate shocks, such as business-cycle fluctuations, that affect migration flows to the United States.

Because enumeration district boundaries change over time, we cannot include district fixed effects. We therefore estimate variants of (4) that include hexagon-by-time fixed effects ($\alpha_{h \times t}$). These absorb time-varying unobserved heterogeneity at sub-county spatial units and purge confounding variation driven by local factors such as economic conditions that evolve within counties. We also report specifications that include municipality-by-time fixed effects ($\alpha_{m(i) \times t}$). These absorb time-varying heterogeneity at the origin level, including population growth, which mechanically generates larger emigrant flows, as well as local economic conditions in Italy. In addition, they account for the expansion over time of origin-specific migrant networks.

In various specifications, we include district-level controls (X_d) capturing demographic characteristics (population, the share of women, the share of African Americans, the share of immigrants, and the share of Italians) and economic characteristics (employment shares across occupations). In addition, some specifications include individual-level controls (X_i), namely occupation, gender, birth year, and year of immigration.

The coefficient of interest (β) captures how the outcome y responds to changes in the diversity of the local Italian immigrant community. We consider four classes of outcomes: the local name index and its distance-adjusted variant; the foreign name index; and a set of indicators capturing the probability that individual i married an Italian migrant originating from a different municipality, province,

²²To benchmark observed segregation against random sorting, we construct counterfactual neighbor-distance distributions via permutation within districts. For each individual i and district d , we define the pool of potential neighbors as all Italian immigrants observed in the same census year and district, excluding members of i 's household. Let n_{id} denote the number of such neighbors in i 's observed district. In each simulation draw, we sample n_{id} individuals at random from the pool and compute the mean distance between i 's municipality of origin and those of the sampled individuals. We run 100 simulations per individual, yielding a counterfactual distribution under random assignment that preserves group sizes and household structure.

or region.

IV.A.3 Identification Challenge and Instrumental Variable Design

Identification Challenges. Identification of the causal effect of diversity in (4) is potentially problematic for two reasons.

First, the origin mix within a district may reflect selective sorting of immigrants from specific municipalities across US locations. If these groups differ in unobserved traits related to identity and assimilation, then $\hat{\beta}$ may partly capture omitted origin- and destination-level heterogeneity.

Second, $H_{d(t)}$ may respond endogenously to time-varying local conditions. Destination shocks can simultaneously affect outcomes and the composition of the Italian community by shifting migration flows and cohort retention. Relatedly, residential mobility and marriage-market choices may feed back into measured diversity, inducing simultaneity.

Instrumental Variable Design. To address these concerns, we construct an instrument based on quasi-random exposure to Italian diversity induced by ship-level composition. Building on Dippel and Heblich (2021), we hypothesize that interactions during the transatlantic voyage, which could last more than 50 days, as documented by Hatton (2024), shaped subsequent settlement patterns. Under this hypothesis, we can isolate plausibly exogenous ship-level variation in the origin mix of passengers to estimate the causal effect of exposure to diversity.

Consider individual i from municipality $m(i) = m$ who traveled on ship $s(i) = s$.²³ Let x_{os} denote the number of immigrants from municipality o who traveled on ship s , and let p_{os} be the share of immigrants from municipality o on s , so that $\tilde{p}_{os} = p_{os} / (1 - p_{ms})$ is the associated normalized share.²⁴ The instrument for individual i is then given by

$$\hat{H}_i = - \sum_{o \neq m} \tilde{p}_{os} \log \tilde{p}_{os}. \quad (5)$$

In practice, we compute $\hat{H}_i$ only on ships with at least 100 immigrants (95% of the total) to avoid numerical instability. We estimate regression (4) using the instrument in (5) via two-stage least squares (2SLS).

The instrument associated with each individual excludes shipmates born in the same municipality. This leave-out methodology ensures that the instrument does not mechanically reflect pre-existing social networks. Results remain quantitatively unchanged when excluding immigrants from the same province, indicating that residual social networks not captured by excluding own-municipality shipmates are unlikely to drive the results.

²³A “ship” is defined as a unique ship-year pair. We omit the year variable and simply call it a “ship” to avoid notational burden.

²⁴Municipality shares on ship s are computed as $p_{os} = (x_{os} + \lambda) / (N_s + \lambda K)$, where x_{os} is the number of passengers from municipality o , $N_s = \sum_o x_{os}$ is the total number of passengers on the ship, and K is the number of distinct municipalities represented. Absent smoothing, these shares are the empirical frequencies from a multinomial distribution over municipalities induced by the ship’s passenger list. Because $\log(p_{os})$ appears in Shannon entropy, small counts generate extreme or undefined values, especially in leave-one-out transformations. We therefore apply Laplace smoothing ($\lambda = 1$), which amounts to adding one observation per municipality. This regularizes entropy in finite samples without affecting large ships, as the influence of smoothing vanishes as N_s grows.

Instrument Relevance. We implement two analyses to verify that ship diversity predicts diversity exposure at the neighborhood level.

First, we regress the endogenous variable (3) on the instrument (5), conditional on the controls we include in the baseline regression specification. Figure IIIa displays the result. We find a positive, strongly statistically significant association between the two variables, indicating that ship-entropy is a relevant predictor of Italian community diversity.

Second, we implement a simple simulation approach building on Basse et al. (2024), who developed a randomization test for peer effects in group formation.²⁵ We test whether shipmates from different municipalities are disproportionately likely to co-locate after arrival. For each ship, we compute the number of passengers who end up in the same district. We compare this probability to a benchmark in which the same passengers are assigned to districts at random. We compute the benchmark distribution by simulation. Excess cross-municipality co-location relative to this random benchmark indicates that ship-level interactions predict subsequent residential clustering beyond what is implied by local population sizes.

Figure IIIb reports the distribution of the ship-level excess co-location statistics against its asymptotic distribution under the null of no co-location (in blue). We find that passengers of the same ship are substantially more likely to settle in the same district. For example, we find more than four times as many ships with a p -value beyond 5% than what would be implied by the null of no co-location.

Evidence in Support of the Identification Assumption. Identification relies on the assumption that, conditional on fixed effects and controls, ship composition influences outcomes only by shifting post-arrival exposure to other Italian origins. Under this exclusion condition, $\hat{H}_i$ provides plausibly exogenous variation in $H_{d(t)}$, and the resulting IV estimates recover the causal effect of exposure to a more diverse Italian origin mix on identity formation.

We implement two exercises to assess the empirical plausibility of this assumption. First, we examine balance by regressing predetermined area characteristics on the instrument, conditioning on the same fixed effects and controls as in the baseline specification. Figure D.7a reports the resulting coefficients. Because enumeration districts cannot be tracked consistently over time, we construct these predetermined characteristics at broader geographic levels and report results at the county level (in blue) and the hexagon level (in red). Across a wide set of variables capturing the diversity of the Italian community, local demographics, immigrant composition, and employment shares by industry and occupation, the estimates are small and statistically indistinguishable from zero, indicating no systematic correlation between the instrument and pre-migration area characteristics.

Second, we conduct a placebo exercise by regressing the instrument on one-decade leads of individual characteristics that should not be affected by ship diversity. These include demographic variables, which are fixed by construction, and measures of employment status by industry and occupation. Figure D.7b reports the resulting estimates. We find no statistically significant associations, suggesting that the instrument is not systematically related to individual characteristics that may in-

²⁵Appendix A.IV discusses in greater detail the simulation-based approach.

fluence location choices or to forward-looking outcomes that may indicate residual correlation with unobserved determinants of identity formation.

IV.B Local, National, American Identity and Neighborhood Diversity

IV.B.1 Descriptive Patterns of Assimilation After Arriving in the US

We begin our analysis of the effect of neighborhood diversity on national identity formation with a descriptive exercise. In Figure IV, we display the evolution of the local name index (Figure IVa) and the foreign name index (Figure IVb) for children of immigrants by the number of years since migration to the United States. We split the baseline sample into two groups: red dots represent immigrants who settle in neighborhoods with above-median Italian community diversity, whereas blue dots represent immigrants who settle in neighborhoods with below-median Italian community diversity.

Importantly, migrants were plausibly similar in terms of their LNI and FNI at the time of migration, regardless of the neighborhood they settled in. We detect no statistically significant difference in LNI or FNI for migrants who had been living in the United States for just one year. This indicates that we are comparing individuals who were not differentially selected on levels of particularism or assimilation at arrival.

The trajectories of the LNI and FNI, however, diverge markedly over time. Migrants who settle in low-diversity areas exhibit no change in name particularism and experience substantial name-based assimilation. Ten years after arrival in the United States, their children bear names that are no longer differentially local, but are 5% more similar to English-sounding names. This pattern of gradual assimilation is consistent with existing evidence (Abramitzky et al., 2020).

Migrants who settle in high-diversity areas, by contrast, experience a pronounced decline in name-based particularism. Ten years after migration, their children bear names that are 11% less local. Differences in name particularism emerge as early as two years after arrival and remain relatively stable thereafter. In contrast, even ten years after arrival, the children of migrants in high-diversity areas do not display significantly more foreign-sounding names than at arrival. This pattern suggests substantially slower, or even absent, cultural assimilation into US society among migrants exposed to more diverse Italian communities.

IV.B.2 Name-Based Assimilation: OLS and 2SLS Results

To provide more formal evidence for these descriptive patterns, Table II reports estimates of specification (4). Panels A, B, and C present results using the local name index (1), the distance-adjusted local name index (2), and the foreign name index as dependent variables, respectively. Columns (1–6) report OLS estimates. Columns (7–8) present the reduced-form effects of the instrument in (5) on each outcome, while columns (9–10) report the corresponding 2SLS estimates.

Depending on the specification, the OLS estimates indicate that a unit increase in the entropy of the Italian community is associated with a 0.12–0.17 decrease in the local name index. To put these magnitudes into perspective, a one-standard-deviation increase in Italian community diversity corresponds to a 4%–5%-standard-deviation decline in the local name index. Panel B reports results using the distance-adjusted LNI. Because higher values of the DLNI indicate names that are more

geographically widespread within Italy, increases in this index capture stronger national identity. We find that a unit increase in Italian community diversity is associated with a 0.02 increase in the DLNI, implying that a one-standard-deviation increase in diversity yields a 10%-standard-deviation increase in the distance-adjusted local name index. Column (3) reports the complementary effect of Italian diversity on the foreign name index, which, following Abramitzky et al. (2020), we interpret as a proxy for cultural assimilation in US society. A unit increase in Italian diversity is associated with a 0.02 increase in the FNI. Analogously, a one-standard-deviation increase in Italian diversity corresponds to an 11%-standard-deviation increase in the foreign name index.

These estimates are remarkably stable across a range of specifications. In column (2), we replace county fixed effects with hexagon fixed effects to control for more granular time-invariant local characteristics. In column (3), we include municipality-of-origin-by-time fixed effects, which absorb time-varying unobserved heterogeneity at the origin level. For instance, population growth may increase migrant outflows from a given municipality, potentially affecting the likelihood of settling near migrants from the same origin. In column (4), we add individual-level time-invariant controls from the US census, including occupation, gender, birth year, and year of immigration. Finally, columns (5) and (6) incorporate neighborhood-level time-varying controls to account for demographic factors—population size, the share of African Americans, immigrants, and Italian immigrants—and economic conditions, measured by employment shares across industries, that may independently influence assimilation.

As discussed in Section IV.A.3, OLS estimates may not recover the causal effect of exposure to Italian diversity due to selective sorting and simultaneity concerns. We therefore report reduced-form estimates based on the ship-level instrument defined in (5) in columns (7–8), and the corresponding 2SLS estimates in columns (9–10). The instrumental-variable estimates corroborate the OLS results and are substantially larger in magnitude. Specifically, the 2SLS coefficients are approximately an order of magnitude larger than their OLS counterparts.

The difference in magnitude between the OLS and IV estimates is consistent with the instrumental-variable strategy identifying a local average treatment effect (LATE). The ship-based instrument shifts neighborhood diversity for a subset of migrants whose settlement locations were shaped by interactions with shipmates upon arrival. For these compliers, quasi-random variation in ship composition translated into meaningful differences in post-arrival exposure to Italians from other municipalities.

There are two reasons why the causal effect for this group may exceed the average effect captured by OLS. First, migrants whose residential choices were influenced by shipmate networks were likely more socially embedded within Italian communities, making them particularly responsive to within-group diversity. Second, selective sorting and measurement error in neighborhood composition are likely to attenuate OLS estimates, as migrants with stronger pre-existing identity or assimilation preferences may self-select into particular locations. By contrast, the IV isolates variation in exposure that is orthogonal to such preferences and therefore recovers the effect for individuals whose social environment was plausibly exogenously assigned.

Under this interpretation, the IV and OLS estimates are not directly comparable: the latter cap-

ture an average association in the population, whereas the former identify a treatment effect for a specific subpopulation. Nonetheless, the IV results indicate that exogenous exposure to Italian diversity causally increased national identity formation and reduced cultural assimilation into US society, under the maintained exclusion restriction.

These results highlight two findings. First, the composition of the immigrant community matters for assimilation into the host country's society: exposure to greater within-origin diversity strengthens national identity and slows assimilation. As a result, focusing solely on the size of immigrant groups by country of origin—as is standard in economics—misses the role played by the *composition* of diasporic communities. Second, inter-group contact among migrants reallocates identity within the immigrant population, fostering a shared national identity rather than accelerating convergence toward native cultural norms, as predicted by the theoretical framework in Section I.

IV.B.3 Out-Group Marriage Patterns

Marriage patterns provide an additional proxy for group identity formation (Abramitzky et al., 2020). In our context, Gordon (1964) documents the central role of marriages between migrants and natives, as well as between migrants from different origin groups, as indicators of immigrant assimilation. We distinguish three types of marriages, based on whether spouses originate from different municipalities, different provinces, or different regions.

In more diverse immigrant communities, individuals are more likely to marry outside their municipality of origin simply because fewer potential spouses from the same origin are available. As a result, raw inter-municipality marriage rates may mechanically increase with diversity, even in the absence of changes in identity or preferences. To address this issue, we follow Bazzi et al. (2019) and construct supply-adjusted measures of inter-origin marriage that net out the role of local marriage-market composition. Specifically, we compare observed inter-origin marriages with the intergroup marriage probabilities that would arise under random matching between spouses irrespective of origin. We interpret the residual as reflecting identity-driven integration rather than mechanical supply effects.²⁶

Figure Va displays the results. Each line reports estimates from equation (4), comparing specifications that do not correct for the mechanical association between diversity and intergroup marriage probabilities (red dots) with those that account for it (blue dots). Exposure to more diverse Italian communities is associated with a higher probability of intergroup marriage. When controlling for supply-side effects, a unit increase in Italian community diversity raises the probability of marrying an immigrant from another municipality by 4%. Given that such marriages were relatively rare, this effect corresponds to approximately 22% of the mean. Exposure to more diverse communities also has a positive, statistically significant effect on the probability of marrying an Italian from another province. By contrast, once we correct for supply-side effects, we find no effect on marriages with

²⁶For each married individual i , define the marriage market $\mathcal{M}(i)$ by census year and enumeration district, and let $\mathcal{S}_{-i}$ denote the set of opposite-sex individuals in $\mathcal{M}(i)$. The supply-only benchmark is constructed by repeatedly drawing a spouse $j \in \mathcal{S}_{-i}$ uniformly at random and computing $\mathbb{E}[I\{g(i) \neq g(j)\} \mid \mathcal{M}(i)]$, where $g(\cdot)$ denotes origin (municipality, province, or region). We net out this benchmark from observed inter-origin marriage outcomes in the estimating equation.

migrants from other regions.²⁷

These patterns provide evidence consistent with previous findings that diversity within the Italian community contributed to the formation of a shared “national” group identity.

IV.C Return Migration and Neighborhood Diversity

A final indicator of group identity formation is the propensity of migrants to return to their origin country. Among all emigration countries, Italians had the highest return rates (Bandiera et al., 2013). However, if exposure to diversity within immigrant communities fostered a shared national identity that was distinct from local particularism in Italy, migrants in more diverse communities should have been less inclined to return.

Data on return migration are notoriously scarce. Neither the US nor Italian authorities collected systematic, spatially disaggregated data on return migrants. We propose a methodology to partially address this limitation that uses longitudinal links of US census records. We derive a differential return estimator and use it to examine whether return rates among Italian immigrants varied with the diversity of the Italian communities they were exposed to.

Let $I \in \{0, 1\}$ indicate nationality of an individual, where $I = 1$ denotes Italians and $I = 0$ denotes natives, and let $D \in \{H, L\}$ indicate whether the individual lives in a high- or low-diversity neighborhood. We observe whether an individual present in the census at time t can be linked forward to the US census at $t + h$; let $L_{t+h} = 1$ if the individual is successfully re-observed, and $L_{t+h} = 0$ otherwise. We define the forward-link probability as $\ell_{I,D} = \Pr(L_{t+h} = 1 \mid I, D)$.

Linking forward in the US census requires not returning to Italy and, separately, not disappearing due to other factors such as death, internal migration within the US, or linkage error. Let $R = 1$ denote a return to Italy between t and $t + h$, and let $M = 1$ denote disappearance without return. An Italian individual in a D -neighborhood remains observable in $t + h$ if and only if $R = 0$ and $M = 0$. The forward-link probability for Italians is therefore $\ell_{1,D} = 1 - r_{1,D} - m_{1,D}$, where $r_{1,D} = \Pr(R = 1 \mid I = 1, D)$ is the (latent) return probability and $m_{1,D} = \Pr(M = 1 \mid I = 1, D)$ captures all other forms of attrition. For natives, $r_{0,D} = 0$, so $\ell_{0,D} = 1 - m_{0,D}$.

A naïve estimator for the return probability is the difference in forward-link rates between Italians and natives within neighborhoods, $\ell_{1,D} - \ell_{0,D}$. Substituting the expressions above yields $\ell_{1,D} - \ell_{0,D} = -r_{1,D} - (m_{1,D} - m_{0,D})$, which implies that this difference would identify the return rate $r_{1,D}$ only if $m_{1,D} = m_{0,D}$, i.e., Italians and natives must have the same non-return disappearance rate. This is a demanding condition: if, as is plausible, Italians and natives have different linkage success rates, then $\ell_{1,D} - \ell_{0,D}$ cannot be interpreted as a return probability.

To relax this assumption, we compare how much more Italians disappear relative to natives in high-diversity areas versus low-diversity ones. Consider the “difference-in-differences” object ($\ell_{1,H} -$

²⁷Two reasons may explain why we do not find an effect of diversity on out-of-region marriages. First, regions may be too coarse to capture the identity. Given that there are only 21 regions in Italy and that most migrants originated from Southern regions, there is limited variation in marriages with individuals from other regions. Second, the supply-side correction is strongest at coarse aggregation levels, implying that once this benchmark is netted out, there is little residual variation in marriage patterns left.

$\ell_{0,H}) - (\ell_{1,L} - \ell_{0,L})$. Plugging in the decomposition above yields our differential return estimator:

$$\varrho \equiv (\ell_{1,H} - \ell_{0,H}) - (\ell_{1,L} - \ell_{0,L}) = -(r_{1,H} - r_{1,L}) - [(m_{1,H} - m_{0,H}) - (m_{1,L} - m_{0,L})] \quad (6)$$

Estimator ϱ recovers the difference in return rates between high- and low-diversity neighborhoods, $r_{1,H} - r_{1,L}$, provided that the Italian-native attrition gap unrelated to return, $m_{1,D} - m_{0,D}$, is stable across neighborhood types. This parallel attrition bias condition, because it only requires that any Italian-native difference in non-return disappearance behaves similarly in high- and low-diversity areas, is weaker than demanding $m_{1,D} = m_{0,D}$ in levels. Under this assumption, ϱ isolates the extent to which the additional disappearance of Italians in high-diversity areas reflects true return migration rather than differential mortality, internal migration, or linkage failure.

Figure Vb displays the estimated values of ϱ across age bins, where H and L denote, respectively, districts with above- and below-median diversity of Italian migrants.²⁸ We find that return rates are between 10% (for ages 0–10) and 5% (for ages 25–29) lower in high-diversity neighborhoods relative to low-diversity neighborhoods. Bandiera et al. (2013) estimate that more than 60% of Italians returned over the course of a single decade, implying that exposure to relatively more diverse communities reduced the propensity to return by approximately 8%.²⁹

Lower return rates among migrants in high-diversity districts are consistent with the interpretation that diversity fostered a durable sense of national belonging, thereby weakening ties to local communities in Italy.

IV.D Movers Design

We complement the analysis with a design along the lines of Chetty and Hendren (2018), which leverages individuals who move across neighborhoods (“movers”) to identify the effect of exposure to diversity of the Italian community on national identity formation.

We employ the dataset described in Section III.E.2, in which each individual is observed whenever they have a child. Consider individual i , who has a child at times t and t' . We define a mover as anyone who moves between two hexagons between t and t' .³⁰ We denote the “diversity shock” i is exposed to upon moving as $D_i \equiv H_{d'(i)} - H_{d(i)}$, which is simply the difference in diversity between the new and the old neighborhood where i lives. We binarize D_i into a categorical variable, call it *Diversity Shock* $_i$, equal to one if D_i is in the top 20% of the distribution and zero otherwise, and denote t_i^* as the first year when i moves between two hexagons.

²⁸To estimate the differential return ϱ , we construct 30-year forward links. Results are qualitatively similar when using alternative horizons. Standard errors are computed using a district-level cluster bootstrap with 100 replications. For each age group, we resample districts with replacement and recompute the collapsed difference-in-differences estimand; reported standard errors are the bootstrap standard deviation.

²⁹This comparison should be interpreted with caution. The methodology in Bandiera et al. (2013) is not directly comparable to our pseudo-return estimator, and we focus on long-run return decisions over a 30-year horizon, whereas their estimates concern ten-year return spells.

³⁰We use hexagons rather than districts because we cannot consistently follow districts over time. The diversity of the local community, however, is measured at the district level, as in the rest of the analysis.

We estimate variations of the following specification:

$$LNI_{idht} = \alpha_i + \alpha_{h \times t} + \sum_{\substack{k=-10 \\ k \neq 0}}^{+10} \beta_k \cdot I(t - t_i^* = k) \cdot \text{Diversity Shock}_i + \varepsilon_{idht}, \quad (7)$$

where α_i denote individual fixed effects and α_{ht} denote hexagon-by-time fixed effects. The treatment effect coefficients $\{\beta_k\}$ capture differences in the local name index between movers and non-movers before and after movers change their neighborhood.

Figure VI presents the estimates from equation (7). Identification relies on a standard parallel trends assumption requiring that, conditional on fixed effects, treated and control individuals would not have experienced diverging LNI trajectories if the former had not moved to a different neighborhood. While this assumption is untestable, we find that pre-move coefficients $\{\beta_k\}_{k=-10}^{-1}$ are never statistically significantly different from zero, indicating that movers and non-movers were not systematically different before the former moved to a different neighborhood.

Over the decade following the move, we estimate that migrants who experienced a “diversity shock” gave their children names that were 10% less local compared to individuals who did not move or moved to neighborhoods with diversity comparable to their origin district. This effect builds over time and is largest 6 to 7 years after migrants move. Quantitatively, exposure to diverse neighborhoods upon moving decreases LNI by approximately 9% of a standard deviation.

In Table III, we report estimates from a mean-shift specification in which the event-time indicators are replaced by a categorical term $I(t \geq t_i^*)$ that equals one for all observations after the treatment period and zero otherwise. In the baseline specification, shown in column (1), we include individual, area, and event-time fixed effects. Individual fixed effects imply that identification relies on within-individual changes in LNI before and after treatment. In column (2), we add birth-year fixed effects to control for cohort-specific shocks that may affect trends in name choice and correlate with the local name index. In column (3), we replace area and cohort fixed effects with area-by-cohort fixed effects, absorbing neighborhood-level time-varying shocks such as changes in local labor market conditions or immigrant inflows. Column (4) replaces the binary diversity indicator with a continuous treatment measure and confirms the qualitative results.³¹

In column (5), we add an interaction between the age at which migrants moved and the baseline treatment to test whether the effect of exposure to diversity is age-specific. We do not find evidence in this direction. The estimated difference-in-differences coefficient remains negative, although it is less precisely estimated, while the interaction term between the treatment and age at move is not statistically significant.

Last, in column (6), we replicate the analysis using the distance-adjusted LNI metric, which confirms the result obtained from the unweighted LNI indicator. Exposure to more diverse neighborhoods has a positive and statistically significant effect on the distance-adjusted local name index, indicating *less* name particularism, equal to approximately 15% of its standard deviation.

³¹Because non-movers do not experience a cross-sectional diversity shock, we apply a monotone increasing transformation to the continuous diversity measure (exponential) and impute zero exposure for non-movers.

These results complement evidence presented in the previous sections that exposure to diversity within the Italian community contributed to in-group “national” identity formation.

IV.E Robustness Analysis

A first concern regarding our name-based indicators is that naming decisions for firstborn children may reflect family tradition rather than parents’ identity choices. In Table C.2 and Table C.5, we therefore recompute all name-based measures excluding firstborn children. The results remain quantitatively unchanged.

The local name index is computed at the municipal level to maximize the granularity of information contained in first names. However, one may be concerned that municipalities are too small to meaningfully capture identity. In column (1) of Panels A and B in Table C.4 and Table C.7, we therefore recompute the LNI at the province-of-origin level. Provinces are substantially larger administrative units—Italy has only 107 provinces, compared to more than 8,000 municipalities—so the province-level index reflects a coarser layer of identity. The results based on province-level and municipality-level LNI are quantitatively similar.

In the baseline definition of Italian community diversity in (3), we measure diversity at the enumeration district and municipality levels. Enumeration districts are small and, we argue, better suited than counties to proxy for the local communities immigrants interacted with in the US. Municipalities, in turn, provide an appropriate proxy for local social networks in the origin country. In Table C.4 and Table C.7, we replicate the analysis along two dimensions. First, in columns (3–4) and (5–6), we recompute the diversity measure at the province and region levels, respectively. Second, in columns (2), (4), and (6), we recompute diversity at the US county level. Across these alternative definitions, we consistently find a negative association between Italian diversity and name-based particularism. Estimates are less precise when diversity is measured at the county level, which is expected: because counties are substantially larger than the relevant social units migrants interacted with, aggregating diversity at this level introduces attenuation bias due to measurement error.

A related concern regarding the measure of the diversity of the Italian immigrant community in (3) is that Shannon entropy is not the most commonly used diversity metric in economics. In Table C.3–Table C.6, we therefore replicate the baseline estimates using the fractionalization index to measure the diversity of the Italian immigrant community and obtain qualitatively unchanged results.³²

Italian immigrants were concentrated in a small number of states—most importantly New York, Pennsylvania, New Jersey, and Massachusetts—raising the concern that our results may be driven by idiosyncratic characteristics of specific locations rather than by the diversity of Italian communities per se. In Figure D.8 and Figure D.9, we therefore replicate the baseline analysis while sequentially excluding one state at a time from the estimation sample. The resulting OLS and 2SLS estimates remain stable and statistically indistinguishable from the baseline coefficient regardless of the state

³²The fractionalization index is defined as one minus the Hirschman-Herfindahl index. Formally, let s_g denote the share of migrants originating from municipality g in a given district. The fractionalization index is given by $F = 1 - \sum_g s_g^2$. This measure captures the probability that two randomly drawn individuals belong to different origin groups and is widely used as a measure of ethnic or group diversity (e.g., see Alesina et al., 2003).

we exclude.

Similarly, selective sorting across US locations by migrants from specific areas of Italy may induce spatial autocorrelation and potentially inflate statistical significance. To address this concern, in Figure D.10 and Figure D.11 we report confidence intervals obtained using a range of alternative variance estimators. In particular, we show that statistical significance is unchanged when adopting different clustering strategies across US locations and when applying the Conley (1999) spatial correction with varying distance cutoffs. Similarly, the estimates remain statistically significant when clustering the standard errors by the origin of immigrants, at different layers of spatial granularity, and when accounting for spatial autocorrelation at the origin level.

V EMIGRATION AND THE RISE OF NATIONAL IDENTITY IN ITALY

We now turn to Italy. We explore how the sentiment of national identity that emerged among US emigrants contributed to the rise of national identity in Italy. Section V.A reports the results obtained using names as a measure of identity. In Section V.B and Section V.C, we present the results using street names and monuments as ancillary proxies. Section V.D discusses the robustness analyses.

V.A Name-Based National Identity

Our primary measure of national identity formation in Italy relies on given names, as described in Section III.C.1. The idea is to trace how name-based particularism in each municipality evolved after migrants from that origin were exposed, while in the United States, to Italians from other parts of the country. To this end, we implement two empirical strategies.

V.A.1 Panel Regression

First, we estimate a set of panel regressions of the following form:

$$y_{mt} = \alpha_m + \alpha_t + X'_{mt}\Gamma + \sum_{\tau=1880}^{1930} \beta_{\tau} \cdot \text{Diversity Exposure}_m \cdot I(t = \tau) + \varepsilon_{mt} , \quad (8)$$

where m indexes municipalities and t indexes years, with municipality and year fixed effects given by α_m and α_t . The vector X_{mt} collects time-varying municipality-level controls. In some specifications, X_{mt} additionally includes the number of emigrants from m during the Mass-Migration period interacted with year dummies, allowing outcomes to flexibly evolve with emigration intensity. The coefficients of interest are β_{τ} , which trace the differential evolution of y_{mt} over time as a function of diversity exposure. We define $\text{Diversity Exposure}_m$ as the share of migrants from municipality m during the Mass-Migration period who settled in US neighborhoods in the top 20% of the baseline entropy distribution in (3).

Because Italian emigration to the United States was concentrated in the period 1895–1921, the coefficients for the pre-migration years in (8) provide a natural falsification test. If the treatment effect coefficients capture the effect of exposure to diversity through migration, there should be no systematic differential evolution between high- and low-exposure municipalities before large-scale emigration began, that is, in 1880–1895. This logic mirrors the interpretation of pre-treatment placebo coefficients

in standard event-study designs.

V.A.2 Difference-in-Differences Regression

To implement a formal difference-in-differences design, we define a municipality's treatment time as the first year in which its annual emigration rate exceeds 1‰ of the 1880 population. This choice is motivated by Spitzer and Zimran (2023), who show that US-bound emigration followed an S-shaped diffusion pattern. We interpret the 1‰ threshold as the point at which out-migration gains momentum. While the threshold is somewhat arbitrary, results are robust to alternative cutoffs between 1‰ and 5‰ (see Table C.8).

Formally, we estimate variations of:

$$y_{mt} = \alpha_m + \alpha_t + X'_{mt}\Gamma + \sum_{\substack{\tau=-10 \\ \tau \neq 0}}^{20} \beta_\tau \cdot \text{Diversity Exposure}_m \cdot I(t - t_m^* = \tau) + \varepsilon_{mt}, \quad (9)$$

where t_m^* denotes the first year in which the emigration rate exceeds 1‰. The control vector X_{mt} always includes an indicator $I(t \geq t_m^*)$ that equals one once municipality m enters the mass-migration period. The coefficients β_τ therefore trace event-time differences in outcomes for municipalities with higher $\text{Diversity Exposure}_m$ relative to those with lower exposure, before and after the onset of mass emigration.

Identification in (9) relies on a parallel-trends assumption. In particular, absent exposure to diverse Italian communities in the United States, municipalities with higher $\text{Diversity Exposure}_m$ would have exhibited the same evolution in y_{mt} around t_m^* as other municipalities. We assess this implication by reporting the pre-treatment coefficients β_τ for $\tau < 0$, which are small in magnitude and statistically indistinguishable from zero; we discuss this evidence in the next section.

V.A.3 Results

In Italy, the mass migration period coincided with persistent localism and fragmentation, notwithstanding sustained efforts by Fascist authorities to promote national identity formation (De Mauro, 1963). Figure VIIa plots the average local name index of Italian men born between 1880 and 1920. Over this period, the LNI increased by more than 30%, indicating a shift toward more locally concentrated naming practices.

Figure VIIb reports the estimates from equation (8). Italian municipalities whose migrants later settled in more diverse US locations do not differ from other municipalities between 1885 and 1900. Because large-scale emigration began in the mid-1890s, this pattern indicates that treated and control municipalities were comparable in terms of name particularism prior to the onset of mass migration. Beginning in the 1910s, however, municipalities whose migrants were exposed to relatively more diverse Italian communities in the United States exhibit significantly lower levels of local naming than control municipalities. The magnitude of this difference increases over time and is largest by the late 1920s.

In Table IV, we report a set of alternative specifications. Column (1) includes municipality and year fixed effects. Column (2) additionally controls for interactions between the total number of US

emigrants during the mass migration period and year fixed effects. This specification thus compares municipalities with similar emigration intensity and exploits variation in the destinations where migrants settled in the United States. The results confirm that the LNI declined in municipalities whose migrants were exposed to more diverse Italian communities in the US, while the pre-migration coefficients remain small in magnitude and statistically indistinguishable from zero. Column (3) documents a similar pattern when using the distance-adjusted local name index as the outcome.

Figure VIIc presents the dynamic treatment effects estimated from equation (9). Prior to the onset of mass emigration, defined as the year in which a municipality's emigration rate to the United States first exceeds 1% of the 1881 population, differences in the average LNI between treated and control municipalities are small and statistically insignificant. Following the start of the mass migration, naming practices in treated municipalities became progressively less local relative to those in control municipalities. As in Figure VIIa, this gap widens over the subsequent two decades. On average, municipalities whose migrants settled in relatively more diverse areas of the United States experience a 52% decline in the local name index relative to the mean.

In Table IV, we report results from alternative specifications of equation (9). Column (4) includes municipality and year fixed effects. Column (5) further controls for the total number of US emigrants between 1892 and 1924, interacted with period dummies, thereby exploiting variation in emigrants' settlement destinations among municipalities with comparable emigration intensity. Column (6) documents a similar pattern when we employ the distance-adjusted local name index as the outcome variable.

Across all specifications, we also report the coefficient on a secondary treatment indicator that equals one after a municipality enters the mass migration period. The coefficient of this term remains consistently small in magnitude and statistically insignificant, indicating that transatlantic emigration per se did not affect local identity. Instead, this result suggests that increased national identity in treated municipalities was driven by intergroup contact between emigrants from each municipality and Italians from other parts of the country.

In Figure VIId, we examine heterogeneity in the effect of exposure to diversity through migrant communities by the size of the origin municipality. We estimate the baseline mean-shift specification in (9) and interact the difference-in-differences term with indicators for quintiles of municipal population.³³ The gray bars report the average local name index in the pre-treatment period for municipalities in each quintile. As expected, smaller municipalities exhibit higher levels of name particularism, reflecting stronger local identities in smaller and more isolated communities. Exposure to more diverse Italian communities in the United States reduces the LNI across all population quintiles. However, the decline is quantitatively larger in larger municipalities, whose estimated treatment effect differs statistically from the ATEs in each other population quintile, indicating that transatlantic emigration amplified pre-existing differences in particularism between large and small centers.

³³In this specification, we additionally include population-quintile-by-year fixed effects to absorb time-varying shocks common to municipalities of similar size that may affect naming practices.

V.B Street Naming Patterns

Street names provide an alternative proxy for national identity formation, as municipal councils exercise discretion over the naming of new streets. We classify street names into two identity-related categories, local and national, and three “placebo” categories, art, literature, and religion. Our dataset consists of a cross-section covering the universe of municipal streets in Italy as of 2025.

We estimate variations of the following specification:

$$y_m = \alpha_{p(m)} + X'_m \Gamma + \beta_1 \cdot Diverse_m + \beta_2 \cdot Homogeneous_m + \varepsilon_m, \quad (10)$$

where $p(m)$ denotes the province of municipality m . The variable $Diverse_m$ measures the number of US emigrants from m who settled in neighborhoods in the top 20% of the entropy distribution, while $Homogeneous_m$ captures the number of emigrants from m who settled in less diverse neighborhoods. The control vector X_m includes population levels before (1881) and after (1991) the mass-migration period, as well as geographic characteristics such as municipal area and altitude. Depending on the specification, we further include province or region fixed effects.

Table V reports the results. In columns (1–3), the dependent variable is the share of streets bearing a national-identity designation, while columns (4–6) focus on streets associated with local identity. Columns (1) and (4) present the baseline specification; columns (2) and (5) add province fixed effects; and columns (3) and (6) further control for municipality characteristics. We find that the number of emigrants who were exposed to diverse Italian communities in the United States is positively associated with the share of streets commemorating national figures or events. By contrast, the number of emigrants exposed to relatively more homogeneous migrant communities is negatively associated with national-identity street naming.

Local-identity naming follows the opposite pattern. Exposure of emigrants to more diverse Italian communities is negatively correlated with the share of streets named after local figures or motives, whereas exposure to more homogeneous communities is associated with a higher prevalence of local-identity street names.

These patterns are quantitatively stable across specifications and underscore sharp differences in commemorative practices across municipalities that experienced similar emigration intensity but whose emigrants settled in more *vis-à-vis* less diverse communities. Table V separately reports the p -value of a test of equality between the coefficients of emigration flows to diverse and homogeneous communities. Across all models, we find strong evidence that the two coefficients are statistically different ($p < 0.01$).

The estimates from (10) are correlational. However, the inclusion of placebo categories provides a diagnostic for the plausibility of a causal interpretation. While we find a large and statistically significant association, with opposite signs, between historical exposure to diversity and national versus local identity in street names, we detect no systematic relationship with placebo categories. In columns (7–9), we employ the share of streets named after artistic, literary, and religious figures as the outcome variables. None of these outcomes is significantly associated with either the number of US emigrants who settled in high or low-diversity communities. These patterns suggest that exposure to diversity is

not merely correlated with a general propensity for street naming or commemorative practices, thus restricting the scope for alternative explanations based on unobserved municipal characteristics.

V.C National Monuments

We use public monuments as a third proxy for national identity sentiment. We rely on a municipality-level dataset and aggregate the number of monuments to the decade level.

We estimate the following specification:

$$y_{mt} = \alpha_m + \alpha_{r(m) \times t} + X'_{mt}\Gamma + \sum_{\tau=1870}^{1990} \beta_{\tau} \cdot \text{Diversity Exposure}_m \cdot I(t = \tau) + \varepsilon_{mt}, \quad (11)$$

where, as in the rest of the analysis, $\text{Diversity Exposure}_m$ denotes the share of US emigrants from municipality m who settled in districts in the top quintile of the entropy distribution. Because this analysis spans the entire twentieth century, we include region-by-decade fixed effects ($\alpha_{r(m) \times t}$) to absorb time-varying shocks at a relatively coarse geographic level, such as differential exposure to wars, reconstruction, and institutional change.

As in (8), because US migration began in the late 1890s, we interpret the coefficients for the 1870s, 1880s, and 1890s as evidence on pre-treatment differences between treated and control municipalities. In this analysis, we restrict the sample to Southern municipalities for two reasons. First, national identity monuments were largely erected to commemorate casualties of the World Wars, and Northern regions were substantially more exposed to combat, particularly during World War I, so monument construction there is more likely to reflect proximity to warfare rather than identity formation. Second, given the long time horizon of the analysis, comparisons between Northern and Southern regions would confound the estimates with multiple region-specific developments that unfolded over the twentieth century. This restriction is also empirically relevant, as the large majority of emigrants originated from the South (75%, as shown in Panel A of Table C.1).

Figure VIII reports the estimated effects. In both panels, the left y -axis plots the coefficients from equation (11), while the gray line on the right y -axis shows the number of monuments by year of construction.

Figure VIIIa focuses on national monuments. Most national monuments were erected in the 1920s and 1930s and commemorate soldiers who died during the First World War, with a second, quantitatively smaller wave following the Second World War. Prior to the onset of mass emigration, we find no statistically significant differences between municipalities whose US emigrants later settled in more versus less diverse Italian communities. This pattern suggests that the estimates do not conflate pre-existing differences across origin municipalities correlated with subsequent exposure to diversity in the US.³⁴

Beginning in the interwar period, municipalities whose migrants settled in relatively more diverse Italian communities in the United States exhibit more national monuments than control municipalities. Importantly, we also estimate a positive and statistically significant effect of exposure to diversity

³⁴This pre-trends test should be interpreted with caution, given that relatively few monuments were erected before the mass migration period.

on the number of national monuments erected between 1950 and 1959, when most monuments commemorate WW2 casualties. This persistence suggests that national-identity values formed through emigrants' historical exposure to Italians from other parts of the country while in the United States were subsequently activated by the experience of the Second World War.

Figure VIIIb replicates the analysis for non-national monuments, which we interpret as a placebo category. Non-national monuments were erected throughout the sample period, with construction peaking between the 1890s and the 1930s and declining thereafter. We find no association between exposure to Italian diversity in the United States and non-national monument construction prior to the mass migration period. Likewise, exposure to diversity has no effect on the construction of non-national monuments in the post-migration period. This pattern suggests that our estimates are unlikely to capture general shifts in commemorative activity unrelated to exposure to diversity, unless such shifts differentially affected national monuments relative to other categories.

Table C.9 reports the estimates in tabular form under a range of alternative specifications where we control for time-varying shocks at the region (columns 2 and 7) and province (columns 3 and 8) levels. In columns (5) and (9), we control for the number of US emigrants over the migration period interacted with period dummies to test whether the results are driven by the overall intensity of the migration outflows rather than by its direction. In columns (6) and (10), we add municipality-level time-invariant controls—most importantly, the number of deaths during WW1 and WW2. The baseline results remain quantitatively robust across all specifications.

V.D Robustness Analysis

Most Italian migrants in the United States (approximately 70%) originated from Southern regions. A potential concern, therefore, is that our results may be driven by specific origin areas rather than reflect a general effect of exposure to diversity on national identity formation. In Figure D.12, Figure D.13, and Figure D.14, we replicate the baseline analyses for names, streets, and monuments while sequentially excluding one region at a time from the estimation sample. In each panel, the first row reports the full-sample baseline estimate, and subsequent rows report the treatment effects obtained when excluding each region. Across all outcomes and specifications, the estimates remain statistically comparable to the baseline, indicating that the results are unlikely to be driven by idiosyncratic characteristics of particular regions.

A related concern is that spatial clustering of emigrants across regions or municipalities may induce spatial autocorrelation, potentially inflating statistical significance. In Figure D.15, Figure D.16, and Figure D.17, we therefore replicate all analyses using a range of alternative standard error estimators, including the spatial correction proposed by Conley (1999) with different distance cutoffs. Statistical significance remains largely unchanged across outcomes, specifications, and standard error estimators.

We assess the sensitivity of our naming and monument results to potential violations of the parallel trends assumption using the framework developed by Rambachan and Roth (2023), which formalizes how pre-treatment deviations from inform post-treatment violations rather than assuming

exact parallel trends. In our application, robust confidence intervals reported in Figure D.18 and Figure D.19 remain statistically significant for values of the tuning parameter up to one, meaning that the post-treatment departure from parallel trends would have to be at least as large as the largest pre-treatment deviation to overturn the results. These patterns indicate that the results remain robust to sizable departures from the parallel trends assumption.

VI CONCLUSIONS

Italian cuisine is a defining element of Italian culture. It is thus surprising that several world-famous dishes, including pizza, acquired their modern form within Italian immigrant communities in the US during the Age of Mass Migration (1890–1924). Others, including spaghetti with tomato sauce, were invented by immigrants. Historical qualitative evidence indicates that the nation-building role of intergroup contact in these communities extended beyond cuisine, from the homogenization of religious practices to the gradual abandonment of local dialects in favor of standardized Italian.

We exploit this unique setting to study how intergroup contact among members of a fractionalized country can foster the emergence of national identity. We construct a novel, highly detailed dataset that allows us to observe the municipality of origin of half a million Italian migrants in the United States by linking the US census records and the Ellis Island ship passenger lists. We document that American neighborhoods were areas where Italians from different parts of the country met and interacted. These contacts contributed to the emergence of a sense of shared national identity even in the absence of coordinated state interventions. Increased national identity among migrants, however, hampered cultural assimilation into US society.

We then show that the national identity forged within immigrant communities diffused back to Italy. Municipalities more exposed to interregional interaction through their emigrants experienced an increased sense of national identity. These patterns are difficult to reconcile with purely domestic explanations and instead point to a transnational process through which identity formation abroad reshaped cultural outcomes at home.

Taken together, our findings suggest that today’s diasporic communities can play an important and underappreciated role in fostering nation-building and contribute to social cohesion in migrants’ countries of origin.

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TABLES

Table I. Descriptive Statistics of Selected Variables

	Mean	Std. Dev.	Median	Min.	Max.	Observations
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. United States						
<i>Panel A.1 Neighborhood-Level Data</i>						
Italian Diversity	1.911	1.222	1.906	0.000	5.955	39,477
Population	1,679.481	843.645	1,553.000	13.000	32,766.000	39,477
Immigrants	24.232	13.691	22.607	0.000	84.400	39,477
Italian Immigrants	6.644	9.156	3.122	0.000	76.915	39,477
<i>Panel A.2 Individual-Level Data</i>						
Local Name Index	4.219	3.905	3.241	0.014	99.827	335,569
Distance-Adjusted Local Name Index	9.593	0.495	9.743	0.000	10.002	335,569
Foreign Name Index	0.759	0.153	0.786	0.000	1.000	330,247
Married to Italian from Other Municipality	0.180	0.384	0.000	0.000	1.000	317,885
log(Occupational Income)	2.458	1.356	3.045	0.000	4.394	335,569
Employed in Manufacturing	0.244	0.430	0.000	0.000	1.000	335,569
Skilled Manufacturing Job	0.154	0.361	0.000	0.000	1.000	335,569
Unskilled Manufacturing Job	0.165	0.371	0.000	0.000	1.000	335,569
Panel B. Italy						
Population	5,052.860	28,469.863	2,231.000	0.000	2,839,638.000	104,299
Number of US Emigrants	129.247	433.492	22.000	1.000	19,225.000	5,846
Share US Emigrants in Diverse Areas	0.175	0.164	0.168	0.000	1.000	5,846
Local Name Index	0.009	0.017	0.004	0.000	0.679	68,568
Distance-Adjusted Local Name Index	9,910.692	194.009	9,964.916	3,174.663	10,001.901	68,568
Number of Streets	107.014	281.237	53.000	0.000	17,873.000	8,176
Number of Streets by Designation:						
National Identity	19.327	27.333	10.000	0.000	680.000	8,176
Local Identity	35.738	118.161	14.000	0.000	5,905.000	8,176
Arts	4.396	14.274	1.000	0.000	774.000	8,176
Literature	5.807	13.449	1.000	0.000	548.000	8,176
Religious Figure	10.791	27.654	5.000	0.000	1,527.000	8,176
Number of Monuments	0.301	6.591	0.000	0.000	1,288.000	113,400
Number of National Identity Monuments	0.123	5.707	0.000	0.000	1,282.000	113,400

Notes. This table reports descriptive statistics for selected variables for the United States (Panel A) and Italy (Panel B). Panel A.1 uses census enumeration districts observed at decennial frequency from 1900 to 1930. Panel A.2 is the baseline individual-level dataset of Italian immigrants, observed in the first census in which they appear and, when possible, linked across subsequent censuses. Panel B consists of municipality-level data for Italy: population, name-based indices, and monuments are observed decennially, while all other variables are measured once per municipality. Referenced on pages: [18](#), [19](#).

Table II. Exposure to Immigrant Heterogeneity and National Identity Formation

	Ordinary Least Squares						Reduced Form		Two-Stage Least Squares	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A: Local Name Index										
Italian Diversity	-0.170*** (0.014)	-0.164*** (0.015)	-0.154*** (0.016)	-0.162*** (0.014)	-0.121*** (0.017)	-0.126*** (0.020)			-2.210*** (0.686)	-4.939*** (1.330)
Ship Diversity							-0.049*** (0.009)	-0.047*** (0.010)		
Mean Dep. Var.	3.990	3.989	3.989	3.989	3.989	3.989	3.988	3.988	3.988	3.988
Panel B: Distance-Adjusted Local Name Index										
Italian Diversity	0.020*** (0.002)	0.020*** (0.002)	0.019*** (0.002)	0.019*** (0.002)	0.016*** (0.003)	0.015*** (0.004)			0.251** (0.114)	0.560** (0.232)
Ship Diversity							0.006*** (0.002)	0.005*** (0.002)		
Mean Dep. Var.	9.593	9.593	9.593	9.593	9.593	9.593	9.593	9.593	9.593	9.593
Panel C: Foreign Name Index										
Italian Diversity	0.017*** (0.001)	0.016*** (0.001)	0.016*** (0.001)	0.015*** (0.001)	0.015*** (0.001)	0.014*** (0.001)			0.105*** (0.024)	0.215*** (0.055)
Ship Diversity							0.002*** (0.000)	0.002*** (0.000)		
Mean Dep. Var.	0.758	0.758	0.758	0.758	0.758	0.758	0.758	0.758	0.758	0.758
Municipality FE	Yes	Yes	–	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	–	–	–	–	–	–	–	–	–
Year FE	Yes	Yes	–	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Hexagon FE	No	Yes	–	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Municipality-Year FE	No	No	Yes	No	No	No	No	No	No	No
Hexagon-Year FE	No	No	Yes	No	No	No	No	No	No	No
Individual Controls	No	No	No	Yes	No	No	No	No	No	No
Demographic Controls	No	No	No	No	Yes	Yes	No	Yes	No	Yes
Economic Controls	No	No	No	No	No	Yes	No	Yes	No	Yes
# Individuals	223,721	223,531	221,913	223,521	223,531	223,531	208,845	208,845	208,845	208,845
# Observations	334,668	334,303	330,615	334,283	334,303	334,303	311,848	311,848	311,848	311,848
K-P F-stat	–	–	–	–	–	–	–	–	24.327	30.750

Notes. This table reports the effect of exposure to diversity within the Italian community on name-based indices from specification (4). Panel A uses the local name index defined in (1) as the dependent variable; Panel B uses the distance-adjusted local name index defined in (2); Panel C uses the foreign name index of Abramitzky et al. (2020). The unit of observation is an individual observed in the first census in which they appear and, when possible, linked across subsequent censuses between 1900 and 1930. Columns (1–6) report ordinary least squares estimates; columns (7–8) report reduced-form instrumental-variable estimates using the ship-based instrument defined in (5); columns (9–10) report the corresponding two-stage least squares estimates. All specifications include municipality and census-year fixed effects. Column (1) includes county fixed effects; columns (2) and (4–10) replace county with more granular hexagon fixed effects. Column (3) replaces municipality fixed effects with municipality-year fixed effects and hexagon fixed effects with hexagon-year fixed effects. Demographic controls (included in columns 5, 8, and 10) are measured at the district level and include population, the share of women, African Americans, immigrants, and Italian immigrants. Economic controls (included in columns 6, 8, and 10) are measured at the district level and include employment shares by industry. Standard errors, clustered at the district level, are reported in parentheses. Columns (9–10) also report the Kleibergen–Paap *F*-statistics. Referenced on page: 23.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table III. Movers Design and National Identity Formation

	Local Name Index (LNI)					Distance LNI
	(1)	(2)	(3)	(4)	(5)	(6)
Diversity Shock $\times$ Post-Move	-0.336** (0.137)	-0.342** (0.135)	-0.351** (0.144)		-1.619* (0.867)	0.076*** (0.023)
Diversity Shock (Cont.) $\times$ Post-Move				-16.384*** (5.035)		
Age					-0.002 (0.013)	
Diversity Shock $\times$ Post-Move $\times$ Age					0.035 (0.023)	
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Hexagon FE	Yes	Yes	–	–	–	–
Time Since Move FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	No	Yes	–	–	–	–
Hexagon-Cohort FE	No	No	Yes	Yes	Yes	Yes
# Individuals	114,052	114,051	112,155	112,155	112,155	112,155
# Observations	410,343	410,340	402,640	402,640	402,640	402,640
Mean Dep. Var.	3.909	3.909	3.902	3.902	3.902	9.603

Notes. This table reports results from the movers design specification in (7). The unit of observation is an individual migrant observed in each year in which they have a child born in the United States between 1900 and 1930. The dependent variables are the local name index (columns 1–5) and the distance-adjusted local name index (column 6) defined in (1)–(2). The baseline treatment is the interaction between an indicator equal to one after an individual moves to a different hexagon (and zero otherwise) and an indicator equal to one if the difference in diversity between the Italian communities in the destination and origin districts lies in the top 20% of the distribution (and zero otherwise). Column (4) replaces this indicator with the continuous diversity difference measure. Column (5) additionally includes age and its interaction with the baseline treatment. All specifications include individual and event-time (time since move) fixed effects. Column (2) also includes cohort (year) and hexagon fixed effects, while columns (3–6) replace these with hexagon-by-cohort fixed effects. Standard errors, clustered at the district level, are reported in parentheses. Referenced on page: 28.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table IV. Name Particularism in Italy by Exposure to Immigrant Diversity

	Exposure Panel			Difference-in-Differences		
	(1) LNI	(2) LNI	(3) D-LNI	(4) LNI	(5) LNI	(6) D-LNI
Diversity Exposure $\times I(1880 < t \leq 1890)$	-0.002 (0.249)	0.086 (0.284)	0.014 (0.029)			
Diversity Exposure $\times I(1890 < t \leq 1900)$	-0.017 (0.249)	0.054 (0.283)	0.011 (0.029)			
Diversity Exposure $\times I(1900 < t \leq 1910)$	-0.268 (0.265)	-0.039 (0.302)	0.030 (0.031)			
Diversity Exposure $\times I(1910 < t \leq 1920)$	-0.505** (0.250)	-0.273 (0.284)	0.065** (0.029)			
Diversity Exposure $\times I(1920 < t \leq 1930)$	-0.798*** (0.258)	-0.596** (0.293)	0.087*** (0.030)			
Diversity Exposure $\times$ Post Mass Migration				-0.464*** (0.091)	-0.405*** (0.097)	0.045*** (0.010)
Post Mass Migration				0.010 (0.025)	0.055* (0.029)	-0.001 (0.003)
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
US Emigrants $\times$ Year FE	No	Yes	No	No	Yes	No
# Municipalities	7,823	5,715	7,823	7,823	5,715	7,823
# Observations	200,774	155,781	200,774	200,774	155,781	200,774
Mean Dep. Var.	0.879	0.886	9.909	0.879	0.886	9.909

Notes. This table reports the effect of exposure to diversity within the Italian migrant community on name-based indices in Italian municipalities. The unit of observation is a municipality observed at a decennial frequency between 1880 and 1930. The dependent variables are the local name index defined in (1) in columns (1–2) and (4–5) and the distance-adjusted local name index defined in (2) in columns (3) and (6). The baseline treatment is defined as the share of a municipality’s US emigrants who settled in districts in the top quintile of the diversity distribution (3). Columns (1–3) report event-study specifications of (8), where we interact diversity exposure with decade indicators. Columns (4–6) report difference-in-differences specifications of (9), where the post indicator equals one in years at or after the municipality-specific onset of mass migration (defined as the first year with at least one emigrant per 1,000 inhabitants); the table reports the coefficient of interaction terms between the diversity exposure measure and the post indicator. All regressions include municipality and year fixed effects. Columns (2) and (5) additionally include interactions between the number of US emigrants and year fixed effects. Standard errors are clustered at the municipality level and reported in parentheses. Referenced on pages: 31, 32.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table V. Street Names and Exposure to Immigrant Diversity

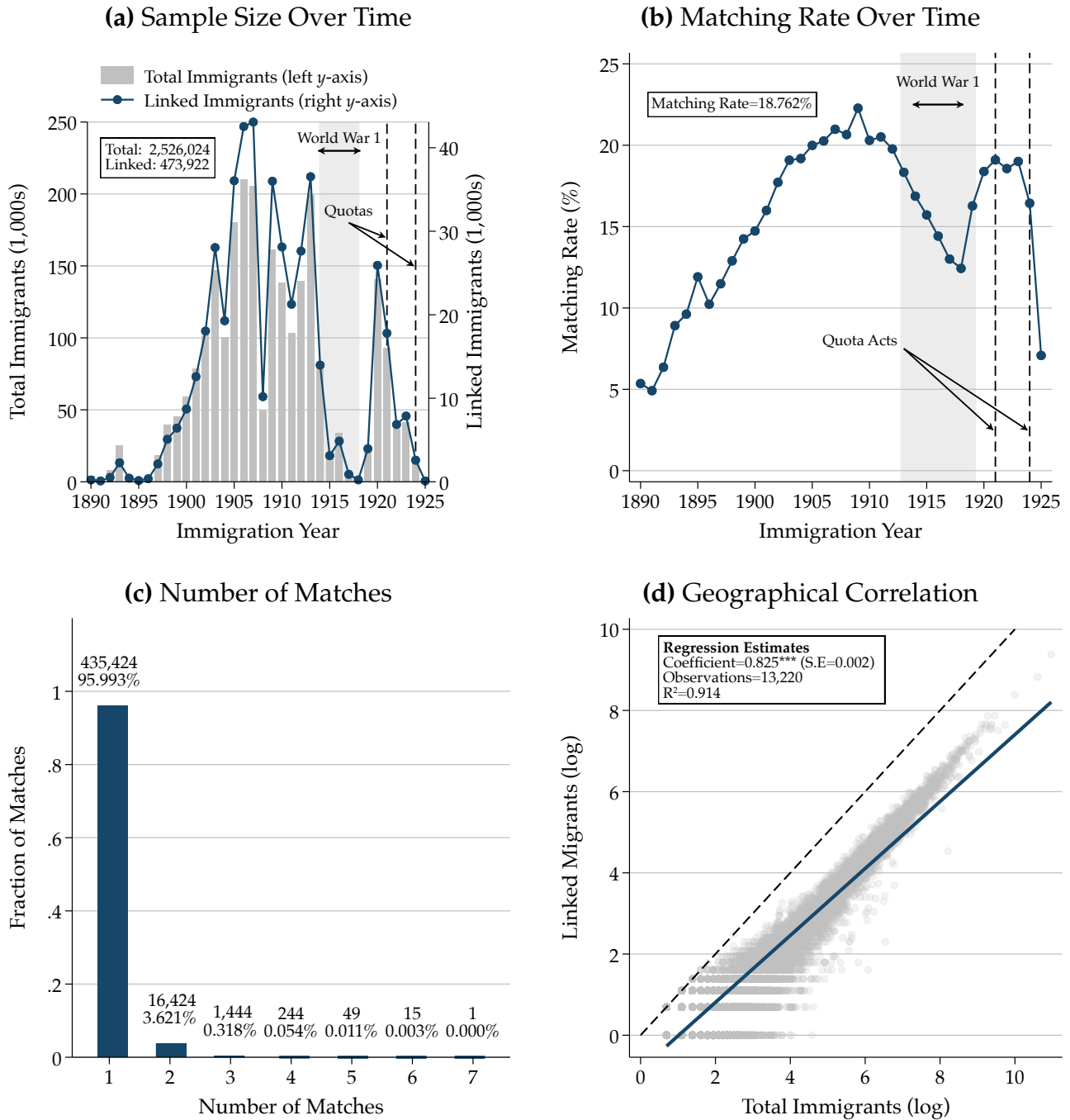
	National Identity Designation			Local Identity Designation			Placebo Designation		
	(1)	(2)	(3)	(4)	(5)	(6)	(7) Arts	(8) Literature	(9) Religious
Number of Emigrants by Neighborhood Diversity:									
Diverse (β_1)	0.326*** (0.079)	0.191** (0.084)	0.344*** (0.095)	-0.251** (0.115)	-0.177 (0.122)	-0.317* (0.192)	0.008 (0.027)	0.048 (0.034)	-0.024 (0.053)
Homogeneous (β_2)	-0.108*** (0.024)	-0.074*** (0.025)	-0.105*** (0.025)	0.108*** (0.036)	0.092** (0.039)	0.098** (0.049)	-0.001 (0.007)	-0.014 (0.009)	0.010 (0.014)
Region FE	Yes	–	Yes	Yes	–	Yes	–	–	–
Province FE	No	Yes	No	No	Yes	No	Yes	Yes	Yes
Controls FE	No	No	Yes	No	No	Yes	No	No	No
p -val: $\hat{\beta}_1 = \hat{\beta}_2$	0.000	0.002	0.000	0.004	0.006	0.001	0.543	0.079	0.266
# Municipalities	8,176	8,176	8,099	8,176	8,176	8,099	8,176	8,176	8,176
Mean Dep. Var.	0.192	0.192	0.194	0.298	0.298	0.299	0.027	0.041	0.105

Notes. This table reports cross-sectional estimates of specification (10) between emigration to diverse Italian communities in the United States and street-naming patterns in Italy. The unit of observation is a municipality. The dependent variables are the shares of streets whose eponyms reflect national identity (columns 1–3) and local identity (columns 4–6); and additional placebo categories (arts, literature, religious, in columns 7–9). The main explanatory variables are the number of emigrants to diverse neighborhoods and the number of emigrants to homogeneous neighborhoods. The outcomes are normalized by the total number of streets in the municipality. Columns (1–3) report national-identity eponyms; columns (4–6) refer to local-identity eponyms; columns (7–9) report placebo outcomes. Columns (1) and (4) include region fixed effects; columns (2) and (5) replace these with province fixed effects; columns (3) and (6) additionally include baseline municipality controls (population in 1881 and 1991, surface area, and elevation). Standard errors are clustered at the municipality level and reported in parentheses. The table also reports p -values from tests of equality between the coefficients on diverse and homogeneous emigration flows. Referenced on page: 33.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

FIGURES

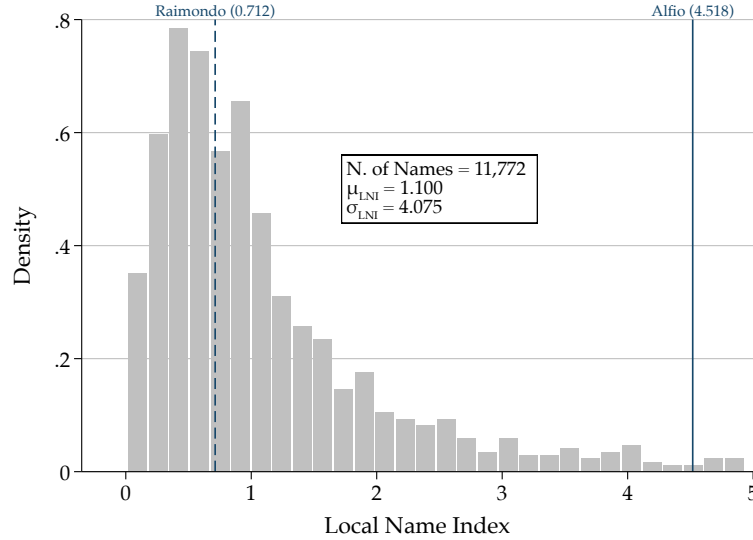
Figure I. Descriptive Statistics on Linked Migrant Sample



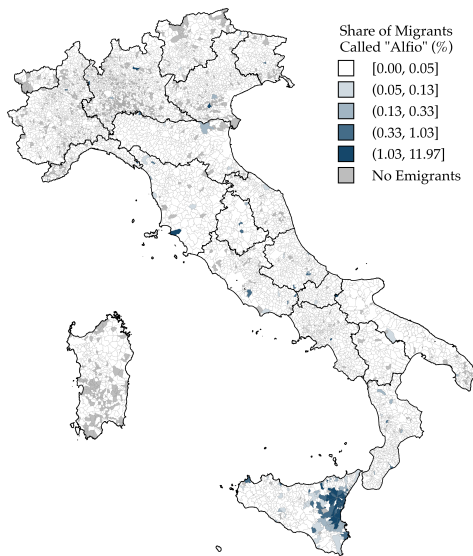
Notes. This figure reports descriptive statistics for the linked sample of U.S. census records of Italian migrants matched to Ellis Island ship records. Panel **Ia** shows the number of migrants in the Ellis Island data (gray bars) and the number successfully linked to U.S. census records (red bars). Panel **Ib** reports the share of U.S. census records that match at least one Ellis Island record. Panel **Ic** displays the distribution of the number of Ellis Island matches per U.S. census record, together with the corresponding shares relative to the total sample. Panel **Id** plots, at the municipality level, the number of migrants recorded in the Ellis Island dataset against the number appearing in the linked dataset, and includes a linear fit, the 45-degree line, and the estimated regression coefficient. Referenced on pages: [12](#), [13](#), [B14](#), [B15](#).

Figure II. Distribution Local Name Index and Sample Names

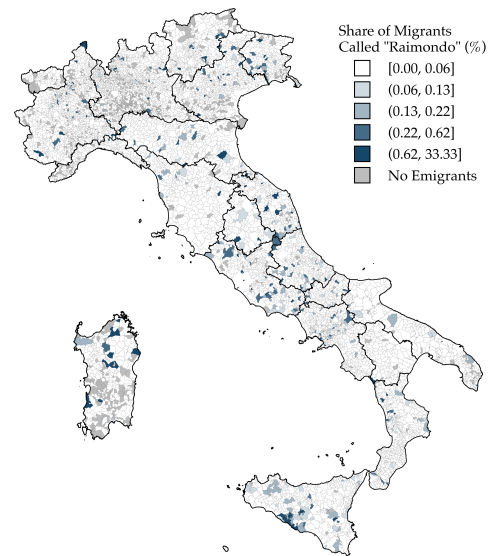
(a) Distribution of Local Name Index



(b) Distribution of Local Name

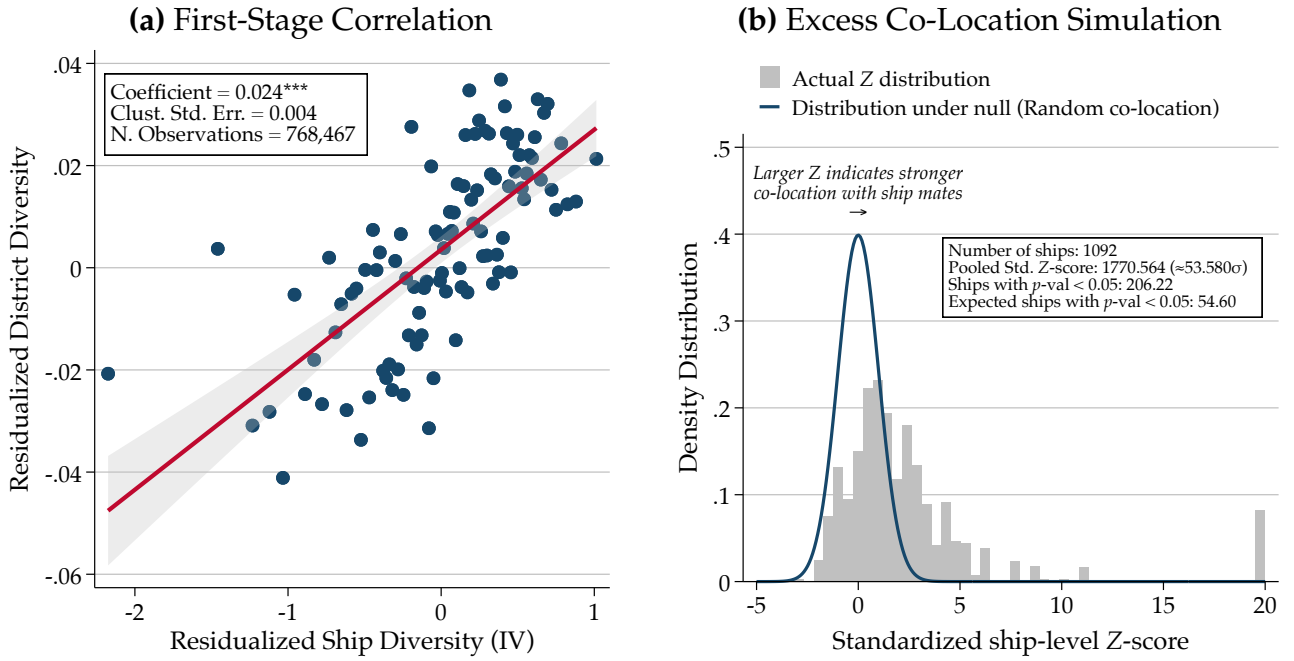


(c) Distribution of National Name



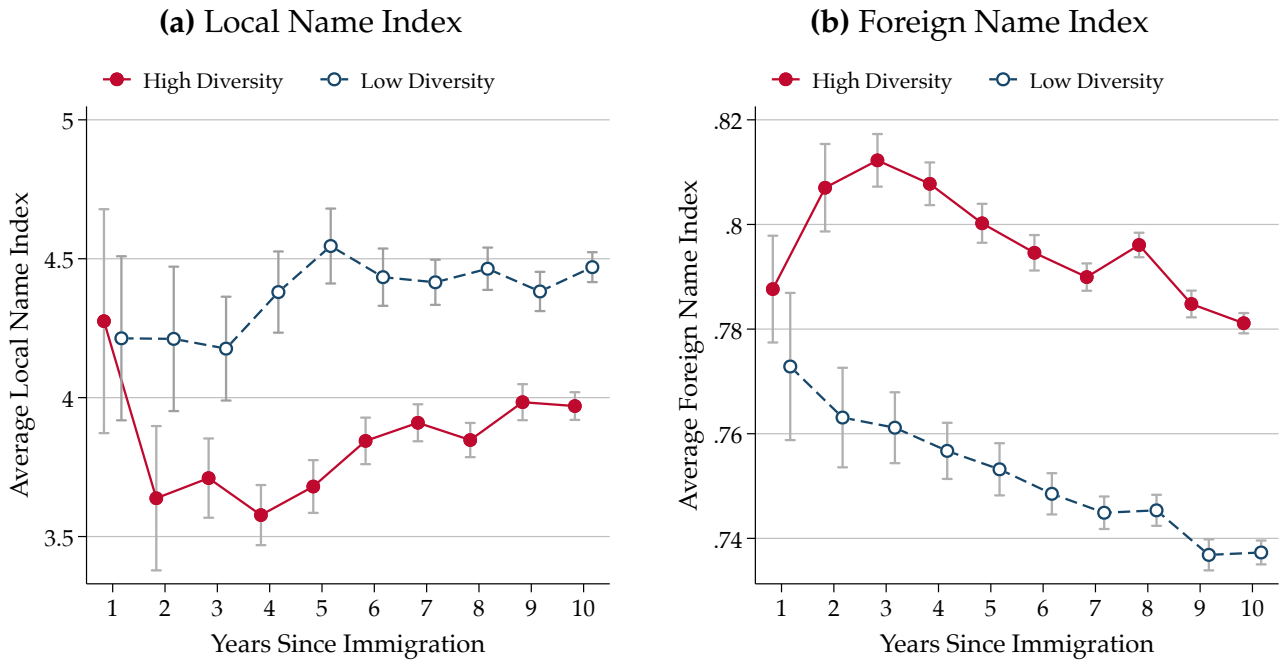
Notes. Panel [IIa](#) reports the distribution of the local name index defined in [\(1\)](#), together with the number of recorded names and the mean and standard deviation of the distribution. The panel also reports the LNI values for two names—"Raimondo" and "Alfio"—chosen because they appear with similar frequency. Panel [IIb](#) (respectively, [IIc](#)) maps the spatial distribution of individuals named "Alfio" (respectively, "Raimondo") across municipalities, expressed as the share of individuals with that name relative to the 1881 population, in percentage terms. The maps superimpose regional borders; municipality and district boundaries are shown using 1991 borders. Referenced on page: [15](#).

Figure III. Instrument Relevance Tests



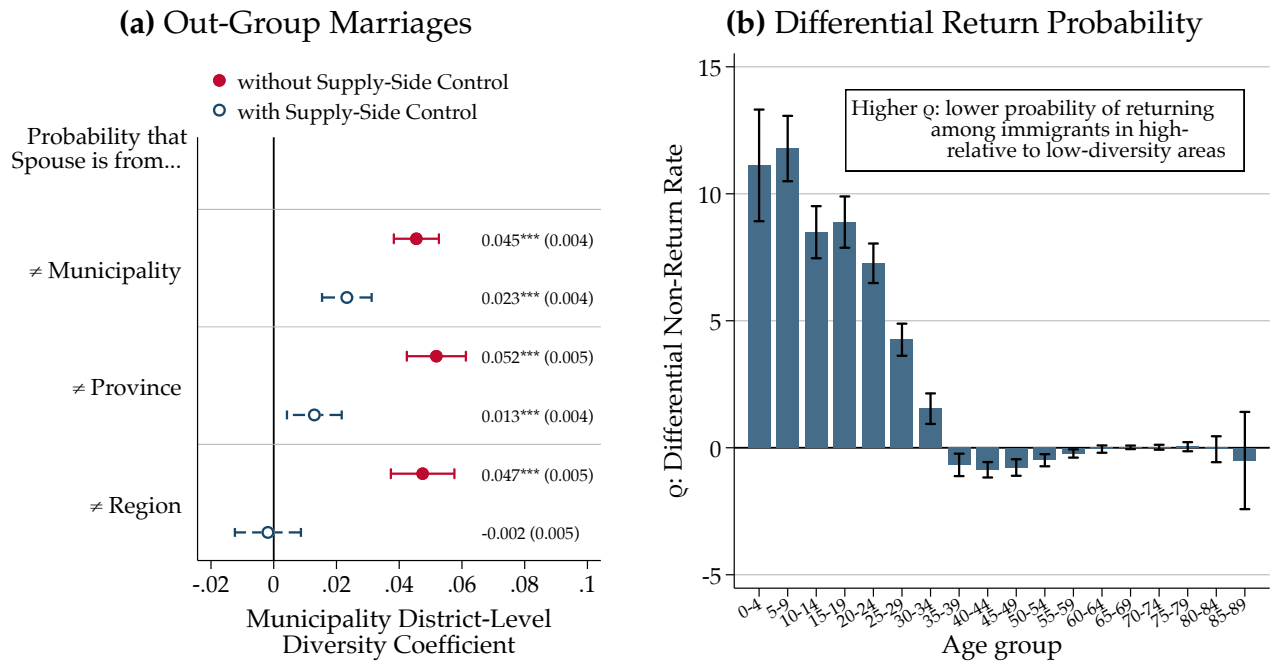
Notes. This figure reports evidence on the relevance of the ship-based instrument defined in (5). Panel IIIa shows the first-stage relationship between the instrument and observed diversity, as defined in (3). The unit of observation is an individual observed in the first census in which they appear and, when possible, linked across subsequent censuses between 1900 and 1930. To construct the binned scatterplot, both variables are residualized with respect to the baseline controls included in (8)—municipality, district, and year fixed effects. The red line displays a linear fit; the corresponding regression coefficient and standard error are reported in the panel. Panel IIIb reports the ship-level distribution of the z-statistic measuring excess co-location with shipmates relative to the benchmark case of random location with shipmates. Under the null hypothesis of no co-location, the z-score distribution would be approximately normal, shown by the blue line. Values to the right indicate excess co-location with shipmates. The panel also reports the number of ships with a p -value below 0.05, together with the number that would be expected under the null of no co-location. Referenced on pages: 22, A6.

Figure IV. Evolution of the Local and Foreign Name Indices



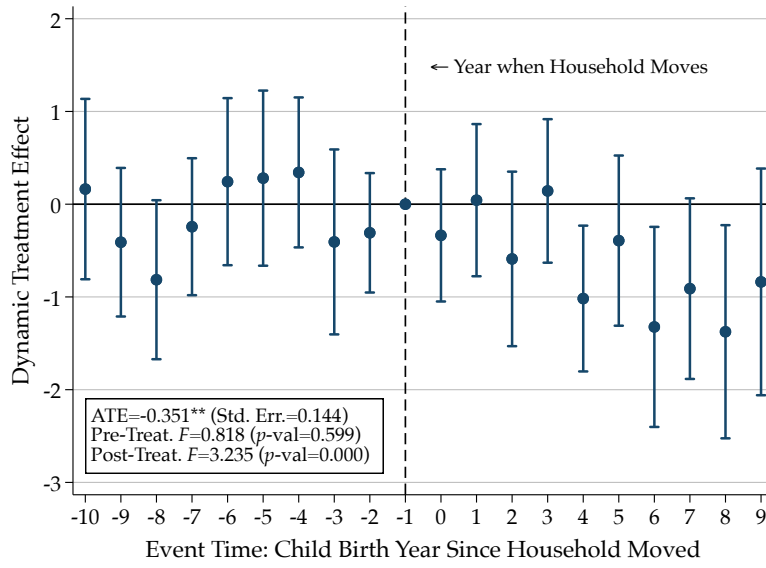
Notes. This figure reports the evolution of name-based indices by time since immigration, separately for immigrants settling in high- versus low-diversity Italian communities. The unit of observation is an individual Italian immigrant observed in the first census in which they appear and, when possible, linked across subsequent censuses between 1900 and 1930. Time since immigration is defined as the difference between year of immigration and census year, and individuals are split into above- and below-median groups based on the municipality-level diversity measure (3). Panel IVa plots the local name index by years since immigration. Panel IVb plots the foreign name index by years since immigration. Each point is the coefficient from a separate regression run at a given event time and estimated separately for the high- and low-diversity groups. All regressions include hexagon and census-year fixed effects. Bands report 90% confidence intervals. Referenced on page: 23.

Figure V. Additional Indicators of National Identity Formation



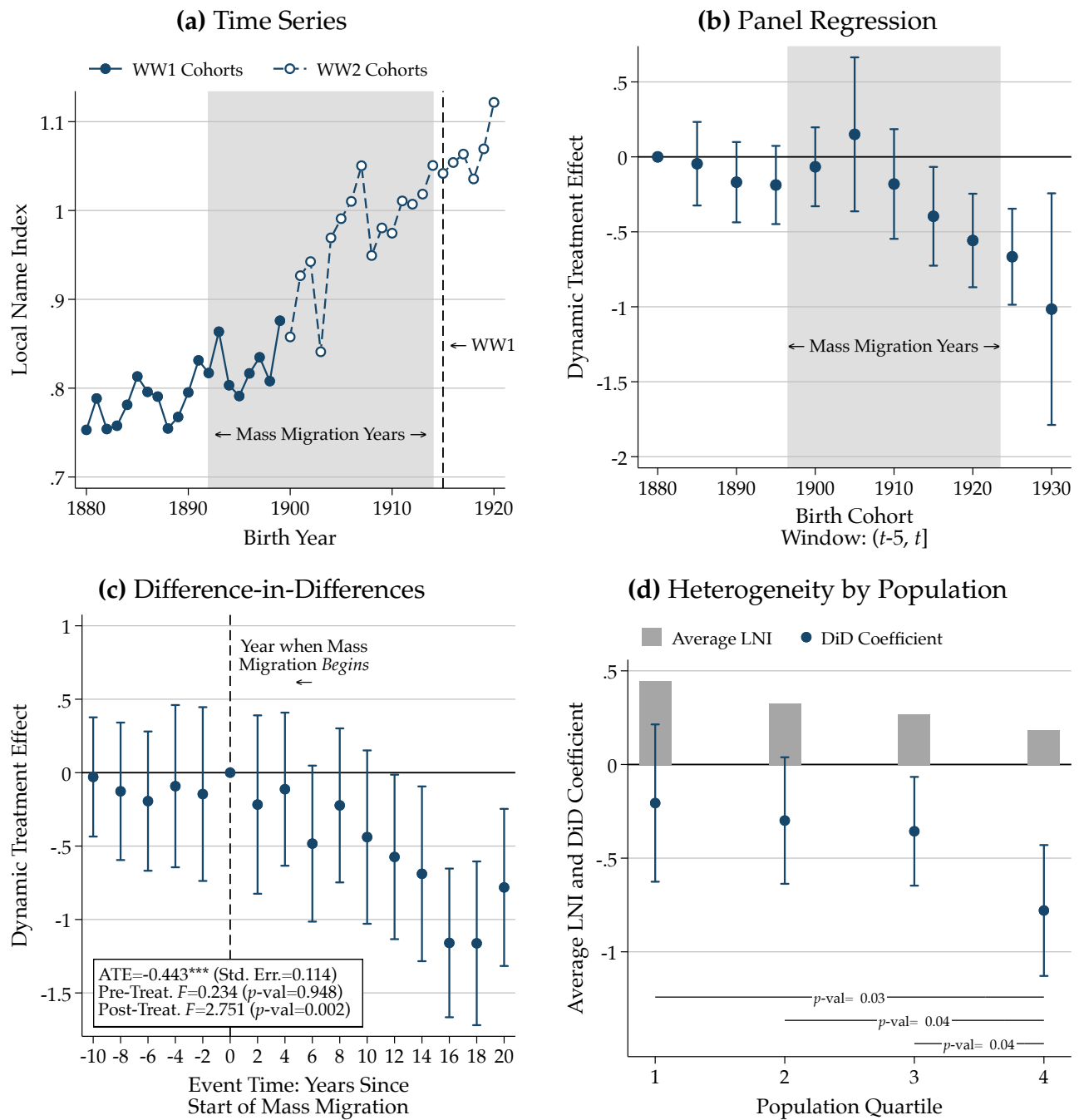
Notes. This figure reports the association between alternative measures of identity formation and exposure to Italian diversity. Panel Va shows regression estimates relating district-level Italian-community diversity to intermarriage outcomes. The unit of observation is an individual Italian immigrant observed between 1900 and 1930. All specifications include municipality, year, and hexagon fixed effects and control for demographic and economic district characteristics. Dots report the coefficient on Italian diversity as defined in (3). Dashed estimates additionally include supply-side controls capturing the predicted availability of same-origin spouses at the municipality, province, or region level, following Bazzi et al. (2019). Standard errors are clustered at the hexagon level and bands report 95% confidence intervals. Panel Vb plots estimates of the differential return statistic defined in (6) by age group at a 30-year horizon. High-diversity districts are those above the median of Italian diversity. For each age bin, the statistic compares the 30-year forward match rate of Italian immigrants with that of natives across high- and low-diversity districts. Standard errors are obtained via bootstrap, and bands report 95% confidence intervals. Referenced on pages: 25, 27.

Figure VI. Movers Design Event-Study Estimates



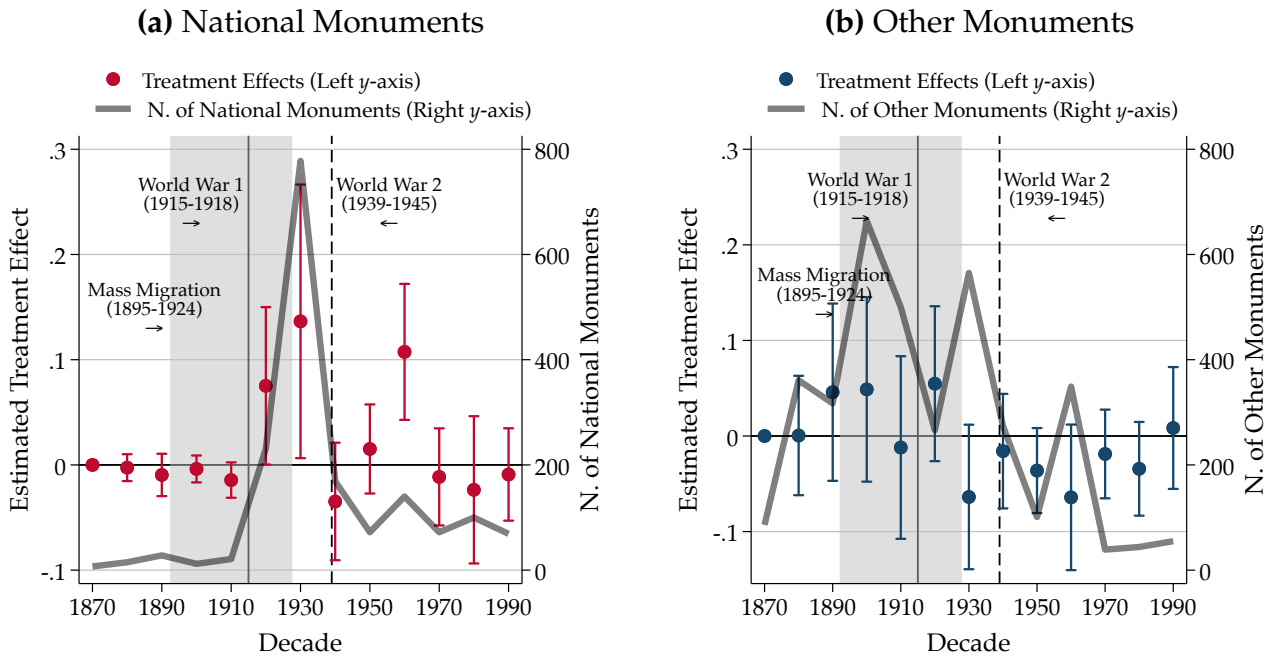
Notes. This figure reports results from the movers design specification in (7). The unit of observation is an individual migrant observed in each year in which they have a child born in the United States between 1900 and 1930. The dependent variable is the local name index defined in (1). The treatment is an interaction between an indicator equal to one for individuals experiencing a large diversity shock, defined as a move to a district where the change in Italian-community diversity between the destination and origin lies in the top quintile of the distribution, and a set of event-time indicators relative to the year of the move. The specification includes individual and event-time fixed effects, as well as hexagon-by-birth-year fixed effects. Standard errors are clustered at the hexagon level. Bands report 95% confidence intervals. We report the average treatment effect as well as joint F -tests for the joint significance of pre- and post-treatment coefficients. Referenced on pages: 25, 27.

Figure VII. Name Particularism in Italy by Exposure to Immigrant Diversity



Notes. This figure reports dynamic estimates of the effect of exposure to diversity within the Italian migrant community on the local name index in Italian municipalities between 1880 and 1930. Panel VIIa reports the average LNI across municipalities over time. In the rest of the figures, the unit of observation is a municipality. Panel VIIb reports an event-study specification (8) that interacts diversity exposure with five-year windows, controlling for municipality fixed effects, time fixed effects, and interactions between the number of US emigrants and time indicators. Panel VIIc reports a difference-in-differences event-study specification (9) where the treatment time is defined as the first year with at least one emigrant per 1,000 inhabitants. In both panels, time dummies are interacted with a treatment defined as the share of a municipality's US emigrants who settled in districts in the top quintile of the diversity measure (3). Panel VIId, the gray bars report the average LNI by municipality population quartiles; the blue dots report the coefficients of interaction terms between the baseline DiD coefficient (conflated in pre- and post-treatment periods) and population quartile dummies. Standard errors are clustered at the municipality level. Bands report confidence 95% confidence intervals. Referenced on pages: 31, 32.

Figure VIII. Monuments by Designation in Italy



Notes. This figure reports dynamic estimates obtained from (11) of the effect of exposure to diversity within the Italian migrant community on monument construction in Southern Italy. The unit of observation is a municipality-decade for municipalities observed between 1861 and 2000. In panel VIIIa, the dependent variable is an indicator equal to one if the municipality has at least one national-identity monument in a given decade. In panel VIIIb, the dependent variable is an indicator equal to one if the municipality has at least one non-national-identity monument in a given decade. In both panels, each dot reports, on the left y-axis, the coefficient of a set of interaction terms between decade indicators and the share of US emigrants from each municipality who settled in districts in the top quintile of the diversity measure defined in (3). In both panels, the specification includes municipality fixed effects, decade fixed effects, and interactions between the number of US emigrants and decade indicators. The gray line reports, on the right y-axis, the total number of monuments in each category. Standard errors are clustered at the municipality level. Bands report 95% confidence intervals. Referenced on pages: 34, 35.

ONLINE APPENDIX

A Nation of Emigrants: Transatlantic Emigration and the Rise of Italian National Identity, 1890–1940

Davide M. COLUCCIA, Giampaolo LECCE, Laura OGLIARI

January, 2026

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A DATA APPENDIX AND ROBUSTNESS CHECKS

A.I Measurement of Completed Fertility

The decennial US Census provides cross-sectional snapshots of households and therefore does not directly observe completed fertility. To recover completed fertility histories, we combine household structure information within each census year with longitudinal individual links across censuses to reconstruct parent-child relationships over time.

We begin by identifying Italian migrants and second-generation individuals in each census year (1900, 1910, 1920, and 1930), defined as individuals either born in Italy or with at least one parent born in Italy. Using household identifiers and relationship-to-head variables, we classify individuals as parents or children within each household and retain only nuclear-family relationships. For each household, we identify up to two parents (male and female) and link each child to the corresponding parent identifiers. This procedure yields a set of parent-child observations for each census year.

Because children may appear in different households or census years as families age, relying on a single census snapshot would undercount fertility. We therefore exploit longitudinal links across censuses to aggregate children observed in different years (Price et al., 2021; Buckles et al., 2025; Abramitzky et al., 2025). Specifically, for each census year taken as a reference, we employ the longitudinal crosswalks that link individual identifiers across census waves to map parents observed in other census years to the reference year. We then pool all parent-child links across census waves and collapse them at the parent level, removing duplicate children identified by year of birth and first name.

The resulting dataset assigns to each parent the full set of children ever observed co-residing with them in any census between 1900 and 1930. This approach recovers fertility histories that are substantially closer to completed fertility than those obtained from a single census snapshot, while relying solely on observed household relationships and longitudinal links.

A.II Street Data

We collect the universe of street names in Italy from the online repository OpenStreetMaps. Streets are grouped into municipal, provincial, regional, and state roads and highways. The national government is responsible for state roads and highways, while provinces and regions are responsible for provincial and regional roads, respectively. Since our analysis leverages local street naming decisions, we exclude non-municipal roads from the sample. We thus have information on 849,124 streets, which is approximately 93.7% of the entire sample of roads.

The dataset contains the name and coordinates of each road. We geo-reference roads to the municipality where they are located in 1991, i.e., the GIS file we employ throughout the analysis on Italian municipalities.

We employ a large language model (GPT5) to infer the motive for the designation of each road. We employ the following prompt to categorize designation motives into local or national identity. Importantly, we give the LLM both the eponym and the municipality of each road and instruct it to

identify local names using that information. Hence, names of local saints are identified as signals of local identity only where they are the subject of specific veneration.

Second, we ask the LLM to identify three “placebo motives” for designation, namely, artistic, literary, and religious figures. We employ this information as a falsification exercise in the analysis. Importantly, we do not constrain the model to assign road names to a single category. Hence, local saints, for example, are flagged as signifying both local identity and a religious motive. Similarly, we do not impose that each road must be classified in one category.

Overall, the algorithm identifies 33% of streets as signaling local identity, 18% as national identity, 4% as designated after (visual) artistic figures, 5% after literary figures, and 10% after religious figures. Approximately 40% of all roads are not classified under any such motive.

gpt-5 Prompt for Street Classification

You are an expert on Italian cultural and political history and street toponymy.

Input:

- Street name: {street}
- Municipality: {comune}

Task: Interpret what this name commemorates in the context of Italian public toponymy and evaluate how strongly it expresses different collective identities.

Use historical and cultural knowledge about Italy to understand what the name symbolizes to Italians — for example, “Via Garibaldi” signals national identity because Garibaldi is recognized as a hero of unification.

Definitions of dimensions:

1. *national_identity* — reference to the Italian nation, its unification, or explicitly national heroes, institutions, or symbols.
Examples: Garibaldi, Cavour, Mazzini, Vittorio Emanuele II, Italia, Unità d'Italia, XX Settembre, 25 Aprile.
2. *local_identity* — reference to regional or municipal belonging, local saints, local landmarks, or figures of local history.
Examples: San Gennaro (in Naples), Sant’Ambrogio (in Milan).
3. *religious_identity* — reference to religion, spirituality, or ecclesiastical institutions.
Examples: San Pietro, Santa Maria, Papa Giovanni XXIII, Madonna, Convento, Santuario.
4. *literature_identity* — reference to literary figures, authors, poets, or literary works.
Examples: Dante, Petrarca, Manzoni, Leopardi, Pirandello.
5. *art_identity* — reference to artists, painters, sculptors, or art movements.
Examples: Michelangelo, Leonardo, Raffaello, Caravaggio, Bernini.

Rules:

- Base scores on how the name would be understood by an informed Italian observer aware of national culture and history.
- Scores = 1 if the name clearly evokes the dimension; 0 if it does not.
- Multiple 1 for one street are possible (e.g., a saint who is also a local patron has `local_identity = 1` and `religious_identity = 1`).
- If the meaning cannot be inferred confidently, set all scores = 0 and `named_entity = null`.
- If the commemorative meaning is unclear or generic, assign all zeros.

Return:

```
{  
  "street_name": "{street}",  
  "comune": "{comune}",  
  "named_entity": "<person, place, event, or concept commemorated>",  
  "national_identity": float,  
  "local_identity": float,  
  "religious_identity": float,  
  "literature_identity": float,  
  "art_identity": float  
}
```

A.III Monuments Data

We collect buildings of significant cultural significance as identified by the Italian Ministry of Culture. This repository covers the universe of heritage sites across the Italian territory. Overall, the dataset covers 128,086 sites and includes, among other information, construction and designation years, addresses and coordinates, site names and typologies, a historical text documenting the monument's historical significance, and an artistic description of the site's aesthetics.

We exclude 24,369 sites with a missing construction year and 5,702 sites with missing coordinates. Moreover, we exclude sites erected before the Italian unification (1861) and after 2000. We exclude sites older than 1861 because regional coverage before unification is uneven, possibly reflecting the heterogeneous designation criteria among pre-unitary states. This leaves us with 32,369 sites. Among these, 44 are georeferenced in areas that are no longer part of Italy (Fiume, Dalmatia, and various islands that are now part of Greece). We discard these sites, yielding a sample of 32,325 monuments.

As with the street data, we assign monuments to municipalities at the 1991 borders to ensure geographical consistency.

We employ the following prompt to infer whether each monument is associated with a national identity motive. In most cases, national identity monuments commemorate soldiers from the munic-

ipality who died during World Wars 1 and 2.

gpt-5 Prompt for Monument Classification

You are an expert in Italian history and culture.

You will be provided with a description of a monument that commemorates a historical event, figure, or cultural aspect.

Your task is to determine whether the monument commemorates Italian national identity.

Information:

- Monument Category: {category}
- Historical Information: {historical_information}
- Description: {description}

Return a JSON object with the following structure:

```
{  
  "national_identity": 1 if the monument commemorates Italian national identity, 0 otherwise  
}
```

A.IV Instrumental Variable: First-Stage Simulation Test

This appendix describes the logic and implementation of the ship-level co-location diagnostic used to assess the plausibility of the first-stage mechanism underlying the instrumental variable strategy.

The instrumental variable strategy exploits variation in ship-level origin composition to generate quasi-exogenous differences in post-arrival exposure to other Italian origins. A necessary condition for relevance is that interactions formed during the transatlantic voyage affected subsequent settlement decisions. Under this hypothesis, passengers traveling on the same ship should display excess residential co-location after arrival, relative to what would be expected based solely on aggregate destination sizes.

The diagnostic implemented here provides a direct test of this implication. It evaluates whether shipmates from different municipalities are disproportionately likely to settle in the same U.S. enumeration districts, after accounting for the distribution of district sizes and removing the mechanical contribution of the ship itself. This test applies the logic of Basse et al. (2024) to construct an exact finite-sample randomization test under a null of conditional exchangeability, in which post-arrival district assignments are independent across shipmates once aggregate destination capacities are held fixed.

Consider a ship-year s with n_s passengers indexed by $j = 1, \dots, n_s$. Let $d(j)$ denote the enumeration district in which passenger j is observed in the census, and let $m(j)$ denote the municipality of origin. For each district d , define n_{sd} as the number of passengers from ship s residing in district d , and n_{smd} as the number of passengers from ship s residing in district d and originating from municipality m .

We define the observed cross-origin co-location statistic as

$$T_s^X = \sum_d \left[\binom{n_{sd}}{2} - \sum_m \binom{n_{smd}}{2} \right], \quad (\text{A.1})$$

which counts the number of pairs of shipmates who settle in the same district but originate from different municipalities. By construction, this statistic excludes within-municipality pairs and therefore does not mechanically capture pre-existing origin-based networks.

To construct a benchmark under the null of no ship-based settlement ties, we simulate counterfactual district assignments for the passengers of ship s . Let N_{dt} denote the total number of individuals observed in district d in census year t , and let n_{sdt} denote the contribution of ship s to district d . We define a leave-one-out district capacity vector

$$N_{dt}^{(-s)} = N_{dt} - n_{sdt}, \quad (\text{A.2})$$

which removes the mechanical contribution of ship s from the destination distribution.

Under the null hypothesis, the n_s passengers of ship s are assigned to districts without replacement, with probabilities proportional to $N_{dt}^{(-s)}$. Municipality labels $m(j)$ are held fixed across simulations. For each simulated assignment, we recompute the statistic T_s^X , yielding a simulated distribution $\{T_{s,r}^{X,MC}\}_{r=1}^R$.

For each ship-year s , we compute the upper-tail Monte Carlo p -value

$$p_s^{MC} = \Pr\left(T_s^{X,MC} \geq T_s^X\right), \quad (\text{A.3})$$

as well as the associated standardized statistic based on the simulated mean and variance. A systematic excess of low p_s^{MC} values across ship-years indicates that shipmates from different municipalities co-locate more than would be expected based on district sizes alone.

Figure IIIb provides strong empirical support for the maintained first-stage mechanism. The distribution of ship-level standardized co-location statistics is sharply shifted relative to the null distribution implied by random assignment, with a large mass at positive values. In the pooled sample, the aggregate standardized statistic is extremely large, reflecting pervasive excess co-location across ship-years, and the number of ships with p -values below conventional significance thresholds substantially exceeds what would be expected under the null. Importantly, these patterns arise from cross-municipality co-location and are computed using a leave-one-out assignment scheme, ruling out mechanical explanations based on same-origin networks or district size effects. Taken together, the evidence indicates that ship-level interactions shaped post-arrival settlement patterns, thereby validating the relevance of ship-based variation as a source of exogenous shifts in local exposure to Italian diversity.

A.V Spatial Standard Errors with High-Dimensional Fixed Effects

As a robustness check, we estimate standard errors that allow for spatial autocorrelation following Conley (1999). Although Conley standard errors are widely used in applied work, existing implementations, particularly in Stata, become computationally prohibitive in large samples and in the pres-

ence of high-dimensional fixed effects (HDFE, Colella et al., 2023). In our application ($N \approx 300,000$), this constraint is binding. We have developed a fast Python implementation of OLS and exactly identified 2SLS with multi-way HDFE and Conley-type spatial correlation, using blocked neighbor searches and just-in-time compilation to ensure memory safety and computational feasibility, to tackle this software limitation. The code is available upon request at davide.coluccia@bristol.ac.uk.

B LINKED SAMPLE OF ELLIS ISLAND AND POPULATION CENSUS RECORDS

This section provides a detailed discussion of the primary data and the algorithm we developed to link the Ellis Island immigration directories and the federal census records. Then, we assess the linked sample’s reliability and representativeness and present key descriptive statistics.

B.I The Ellis Island Immigration Data

The individual-level dataset on Italian immigrants in the United States is from Coluccia and Spadavecchia (2025). The dataset comprises 4,194,728 uniquely identified records of Italians who entered the United States through the Ellis Island immigration station between 1892 and 1924. The data contain information on the name and surname, age, immigration year, ship name, origin municipality, and literacy of each immigrant. We defer the interested reader to the original paper for a detailed methodological discussion of the data.

Origin municipalities are missing in 12% of cases. The remaining 88% are geo-coded using commercial software. Approximately 32% are geo-coded to coordinates outside Italy. These cases conflate reporting errors and migrants returning from temporary stays in Italy who had previously traveled and settled in the United States. Ultimately, our analysis sample thus comprises 2,526,024 individuals who are located at the coordinates of their municipality of origin in Italy.

B.II Motivation and High-Level Description of the Approach

Linking the population census and the Ellis Island individual records shares several challenges faced by standard intergenerational census-linking algorithms (e.g., see Abramitzky et al., 2021). As is common in (historical) record linkage, frequent misspellings and transcription errors do not allow exact matching. Instead, automated algorithms implement “fuzzy” string matching methods that allow the linking of strings with minor discrepancies.

Our task, however, presents additional, more specific obstacles. First, the Ellis Island records were collected by American officers. Hence, Italian names and surnames display frequent phonetic misspellings, translations, and abbreviations. Second, when completing their census enumeration, Italians commonly Americanized—i.e., translated—their names (e.g., see Biavaschi et al., 2017). Both circumstances are problematic for standard fuzzy string-matching algorithms because translations and phonetic errors frequently result in vastly different string records.

We design a linking algorithm to deal with these and related challenges. Our approach leverages recent advances in Natural Language Processing (NLP) and Large Language Models (LLM) to circumvent the algorithmic shortcomings of common string similarity metrics. More specifically, we capture the similarity between translated versions of the same name through a fine-tuned sentence bidirectional encoder representations from transformers (SBERT) model (Reimers and Gurevych, 2019). SBERT models produce string embeddings that encode the contextual representation of tokens. We complement the SBERT similarity metric with a standard Jaro-Winkler string similarity measure to identify spelling and phonetic mistakes.

We use the SBERT-Jaro-Winkler distances and the immigration year to pair each Ellis Island record

with a set of plausible links in the US Census. We label the resulting sample as “*Sample 1*.” We propose two methodologies to reduce the prevalence of “multiple matches,” i.e., Ellis Island records paired with more than one Census entry. First, in dataset “*Sample 2*,” we employ a state-of-the-art LLM to identify the most likely match(es) among those that appear in *Sample 1*. Second, in “*Sample 3*,” we take the match(es) that maximize a harmonic average of the SBERT-Jaro-Winkler similarities within each match. Our analysis is based on *Sample 1*, but we show that all our results remain unchanged using the other samples. As is standard in the literature, we weigh multiple matches by the inverse of the number of matches to avoid multiple counting.

B.III Technical Description of the Algorithm Implementation

Figure D.1 provides a visual overview of the algorithm. We now describe each component in greater technical detail.

B.III.1 Construction of the Training Data

The starting point is the individual-level data contained in the Ellis Island ($N \approx 2,500,000$) and the Census ($N \approx 4,400,000$). We extract the universe of first names listed in these records and harness the flexibility of a state-of-the-art LLM (OpenAI’s gpt-4o-mini) to assign each name to an existing Italian name. This first step produces a set of approximately 40,000 names that appear in the original data and that the LLM pairs with their correct Italian counterpart. We split the sample into two equal-sized sub-samples. The following panel replicates the prompt that the LLM receives to classify name pairs.

gpt-4o-mini Prompt for Name Cleaning

You are an expert in reconstructing Italian **names** that were altered by U.S. immigration officials.

Task:

Return ONLY a valid JSON object with the most likely original Italian **first name** and a confidence score (-100 and 200).

Instructions:

- You must return an existing Italian first name.
- If you cannot find an Italian name, return the original and set the confidence to -100.
- If the name is already an Italian **real** first name, return as is and set the confidence to 200.

Rules for the confidence level:

- 200 = Name is a perfect match with no discrepancies.
- 90 = Extremely confident match (name clearly matches, considering translation/mis-spelling).
- 80 = Likely match but with minor discrepancies (plausible misspelling).
- 70 = Possible but uncertain match; significant discrepancies exist.
- 60 = Unlikely match; substantial doubts remain.

- 50 = Most likely match, but still very uncertain.
- 0 = Uncertain match; no clear match found.
- -100 = No Italian name found; return the original name.

Output Format:

```
{{
original: <original recorded name>,
inferred italian: <most likely Italian name>,
confidence: <confidence score>
}}
```

We restrict the attention to names with a confidence score of at least 80 to ensure that the training set comprises high-quality matches.

B.III.2 Training of the NLP Embeddings

Using the first subsample, we fine-tune a multilingual pre-trained SBERT model from the huggingface repository (paraphrase-multilingual-mpnet-base-v2). The training data contains both translations as well as minor spelling and phonetic mistakes. The embeddings produced by the fine-tuned SBERT model thus replicate the high contextual similarity between recorded names that map onto the same Italian name.

B.III.3 Filtering Classification Rule

Consider two first names, a and b . Name a appears in the Ellis Island data, and b is recorded in the Census. Let $\mathbf{e}_a$ and $\mathbf{e}_b$ denote the embeddings produced by the fine-tuned SBERT model. We say that b is a candidate match for a if the cosine similarity between the two embeddings is above a threshold $\tau_1 \in [0, 1]$:

$$\sigma_{ab}^{SBERT} \equiv \frac{\mathbf{e}_a^\top \mathbf{e}_b}{\|\mathbf{e}_a\| \cdot \|\mathbf{e}_b\|}, \quad (\text{B.1})$$

or if the Jaro-Winkler similarity between a and b is above a certain threshold $\tau_2 \in [0, 1]$. The Jaro-Winkler similarity is defined as one minus the Jaro-Winkler distance. Formally, it is

$$\sigma_{ab}^{JW} \equiv 1 - \delta_{ab}^{JW} = 1 - \begin{cases} 0 & \text{if } m_{ab} = 0 \\ \frac{1}{3} \left(\frac{m_{ab}}{\|a\|} + \frac{m_{ab}}{\|b\|} + \frac{m_{ab} - t_{ab}}{m_{ab}} \right) & \text{else} \end{cases}, \quad (\text{B.2})$$

where m_{ab} is the number of matching characters in the two names, and t_{ab} is the number of transpositions. The number of transpositions is defined as a number of matching characters that are not in the right order divided by two. The Jaro-Winkler similarity is a widely adopted edit metric in record linkage algorithms because it effectively captures minor typographical errors, transpositions, and abbreviations in short strings, and its prefix-weighted structure prioritizes matches that share the same initial characters, which are typically more stable in historical records. The score is normalized such that zero means no similarity, and one denotes exact matches.

We employ the SBERT and Jaro-Winkler disjoint similarities because they capture different imperfect linking instances. A low cosine similarity between two SBERT embeddings reveals that the two names are likely translations of a single name. Hence, they should be considered as candidate matches. A low Jaro-Winkler similarity between two names indicates that the two names are imperfect transcriptions of the same name and, as such, should be considered candidate matches. Since these are different cases, we do not take an average “SBERT-Jaro-Winkler” similarity. Instead, we populate the set of possible matches as those b ’s that have *either* a “high” SBERT-cosine *or* a “high” Jaro-Winkler similarity with a

B.III.4 Estimation of the Classification Hyperparameters

To estimate the thresholds τ_1 and τ_2 , we employ the remaining half of the names matched to their proper Italian counterpart. Specifically, we split the 20,000 samples into a training set composed of 14,000 pairs and a test set of 6,000 names, over which we evaluate the performance of the composite distance classifier. We augment the train and test sets with a random sample of false matches drawn to reflect the frequency distribution of names in the population. Let $\sim$ denote an equivalence class whereby $a \sim b$ implies that names a and b constitute a match. Given the classification rule

$$a \sim b \Leftrightarrow \sigma_{ab}^{SBERT} \geq \tau_1 \vee \sigma_{ab}^{JW} \geq \tau_2 \quad (\text{B.3})$$

we search for the values $\hat{\tau}_1$ and $\hat{\tau}_2$ that maximize the $F1$ -score over the training set.¹ The $F1$ -score is defined as the harmonic average between the precision and recall of the classifier. The precision is defined as the number of true positive results divided by the number of all samples predicted to be positive, including those not identified correctly. The recall is the number of true positive results divided by the number of all samples that should have been identified as positive. The $F1$ -score is a standard measure to target when fine-tuning classifiers, as it balances the tradeoff between accuracy and coverage.

In our application, the optimization routine selects $\hat{\tau}_1 = .65$ and $\hat{\tau}_2 = .95$. We provide additional details on the performance of the fine-tuned classifier in the next section.

B.III.5 Construction of Sample 1

The first sample comprises all matches obtained using the standard classification rule that satisfies the optimal thresholds. Formally, let B denote the universe of records in the Census. For each Ellis Island record a , the set of matches in *Sample 1* is denoted by

$$m_a^1 = \left\{ b \in B \text{ such that } |t_b - t_a| \leq 1 \wedge a \sim b \wedge \tilde{\sigma}_{ab}^{JW} \geq \tau_3 \right\}, \quad (\text{B.4})$$

where t_b and t_a denote the immigration year recorded in the census and in the Ellis Island registers, and the classification rule $\sim$ is computed using the couple optimal thresholds $(\hat{\tau}_1, \hat{\tau}_2)$ estimated in the previous step. The term $\tilde{\sigma}_{ab}^{JW}$ denotes the *surname* Jaro-Winkler similarity between records a and b . Since census enumerators do not translate surnames, it is not necessary to apply NLP tools on

¹Specifically, we run a grid search over the $[0, 1]^2$ space for τ_1 and τ_2 for evenly spaced values of the thresholds using a 0.05 pass.

surnames because the Jaro-Winkler similarity alone ensures that we include misspellings in the set of potential matches. We select $\tau_3 = \hat{\tau}_2$ to ensure high-quality surname matches.

B.III.6 Construction of Sample 2

In *Sample 2*, we refine the set of matches obtained in the previous step using a state-of-the-art LLM (OpenAI's gpt-4o-mini). We run the following prompt using the API.

gpt-4o-mini Prompt for Match Disambiguation

You're an expert historian matching Italian immigrant records to U.S. census data.

Original immigrant details:

- Name: {original_name}

Census candidates:

- Candidates: {candidates}

Task

Carefully assess if any candidate matches the immigrant, considering:

- Italian-to-English translations (e.g., Giovanni ↔ John, Giuseppe ↔ Joseph).
- Phonetic errors, abbreviations, and misspellings from historical American records.

Prefer translations over phonetic matches.

Provide your best candidate selection clearly reflecting your certainty.

If there is one match that is clearly the best, only return that match.

If there is no clear match, return the most likely match(es) and set confidence to 6.

Rules for assigning confidence:

- 1 = Name is a perfect match with no discrepancies.
- 2 = Extremely confident match (name clearly matches, considering translation/misspelling).
- 3 = Likely match but with minor discrepancies (plausible misspelling).
- 4 = Possible but uncertain match; significant discrepancies exist.
- 5 = Unlikely match; substantial doubts remain.
- 6 = Most likely match, but still very uncertain.

Output Format:

```
{{
best matches: <list of candidate names>,
confidence: <confidence score>
}}
```

In the prompt, the set of candidate names that gpt-4o-mini is asked to disambiguate is given by, for the general record a , the set m_a^1 . Thus, if we let $\succsim$ denote the operator that maps potential matches into GPT-validated matches. The set of matches in *Sample 2* is thus given by

$$m_a^2 = \left\{ b \in m_a^1 \text{ such that } b \succsim b' \ \forall \ b' \in m_a^1 \right\} \subseteq m_a^1, \quad (\text{B.5})$$

which simply implies that, in m_a^2 , all matches must be within m_a^1 , so, in this sense, *Sample 2* is a refinement of *Sample 1*, and they must be valid according to GPT.

B.III.7 Construction of Sample 3

The set *Sample 3* is an alternative refinement of *Sample 1*. Let $\hat{\sigma}_{ab}$ be the similarity metric that determines the name-level match. Thus, if $\sigma_{ab}^{SBERT} \geq \hat{\tau}_1$ and $\sigma_{ab}^{IW} < \hat{\tau}_2$, it shall be $\hat{\sigma}_{ab} = \sigma_{ab}^{SBERT}$, and the reverse if $\sigma_{ab}^{IW} \geq \hat{\tau}_2$ and $\sigma_{ab}^{SBERT} < \hat{\tau}_1$. In the residual case where both similarities are above the threshold, we pick $\hat{\sigma}_{ab} = \sigma_{ab}^{IW}$. Then, the set of matches included in *Sample 3* for the generic record a is

$$m_a^3 = \left\{ b \in m_a^1 \text{ such that } \alpha \cdot \hat{\sigma}_{ab} + (1 - \alpha) \cdot \tilde{\sigma}_{ab} \geq \alpha \cdot \hat{\sigma}_{ab'} + (1 - \alpha) \cdot \tilde{\sigma}_{ab'} \ \forall \ b' \in m_a^1 \right\} \subseteq m_a^1, \quad (\text{B.6})$$

where $\alpha \in (0, 1)$ is a weight that determines the importance assigned to the name relative to the surname. In this case, the refinement relative to *Sample 1* is given by taking the most similar match(es) in the set of plausible candidate matches m_a^1 , according to a weighted average of the name and surname similarities $\hat{\sigma}_{ab}$ and $\tilde{\sigma}_{ab}$.

B.IV Internal Validation and Representativeness of the Linked Sample

This section complements Section III.A.

When constructing candidate matches for each Ellis Island record, we restrict the census population to individuals born within a ± 2 -year window around the reported birth year and who immigrated within a ± 5 -year window around the reported immigration year. Such buffers are standard in historical record linkage as they reflect reporting errors in birth—and plausibly immigration—years, partly due to low literacy and recall issues (e.g., Abramitzky et al., 2021).

Figure D.2 shows the difference in birth year (Figure D.2a) and immigration year (Figure D.2b) between linked census and Ellis Island records. Three patterns emerge. First, in both dimensions, the modal match occurs at zero gap, indicating that a relative majority of links involve identical reported years. Second, the distributions are symmetric around zero, ruling out systematic biases toward matching individuals reported as born or immigrated earlier or later in the census. Third, the linking probability declines with the absolute gap between reported years. This pattern is mechanically expected in a well-behaved linkage procedure: if matches were driven by noise or by the blocking windows alone, the distribution of gaps would be approximately uniform within the admissible range, or even exhibit excess mass in the tails. The observed monotonic decline, therefore, indicates that the linkage algorithm assigns greater weight to records that agree more closely on blocking variables, and that spurious matches at the boundaries of the admissible windows are relatively rare.

An additional indicator of match quality is the similarity between names and surnames in the Ellis Island and census records. For each accepted link, the name similarity must exceed either 0.95 under the Jaro–Winkler metric or 0.6 under SBERT. Figure D.3a reports the distribution of Jaro–Winkler similarities, whereas Figure D.3b reports the distribution of SBERT similarities. There are very few matches with SBERT similarity below 0.6, while approximately 30% of the sample has Jaro–Winkler similarity below 0.95. This pattern is expected. The SBERT similarity is designed to capture semantic equivalence between name translations, which often appear orthographically distant and therefore score poorly under Jaro–Winkler. By contrast, high Jaro–Winkler matches generally also exhibit high SBERT similarity, indicating that the two distances are mutually consistent in identifying plausible matches. Importantly, a substantial share of the linked sample (approximately 40%) is constituted by nearly-perfect matches, i.e., matches with a name-similarity above 0.95. This pattern suggests that a large fraction of links is by near-exact agreement on names, rather than relying on the more permissive translation-aware metric.

Figure D.3c displays the distribution of surname similarities. By construction, accepted matches must have a Jaro–Winkler surname similarity above 0.95. We do not apply the SBERT correction to surnames because it was extremely rare for Italians to translate their surname (Carnevale, 2010). Interestingly, the majority of surnames ($\approx 60\%$) exhibit a near-perfect surname match, i.e., a surname similarity of one. This may reflect the fact that surnames were less likely than given names to be altered or Anglicized upon arrival, and that census enumerators tended to record them more consistently across sources.

B.V External Validation of the Linked Sample

In Figure D.4, we compare the baseline linked migrant sample (Sample 1 in Appendix B.III) with two refinements. In Sample 2, we use GPT 4 to validate the matches produced by the baseline algorithm and retain only those deemed most plausible. In Sample 3, for each Ellis Island record with at least one accepted link, we retain only the match with the highest combined name and surname similarity. Both samples are refinements in the sense that they impose additional restrictions on Sample 1, and therefore are weakly smaller by construction.

In Figure D.4a, we plot the matching rate of Sample 2 (in red) and 3 (in blue) over time. The two series co-move and are qualitatively very similar to the pattern displayed in Figure Ia for Sample 1. The average matching rate of the baseline sample (18.762%) is also similar to Sample 2 (16.142%), indicating that GPT 4 deems 90% of the matches in Sample 1 as extremely plausible.

In Figure D.4b, we report the distribution of the number of matches in Sample 2 (red-contoured bars) and 3 (gray lines). In both cases, as in Sample 1 (Figure Ib), single matches account for the vast majority of the accepted links in the sample.

Finally, in Figure D.4c, we report the correlation between Sample 1 and Samples 2 (in red) and 3 (in blue). To construct this figure, we aggregate the matches at the municipality level; hence, each dot displays the municipality-level share of Ellis Island migrants from each municipality that are paired with at least one valid census record in Sample 1 (on the x -axis) and in Samples 2 and 3 (on

the y -axis). By construction, both scatter plots lie below the 45-degree line because Samples 2 and 3 are refinements of Sample 1. In both cases, the correlation between the matching rates is large (0.87 for Samples 1–2 and 0.98 for Samples 1–3) and highly statistically significant, indicating that both refinements rescale the matching rate without modifying its spatial coverage.

Validating the linked dataset using external sources at the individual level is not feasible, as this is the first dataset to recover individual-level information on Italian immigrants' municipalities of origin. More aggregated bilateral data, such as municipality-to-county migration flows, are also unavailable. In Figure D.5, we compare out-migration flows constructed from the Ellis Island records with official figures compiled by the Italian Commissioner General for Emigration and published in the *Annuario Statistico della Emigrazione Italiana dal 1876 al 1925*. As described in Coluccia and Spadavecchia (2025), these statistics report the annual number of emigrants to the United States by region, based on passports issued for transatlantic travel.

Figure D.5a (resp. Figure D.5c) reports the number of emigrants by year compiled using data from the *Annuario* (solid red line, on the left y -axis) and the Ellis Island full (resp. linked) dataset (dashed blue line, right y -axis). The Ellis Island dataset closely replicates the outflow constructed from Official Statistics data throughout the sample period. The number of emigrants in the Ellis Island dataset is lower than in the official statistics because the municipality of origin contains missing or invalid values, but, as explained by Coluccia and Spadavecchia (2025), this is unlikely to generate systematic selection into the dataset.

In Figure D.5b, we report the correlation between the full Ellis Island dataset and data from the *Annuario*. To construct this figure, we collapsed the full Ellis Island dataset, i.e., irrespective of whether migrants are linked to the census, at the region-year level. Hence, each dot reports the number of US emigrants in each year between 1890 and 1924 according to the official statistics (on the x -axis) and the Ellis Island dataset (on the y -axis). We find a positive and highly statistically significant correlation between the two datasets, as evidenced by the regression coefficient reported separately. In all but one case (Calabria, 1902), the difference between the Ellis Island and the official statistics is small relative to the number of emigrants.

Figure D.5d replicates the exercise using the linked Ellis Island dataset. Since, as we documented in Figure Id, the linked sample preserved the spatial distribution of the full Ellis Island dataset, we expect to recover a positive correlation between the linked dataset and official statistics. The figure confirms this pattern: the regression coefficient remains highly statistically significant (though scaled by a factor approximately equal to the matching rate).

Together with the validation results documented in Section III.A, these exercises indicate that (i) the Ellis Island dataset replicates less granular data compiled from official sources, (ii) the linked Ellis Island dataset preserves the representativeness of the sample, and (iii) the reliability of the linking algorithm is confirmed by multiple internal validation refinements.

C ADDITIONAL TABLES

Table C.1. Comparison Between Linked and Non-Linked Migrants

	Linked	Non-Linked	Difference	
	(1)	(2)	(3)	(4)
Panel A. Comparison with Ellis Island Records				
Age at Migration	22.810	26.225	-3.416	(0.480)
Latitude of Origin Municipality	40.727	41.011	-0.284	(0.075)
Longitude of Origin Municipality	13.837	13.676	0.162	(0.046)
Area of Origin Municipality:				
North-West	0.082	0.090	-0.008	(0.004)
North-East	0.055	0.070	-0.015	(0.002)
Center	0.109	0.130	-0.021	(0.004)
South	0.484	0.467	0.018	(0.013)
Islands	0.270	0.243	0.027	(0.015)
Panel B. Comparison with Census Records				
Age at Migration	22.629	22.387	0.246	(0.351)
Is Male	0.801	0.596	0.205	(0.020)
Is Employed	0.513	0.449	0.063	(0.013)
log(Occupational Income)	3.184	3.157	0.026	(0.004)
Industry:				
Durable Manufacturing	0.119	0.115	0.004	(0.003)
Nondurable Manufacturing	0.179	0.161	0.018	(0.003)
Trade	0.184	0.169	0.014	(0.003)
Transportation	0.054	0.054	0.000	(0.002)
Agriculture	0.076	0.089	-0.013	(0.003)
Occupation:				
Professional	0.018	0.017	0.002	(0.001)
Managerial	0.073	0.065	0.008	(0.003)
Clerical	0.012	0.013	-0.001	(0.001)
Sales	0.034	0.035	-0.001	(0.001)
Skilled Manufacturing	0.281	0.248	0.033	(0.003)
Unskilled Manufacturing	0.372	0.398	-0.026	(0.006)
Census Region of Residence:				
North-East	0.593	0.585	0.008	(0.008)
Midwest	0.026	0.024	0.002	(0.002)
South	0.014	0.022	-0.008	(0.002)
West	0.061	0.057	0.004	(0.003)

Notes. This table compares migrants in the linked sample with those without a valid link. The unit of observation is an individual, observed once. Columns (1) and (2) report mean characteristics for linked and non-linked individuals. In Panel A, the non-linked group consists of all non-linked records in the Ellis Island dataset; in Panel B, it consists of all non-linked census records of Italian immigrants. Characteristics in Panel A are observed in the Ellis Island data, while those in Panel B are observed in the population census. Columns (3)–(4) report differences between linked and non-linked migrants, with standard errors in parentheses. Referenced on pages: [13](#), [34](#).

Table C.2. Exposure to Immigrant Heterogeneity and National Identity Formation, Excluding Firstborn Children

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Local Name Index						
Italian Diversity	-0.240*** (0.054)	-0.213*** (0.052)	-0.164*** (0.052)	-0.224*** (0.053)	-0.092 (0.060)	-0.157** (0.072)
Mean Dep. Var.	10.697	10.697	10.710	10.698	10.697	10.697
Panel B: Distance-Adjusted Local Name Index						
Italian Diversity	0.528*** (0.090)	0.544*** (0.085)	0.614*** (0.080)	0.510*** (0.087)	0.464*** (0.096)	0.350*** (0.096)
Mean Dep. Var.	24.163	24.167	24.206	24.167	24.167	24.167
Panel C: Foreign Name Index						
Italian Diversity	0.065*** (0.007)	0.066*** (0.007)	0.073*** (0.007)	0.060*** (0.007)	0.055*** (0.008)	0.043*** (0.008)
Mean Dep. Var.	1.966	1.966	1.970	1.966	1.966	1.966
Municipality FE	Yes	Yes	–	Yes	Yes	Yes
County FE	Yes	–	–	–	–	–
Year FE	Yes	Yes	–	Yes	Yes	Yes
Hexagon FE	No	Yes	–	Yes	Yes	Yes
Municipality-Year FE	No	No	Yes	No	No	No
Hexagon-Year FE	No	No	Yes	No	No	No
Individual Controls	No	No	No	Yes	No	No
Demographic Controls	No	No	No	No	Yes	Yes
Economic Controls	No	No	No	No	No	Yes
# Individuals	207,673	207,499	205,832	207,484	207,499	207,499
# Observations	308,520	308,196	304,480	308,173	308,196	308,196

Notes. This table reports the effect of exposure to diversity within the Italian community on name-based indices from specification (4). Unlike in the main analysis, we exclude firstborn children when computing parents' name-based indices. Panel A uses the local name index defined in (1) as the dependent variable; Panel B uses the distance-adjusted local name index defined in (2); Panel C uses the foreign name index of Abramitzky et al. (2020). The unit of observation is an individual observed in the first census in which they appear and, when possible, linked across subsequent censuses between 1900 and 1930. All specifications include municipality and census-year fixed effects. Column (1) includes county fixed effects; columns (2) and (4–6) replace county with more granular hexagon fixed effects. Column (3) replaces municipality fixed effects with municipality-year fixed effects and hexagon fixed effects with hexagon-year fixed effects. Demographic controls (included in column 5) are measured at the district level and include population, the share of women, African Americans, immigrants, and Italian immigrants. Economic controls (included in column 6) are measured at the district level and include employment shares by industry. Standard errors, clustered at the district level, are reported in parentheses. Referenced on page: 29.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table C.3. Exposure to Immigrant Heterogeneity and National Identity Formation, Fractionalization Index

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Local Name Index						
Italian Diversity (Fractionalization)	-0.549*** (0.071)	-0.523*** (0.078)	-0.474*** (0.085)	-0.495*** (0.073)	-0.093* (0.055)	-0.108* (0.061)
Mean Dep. Var.	3.990	3.989	3.989	3.989	3.989	3.989
Panel B: Distance-Adjusted Local Name Index						
Italian Diversity (Fractionalization)	0.073*** (0.011)	0.072*** (0.012)	0.064*** (0.013)	0.065*** (0.011)	0.024** (0.010)	0.021* (0.011)
Mean Dep. Var.	9.593	9.593	9.593	9.593	9.593	9.593
Panel C: Foreign Name Index						
Italian Diversity (Fractionalization)	0.080*** (0.004)	0.078*** (0.005)	0.078*** (0.005)	0.068*** (0.004)	0.043*** (0.004)	0.038*** (0.004)
Mean Dep. Var.	0.758	0.758	0.758	0.758	0.758	0.758
Municipality FE	Yes	Yes	–	Yes	Yes	Yes
County FE	Yes	–	–	–	–	–
Year FE	Yes	Yes	–	Yes	Yes	Yes
Hexagon FE	No	Yes	–	Yes	Yes	Yes
Municipality-Year FE	No	No	Yes	No	No	No
Hexagon-Year FE	No	No	Yes	No	No	No
Individual Controls	No	No	No	Yes	No	No
Demographic Controls	No	No	No	No	Yes	Yes
Economic Controls	No	No	No	No	No	Yes
# Individuals	223,721	223,531	221,913	223,521	223,531	223,531
# Observations	334,668	334,303	330,615	334,283	334,303	334,303

Notes. This table reports the effect of exposure to diversity within the Italian community on name-based indices from specification (4). Unlike in the main analysis, we use the fractionalization index ($= 1 - \text{Hirschman-Herfindahl Index}$) to measure the diversity of the Italian immigrant community. Panel A uses the local name index defined in (1) as the dependent variable; Panel B uses the distance-adjusted local name index defined in (2); Panel C uses the foreign name index of Abramitzky et al. (2020). The unit of observation is an individual observed in the first census in which they appear and, when possible, linked across subsequent censuses between 1900 and 1930. All specifications include municipality and census-year fixed effects. Column (1) includes county fixed effects; columns (2) and (4–6) replace county with more granular hexagon fixed effects. Column (3) replaces municipality fixed effects with municipality-year fixed effects and hexagon fixed effects with hexagon-year fixed effects. Demographic controls (included in column 5) are measured at the district level and include population, the share of women, African Americans, immigrants, and Italian immigrants. Economic controls (included in column 6) are measured at the district level and include employment shares by industry. Standard errors, clustered at the district level, are reported in parentheses. Referenced on page: 29.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table C.4. Exposure to Immigrant Heterogeneity and National Identity Formation, Variants of Local Name Index and Diversity Definitions

	Italian Diversity Measured by:					
	Municipality -		Province -		Region -	
	(1) District	(2) County	(3) District	(4) County	(5) District	(6) County
Panel A: Municipality-Level Local Name Index						
Italian Diversity	-0.164*** (0.015)	-0.126** (0.051)	-0.192*** (0.022)	-0.043 (0.110)	-0.200*** (0.029)	0.116 (0.152)
Mean Dep. Var.	3.989	3.989	3.989	3.989	3.989	3.989
Panel B: Province-Level Local Name Index						
Italian Diversity	-0.105*** (0.012)	-0.067 (0.042)	-0.129*** (0.018)	-0.064 (0.096)	-0.152*** (0.025)	-0.045 (0.137)
Mean Dep. Var.	4.804	4.804	4.804	4.804	4.804	4.804
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Hexagon FE	Yes	Yes	Yes	Yes	Yes	Yes
# Individuals	223,531	223,531	223,531	223,531	223,531	223,531
# Observations	334,303	334,303	334,303	334,303	334,303	334,303

Notes. This table reports the effect of exposure to diversity within the Italian community on name-based indices from specification (4). Panel A uses the local name index defined in (1) as the dependent variable; Panel B uses a variant where the name distribution is taken at the province-level instead of the municipality-level. Each column reports estimates from a variant of equation (3) that differs only in the geographic level at which we construct H . Column (1) uses the baseline municipality–district definition. Column (2) aggregates districts to the county level (municipality–county). Columns (3–4) redefine origins at the province level (province–district and province–county, respectively). Columns (5–6) analogously use regions as origins (region–district and region–county, respectively). Provinces are coarser than municipalities; Regions are coarser than provinces. Counties are coarser than districts. The unit of observation is an individual observed in the first census in which they appear and, when possible, linked across subsequent censuses between 1900 and 1930. All specifications include municipality, census-year fixed effects, and hexagon fixed effects. Standard errors, clustered at the district level, are reported in parentheses. Referenced on page: 29.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table C.5. Movers Design and National Identity Formation, Excluding Firstborn Children

	Local Name Index (LNI)					Distance LNI
	(1)	(2)	(3)	(4)	(5)	(6)
Diversity Shock $\times$ Post-Move	-0.430* (0.238)	-0.429* (0.238)	-0.509** (0.242)		-0.059 (1.307)	0.066** (0.029)
Diversity Shock (Cont.) $\times$ Post-Move				-15.172* (7.807)		
Age					0.006 (0.023)	
Diversity Shock $\times$ Post-Move $\times$ Age					-0.013 (0.041)	
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Hexagon FE	Yes	Yes	–	–	–	–
Time Since Move FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	No	Yes	–	–	–	–
Hexagon-Cohort FE	No	No	Yes	Yes	Yes	Yes
# Individuals	72,361	72,361	70,793	70,793	70,793	70,793
# Observations	235,386	235,385	229,669	229,669	229,669	229,669
Mean Dep. Var.	3.913	3.913	3.905	3.905	3.905	9.603

Notes. This table reports results from the movers design specification in (7). Unlike in the main analysis, we exclude firstborn children from the estimation sample. The unit of observation is an individual migrant observed in each year in which they have a child born in the United States between 1900 and 1930. The dependent variables are the local name index (columns 1–5) and the distance-adjusted local name index (column 6) defined in (1)–(2). The baseline treatment is the interaction between an indicator equal to one after an individual moves to a different hexagon (and zero otherwise) and an indicator equal to one if the difference in diversity between the Italian communities in the destination and origin districts lies in the top 20% of the distribution (and zero otherwise). Column (4) replaces this indicator with the continuous diversity difference measure. Column (5) additionally includes age and its interaction with the baseline treatment. All specifications include individual and event-time (time since move) fixed effects. Column (2) also includes cohort (year) and hexagon fixed effects, while columns (3–6) replace these with hexagon-by-cohort fixed effects. Standard errors, clustered at the district level, are reported in parentheses. Referenced on page: 29.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table C.6. Movers Design and National Identity Formation, Fractionalization Index

	Local Name Index (LNI)					Distance LNI
	(1)	(2)	(3)	(4)	(5)	(6)
Diversity Shock $\times$ Post-Move	-0.497*** (0.155)	-0.505*** (0.155)	-0.678*** (0.164)		-2.322 (1.796)	0.098*** (0.021)
Diversity Shock (Cont.) $\times$ Post-Move				-582.492** (227.827)		
Age					0.005 (0.021)	
Diversity Shock $\times$ Post-Move $\times$ Age					0.049 (0.053)	
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Hexagon FE	Yes	Yes	–	–	–	–
Time Since Move FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	No	Yes	–	–	–	–
Hexagon-Cohort FE	No	No	Yes	Yes	Yes	Yes
# Individuals	103,079	103,079	101,214	101,214	101,214	101,214
# Observations	359,898	359,898	352,641	352,641	352,641	352,641
Mean Dep. Var.	3.905	3.905	3.898	3.898	3.898	9.602

Notes. This table reports results from the movers design specification in (7). Unlike in the main analysis, we use the fractionalization index ($= 1 - \text{Hirschman-Herfindahl Index}$) to measure the diversity of the Italian immigrant community. The unit of observation is an individual migrant observed in each year in which they have a child born in the United States between 1900 and 1930. The dependent variables are the local name index (columns 1–5) and the distance-adjusted local name index (column 6) defined in (1)–(2). The baseline treatment is the interaction between an indicator equal to one after an individual moves to a different hexagon (and zero otherwise) and an indicator equal to one if the difference in diversity between the Italian communities in the destination and origin districts lies in the top 20% of the distribution (and zero otherwise). Column (4) replaces this indicator with the continuous diversity difference measure. Column (5) additionally includes age and its interaction with the baseline treatment. All specifications include individual and event-time (time since move) fixed effects. Column (2) also includes cohort (year) and hexagon fixed effects, while columns (3–6) replace these with hexagon-by-cohort fixed effects. Standard errors, clustered at the district level, are reported in parentheses. Referenced on page: 29.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table C.7. Movers Design and National Identity Formation, Variants of Local Name Index and Diversity Definitions

	Italian Diversity Measured by:					
	Municipality -		Province -		Region -	
	(1) District	(2) County	(3) District	(4) County	(5) District	(6) County
Panel A: Municipality-Level Local Name Index						
Diversity Shock $\times$ Post-Move	-0.335** (0.137)	-0.234 (0.169)	-0.423*** (0.140)	-0.236 (0.176)	-0.430*** (0.133)	-0.237* (0.134)
Mean Dep. Var.	3.909	3.909	3.909	3.909	3.909	3.909
Panel B: Province-Level Local Name Index						
Diversity Shock $\times$ Post-Move	-0.348*** (0.124)	-0.123 (0.159)	-0.340*** (0.123)	-0.095 (0.167)	-0.348*** (0.122)	-0.104 (0.124)
Mean Dep. Var.	4.678	4.678	4.678	4.678	4.678	4.678
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Time Since Move FE	Yes	Yes	Yes	Yes	Yes	Yes
# Individuals	114,052	114,052	114,052	114,052	114,052	114,052
# Observations	410,343	410,343	410,343	410,343	410,343	410,343

Notes. This table reports results from the movers design specification in (7). The unit of observation is an individual migrant observed in each year in which they have a child born in the United States between 1900 and 1930. Panel A uses the local name index defined in (1) as the dependent variable; Panel B uses a variant where the name distribution is taken at the province-level instead of the municipality-level. The treatment is the interaction between an indicator equal to one after an individual moves to a different hexagon (and zero otherwise) and an indicator equal to one if the difference in diversity between the Italian communities in the destination and origin districts lies in the top 20% of the distribution (and zero otherwise). Each column reports estimates from a variant of equation (3) that differs only in the geographic level at which we construct H . Column (1) uses the baseline municipality–district definition. Column (2) aggregates districts to the county level (municipality–county). Columns (3–4) redefine origins at the province level (province–district and province–county, respectively). Columns (5–6) analogously use regions as origins (region–district and region–county, respectively). Provinces are coarser than municipalities; Regions are coarser than provinces. Counties are coarser than districts. All specifications include individual and event-time (time since move) fixed effects. Standard errors, clustered at the district level, are reported in parentheses. Referenced on page: 29.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table C.8. Name Particularism in Italy by Exposure to Immigrant Diversity, Alternative Mass Migration Threshold

	Threshold of Emigration Rate for Treatment:					
	1 ‰		2 ‰		5 ‰	
	(1) LNI	(2) D-LNI	(3) LNI	(4) D-LNI	(5) LNI	(6) D-LNI
Diversity Exposure $\times$ Post Mass Migration	-0.464*** (0.091)	0.045*** (0.010)	-0.634*** (0.125)	0.062*** (0.014)	-0.824*** (0.177)	0.077*** (0.020)
Post Mass Migration	0.010 (0.025)	-0.001 (0.003)	-0.008 (0.029)	0.000 (0.003)	-0.060 (0.038)	0.006 (0.004)
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
# Municipalities	7,823	7,823	7,823	7,823	7,823	7,823
# Observations	200,774	200,774	200,774	200,774	200,774	200,774
Mean Dep. Var.	0.879	9.909	0.879	9.909	0.879	9.909

Notes. This table reports the effect of exposure to diversity within the Italian migrant community on name-based indices in Italian municipalities from regression (9). The unit of observation is a municipality observed at a decennial frequency between 1880 and 1930. The dependent variables are the local name index defined in (1) in columns (1), (3), and (5), and the distance-adjusted local name index defined in (2) in columns (2), (4), and (6). The baseline treatment is defined as the share of a municipality's US emigrants who settled in districts in the top quintile of the diversity distribution (3) interacted with a post indicator equals one in years at or after the municipality-specific onset of mass migration (defined as the first year with at least one (columns 1–2), two (columns 3–4), or five (columns 5–6) emigrant per 1,000 inhabitants). The table reports the coefficient of interaction terms between the diversity exposure measure and the post indicator. All regressions include municipality and year fixed effects. Standard errors are clustered at the municipality level and reported in parentheses. Referenced on page: 31.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table C.9. Monuments by Designation in Italy

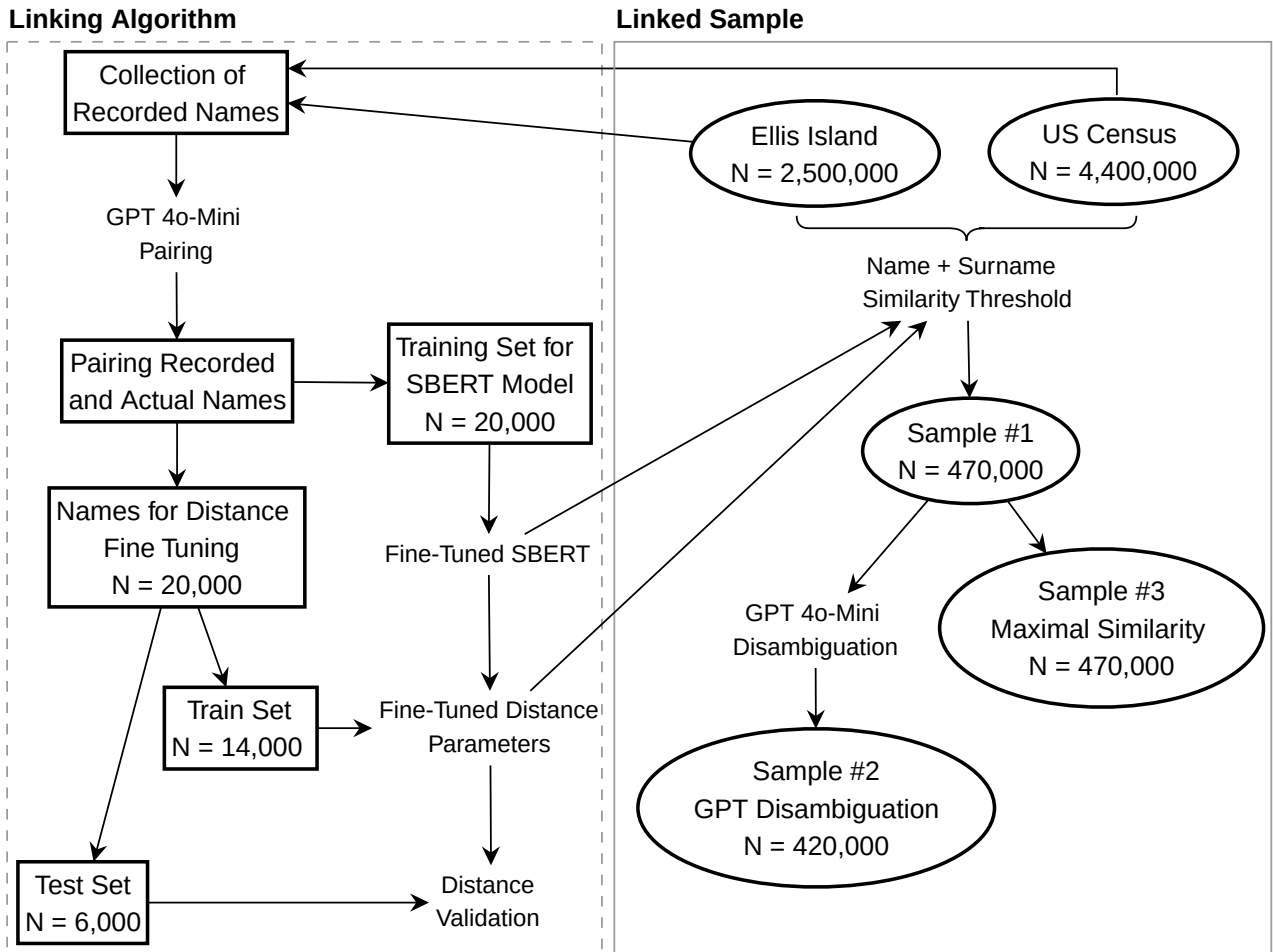
	National Monuments					Other Monuments				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Diversity Exposure $\times I(1870 \leq t < 1880)$	-0.002 (0.007)	0.004 (0.004)	0.000 (0.005)	-0.003 (0.007)	-0.001 (0.008)	0.055 (0.034)	0.074* (0.041)	0.066 (0.041)	0.046 (0.035)	0.036 (0.033)
Diversity Exposure $\times I(1880 \leq t < 1890)$	-0.005 (0.010)	0.005 (0.007)	0.002 (0.008)	-0.010 (0.010)	-0.011 (0.011)	0.040 (0.042)	0.092** (0.046)	0.087* (0.046)	0.047 (0.042)	0.034 (0.043)
Diversity Exposure $\times I(1890 \leq t < 1900)$	-0.004 (0.007)	0.002 (0.004)	-0.000 (0.005)	-0.004 (0.007)	-0.004 (0.007)	0.083* (0.043)	0.108** (0.051)	0.090* (0.051)	0.083* (0.043)	0.068 (0.045)
Diversity Exposure $\times I(1900 \leq t < 1910)$	-0.012 (0.009)	-0.016 (0.010)	-0.016 (0.011)	-0.014* (0.009)	-0.014 (0.009)	0.019 (0.045)	0.090* (0.053)	0.064 (0.053)	0.013 (0.046)	-0.041 (0.042)
Diversity Exposure $\times I(1910 \leq t < 1920)$	0.082** (0.038)	0.019 (0.045)	0.011 (0.047)	0.075** (0.038)	0.094** (0.045)	0.053 (0.040)	0.112** (0.047)	0.100** (0.047)	0.048 (0.041)	0.037 (0.043)
Diversity Exposure $\times I(1920 \leq t < 1930)$	0.139** (0.066)	0.111 (0.072)	0.076 (0.070)	0.136** (0.066)	0.148** (0.075)	-0.038 (0.041)	0.040 (0.048)	0.031 (0.048)	-0.040 (0.041)	-0.067 (0.042)
Diversity Exposure $\times I(1930 \leq t < 1940)$	-0.023 (0.028)	-0.034 (0.028)	-0.039 (0.028)	-0.035 (0.028)	-0.039 (0.033)	-0.032 (0.027)	0.015 (0.028)	0.006 (0.028)	-0.043 (0.028)	-0.048 (0.031)
Diversity Exposure $\times I(1940 \leq t < 1950)$	0.019 (0.021)	0.024 (0.028)	0.001 (0.028)	0.015 (0.022)	0.022 (0.024)	-0.035 (0.024)	-0.012 (0.027)	-0.016 (0.026)	-0.033 (0.024)	-0.042 (0.029)
Diversity Exposure $\times I(1950 \leq t < 1960)$	0.122*** (0.032)	0.109*** (0.036)	0.104*** (0.035)	0.107*** (0.033)	0.131*** (0.037)	-0.043 (0.031)	-0.031 (0.033)	-0.033 (0.033)	-0.037 (0.032)	-0.046 (0.036)
Diversity Exposure $\times I(1960 \leq t < 1970)$	-0.015 (0.023)	-0.004 (0.025)	-0.013 (0.025)	-0.011 (0.024)	-0.004 (0.026)	-0.002 (0.023)	0.004 (0.027)	0.010 (0.027)	0.013 (0.023)	0.013 (0.027)
Diversity Exposure $\times I(1970 \leq t < 1980)$	-0.027 (0.036)	0.000 (0.039)	-0.014 (0.038)	-0.024 (0.036)	-0.021 (0.033)	-0.028 (0.021)	-0.034 (0.021)	-0.028 (0.021)	-0.014 (0.022)	-0.019 (0.023)
Diversity Exposure $\times I(1980 \leq t < 1990)$	-0.010 (0.022)	-0.020 (0.024)	-0.030 (0.024)	-0.009 (0.022)	-0.006 (0.025)	0.023 (0.024)	-0.008 (0.025)	-0.004 (0.024)	0.036 (0.025)	0.023 (0.028)
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Decade FE	Yes	–	–	Yes	Yes	Yes	–	–	Yes	Yes
Region-Decade FE	No	Yes	–	No	No	No	Yes	–	No	No
Province-Decade FE	No	No	Yes	No	No	No	No	Yes	No	No
US Emigrants $\times$ Decade FE	No	No	No	Yes	Yes	No	No	No	Yes	Yes
Controls $\times$ Decade FE	No	No	No	No	Yes	No	No	No	No	Yes
# Municipalities	2,549	2,549	2,549	2,549	2,399	2,549	2,549	2,549	2,549	2,399
# Observations	33,137	33,137	33,137	33,137	31,187	33,137	33,137	33,137	33,137	31,187
Mean Dep. Var.	0.043	0.043	0.043	0.043	0.043	0.041	0.041	0.041	0.041	0.042

Notes. This table reports dynamic estimates obtained from (11) of the effect of exposure to diversity within the Italian migrant community on monument construction in Southern Italy. The unit of observation is a municipality-decade for municipalities observed between 1861 and 2000. In columns (1–5), the dependent variable is an indicator equal to one if the municipality has at least one national-identity monument in a given decade. In columns (6–10), the dependent variable is an indicator equal to one if the municipality has at least one non-national-identity monument in a given decade. The baseline treatment is an interaction term between decade indicators and the share of US emigrants from each municipality who settled in districts in the top quintile of the diversity measure defined in (3). In columns (1), (4), (5), (6), (9), and (10), the specification includes municipality fixed effects and decade fixed effects; in columns (2) and (7), we include region-decade fixed effects; in columns (3) and (8), we add province-decade fixed effects; in columns (4) and (9), we control for the number of US emigrants interacted with decade dummies; in columns (5) and (10), we further control for the population in 1881 and in 1991, the number of WW1 deaths, and an indicator for municipalities on the coast, interacted with decade dummies. Standard errors are clustered at the municipality level and are reported in parentheses. Referenced on page: 35.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

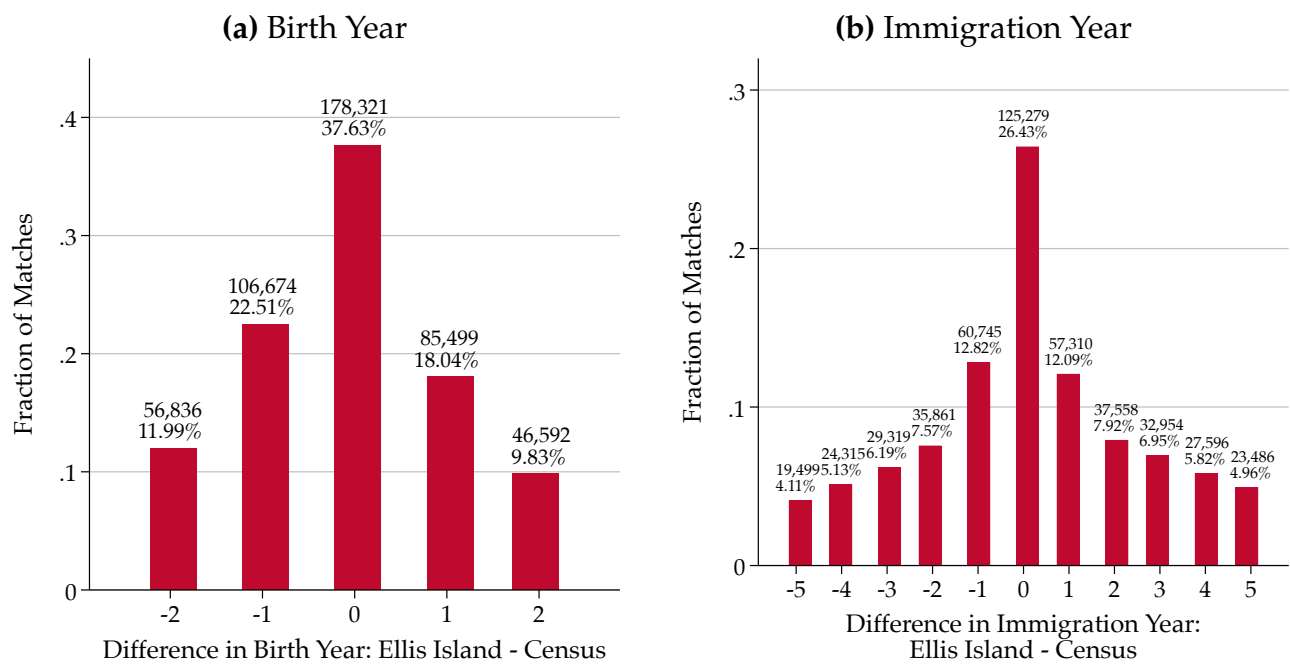
D ADDITIONAL FIGURES

Figure D.1. Visual Representation of the Linking Algorithm



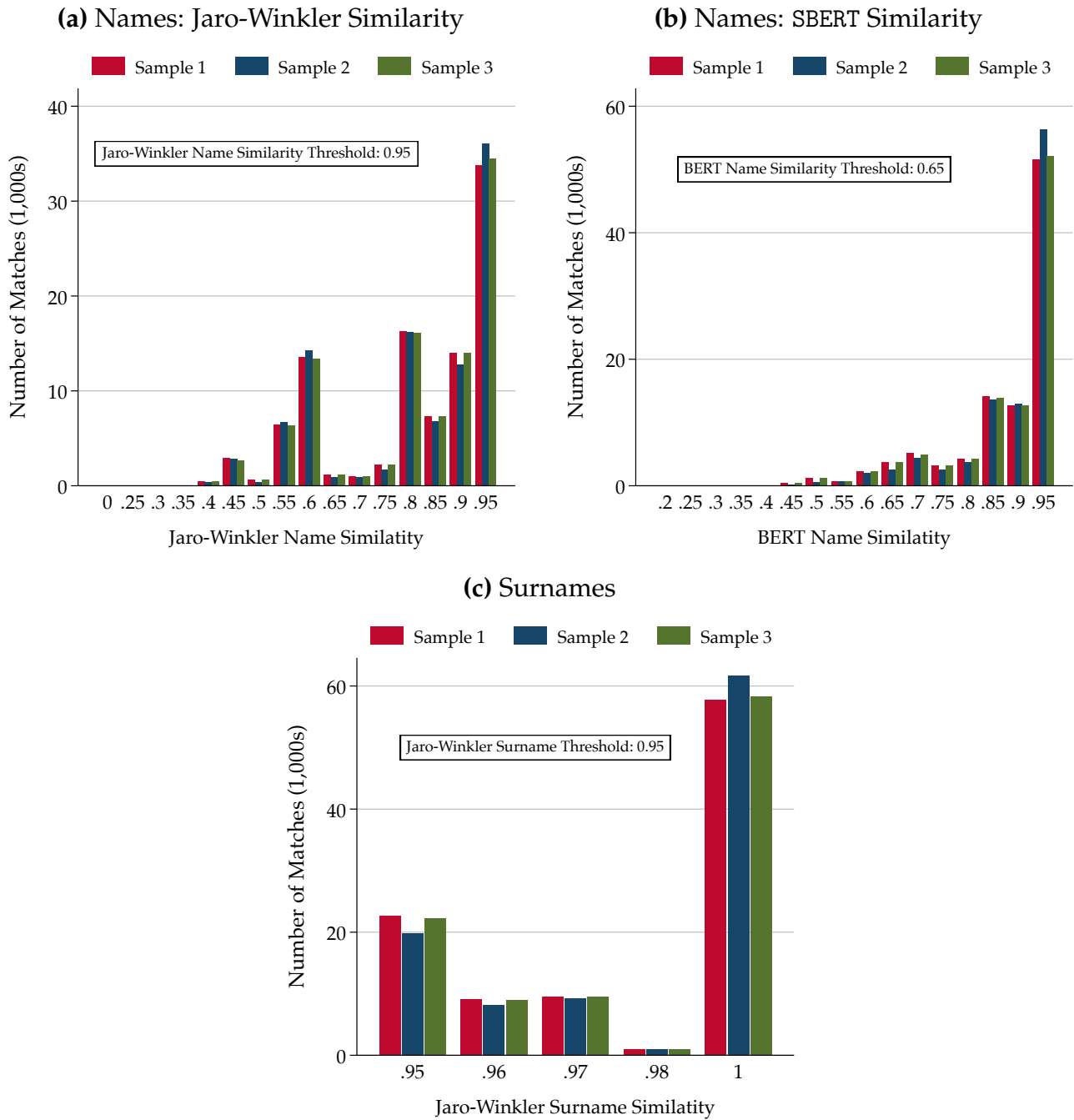
Notes. This Figure provides a graphical overview of the linking algorithm. The round boxes depict the starting data (Ellis Island and US Census) and the final linked sample output. The square boxes denote the dataset we constructed from the original datasets to implement the training and validation of the SBERT and Jaro-Winkler name similarities. The text not embedded in squares or circles denotes logical operations applied to the source dataset to reach the target output. Sample sizes are approximate. Referenced on pages: [12](#), [B9](#).

Figure D.2. Linked Sample: Blocking Variables Discrepancy



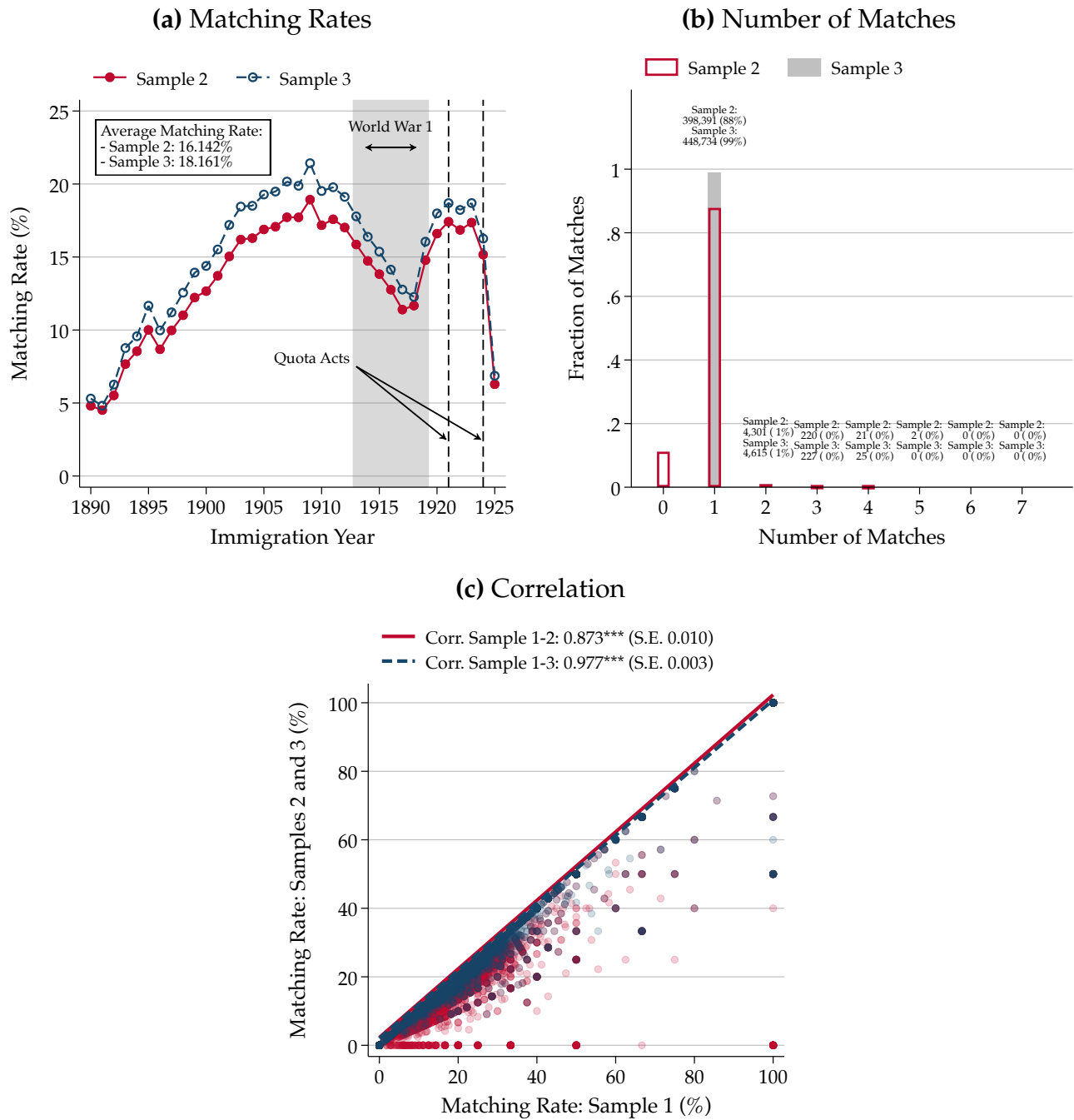
Notes. This figure displays differences between birth year (Panel D.2a) and immigration year (Panel D.2b) as recorded in the census and the Ellis Island datasets. To account for misreporting in the linkage procedure, matches are allowed to differ by up to two years in birth year and up to five years in immigration year. Each bar reports the number of matches in each difference bin and the corresponding share of the total sample. Referenced on page: [B13](#).

Figure D.3. Linked Sample: Match Quality



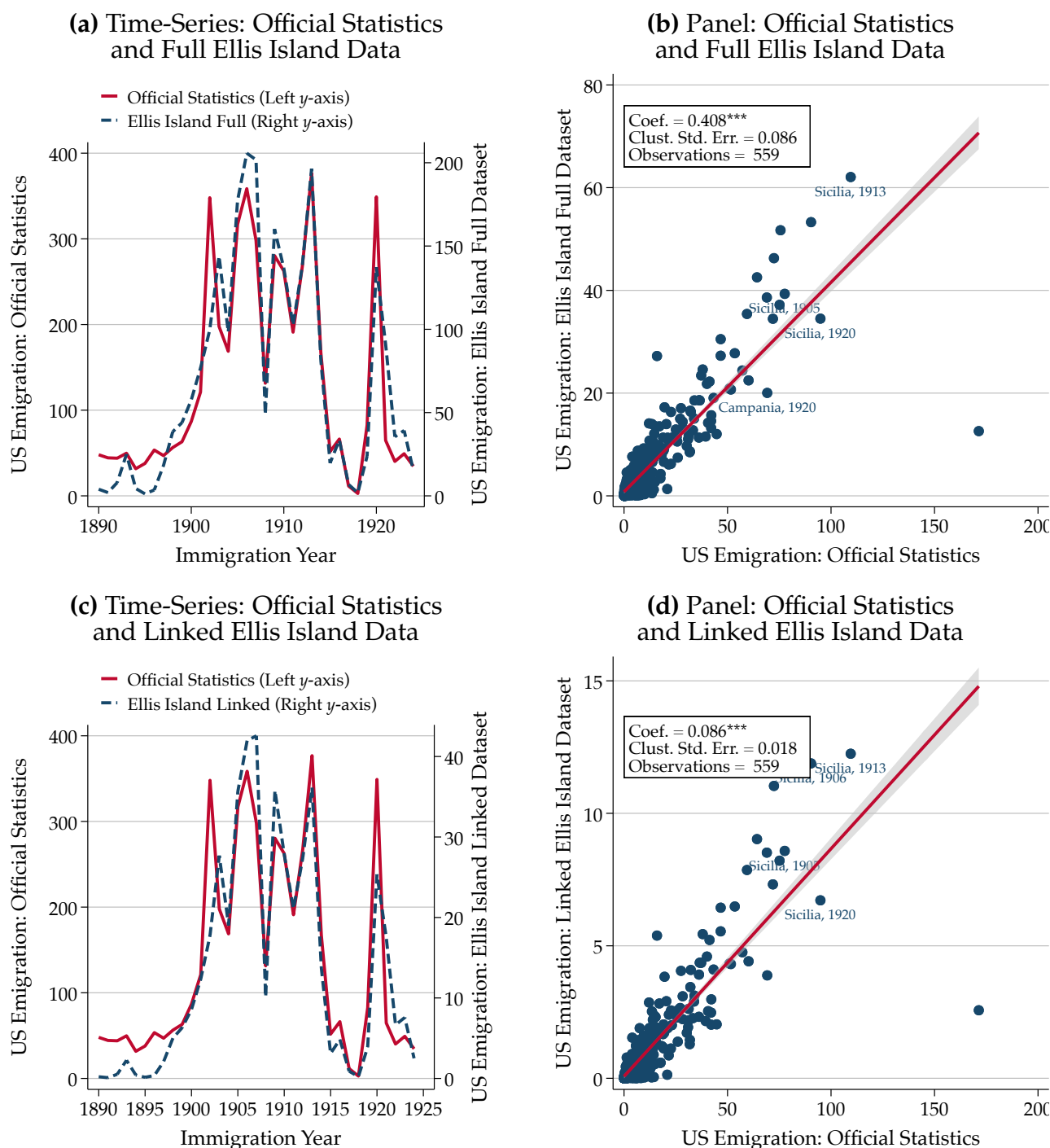
Notes. This figure reports the quality of matches produced by the linking algorithm. Red bars correspond to the baseline linked sample (Sample 1), while blue and green bars correspond to refinements 2 and 3. Panel D.3a shows the distribution of Jaro–Winkler similarities between first names recorded in the census and in the Ellis Island data; matches below the Jaro–Winkler threshold (0.95) satisfy the SBERT similarity criterion. Panel D.3b reports the distribution of SBERT similarities; matches below the SBERT threshold (0.65) satisfy the Jaro–Winkler rule. Panel D.3c reports the distribution of Jaro–Winkler similarities for last names; all matches exceed the Jaro–Winkler threshold (0.95). The number of matches is expressed in thousands. Referenced on page: B13.

Figure D.4. Linked Sample: Comparisons Between Samples



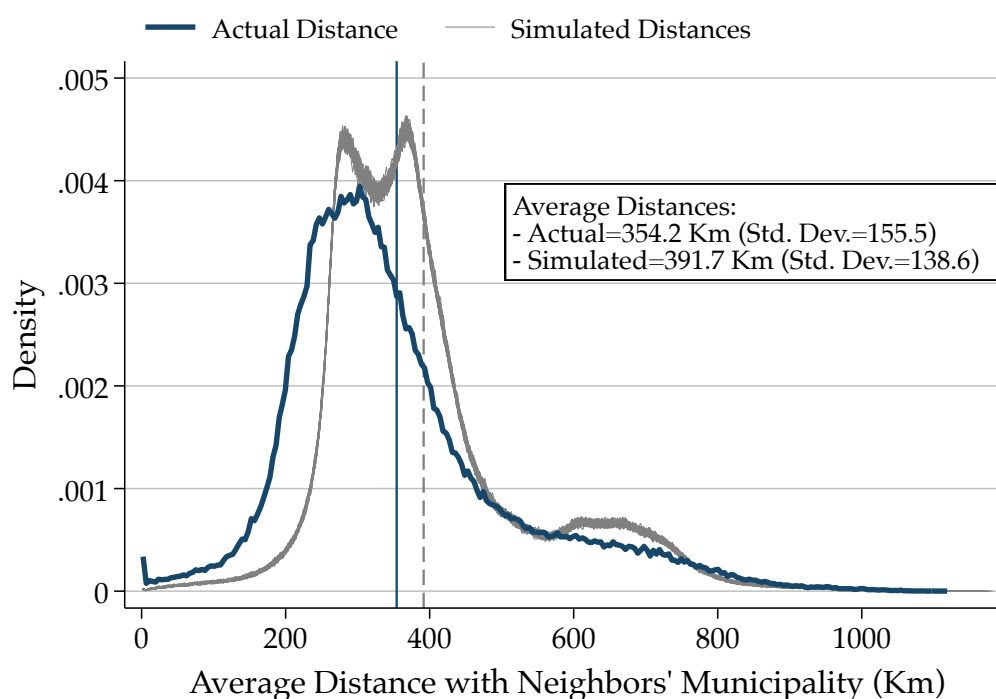
Notes. This figure compares the baseline linked sample with two refinements described in Appendix B.III. Red dots correspond to the refinement based on GPT 4-mini ("Sample 2"), and blue dots to the maximal-similarity refinement ("Sample 3"). Panel D.4a reports matching rates over time and on average. Panel D.4b shows the distribution of the number of census matches per Ellis Island record and the corresponding shares of the total sample. Panel D.4c plots, at the municipality level, the correlation between the (log) number of migrants in the baseline sample ("Sample 1," x-axis) and in Samples 2 and 3 (y-axis). Reported values give the correlation and its standard error. The solid red and dashed blue lines report the fitted values from a linear relationship between the two series in the two samples. Referenced on pages: 14, B14.

Figure D.5. Linked Sample: Validation with Official Statistics



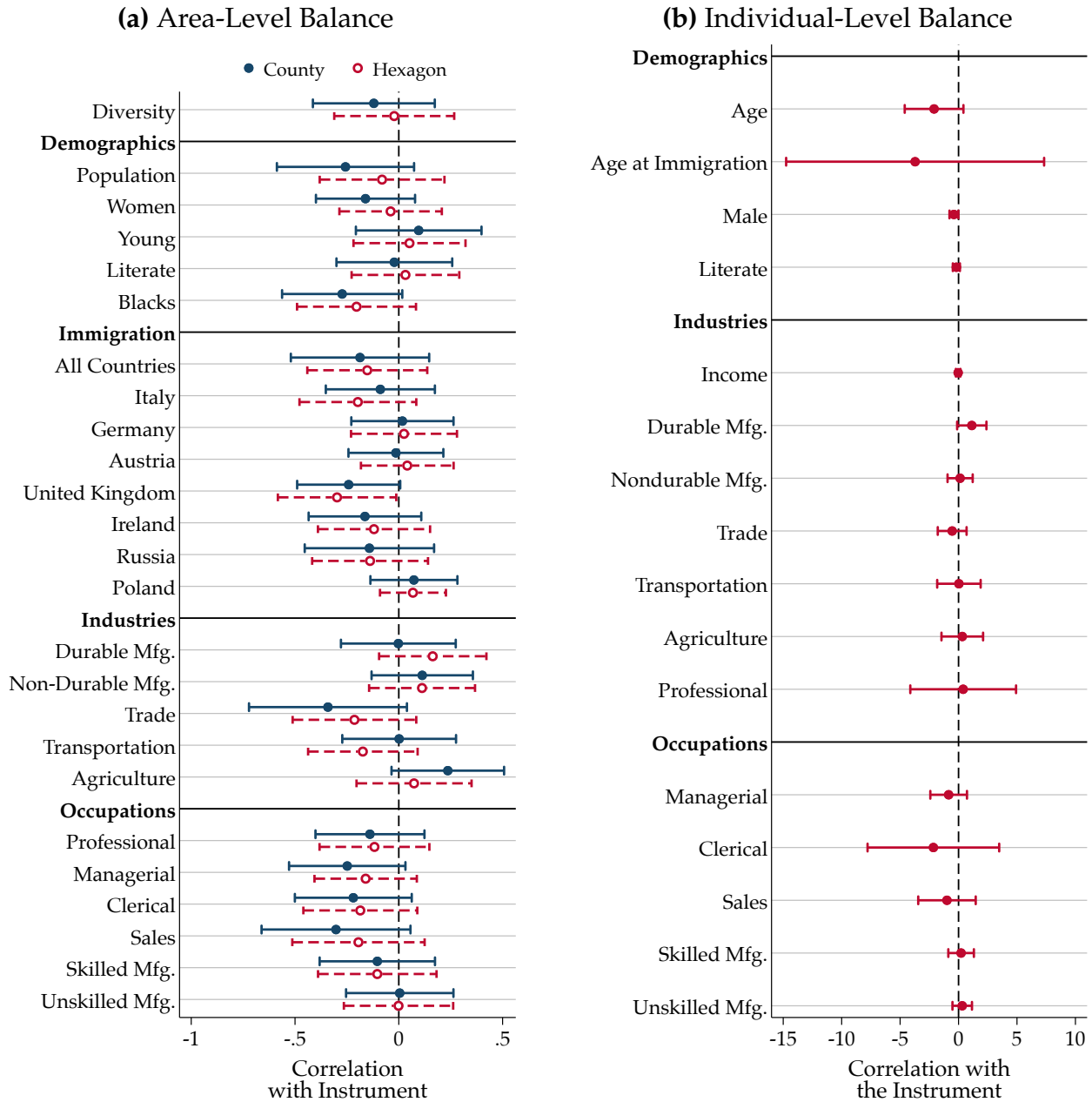
Notes. This figure reports the correlation between the Ellis Island dataset and data compiled from the *Annuario Statistico della Emigrazione Italiana dal 1876 al 1925* published by the General Commissioner for Emigration and containing regional data on US emigration between 1876 and 1925. Panels D.5a and D.5c display the total number of US emigrants recorded in the *Annuario* and the immigrants recorded in the full and linked Ellis Island dataset, respectively. Panels D.5b and D.5d display the correlation between the three datasets. In these figures, a dot is a region-year pair, and each panel separately reports the correlation between data in the *Annuario* and in the Ellis Island full and linked datasets, along with their standard errors clustered by region. The red lines overlay the fitted values from the linear regression and the gray bands display 95% confidence intervals. Referenced on page: B15.

Figure D.6. Distance from Neighbors' Municipality of Origin



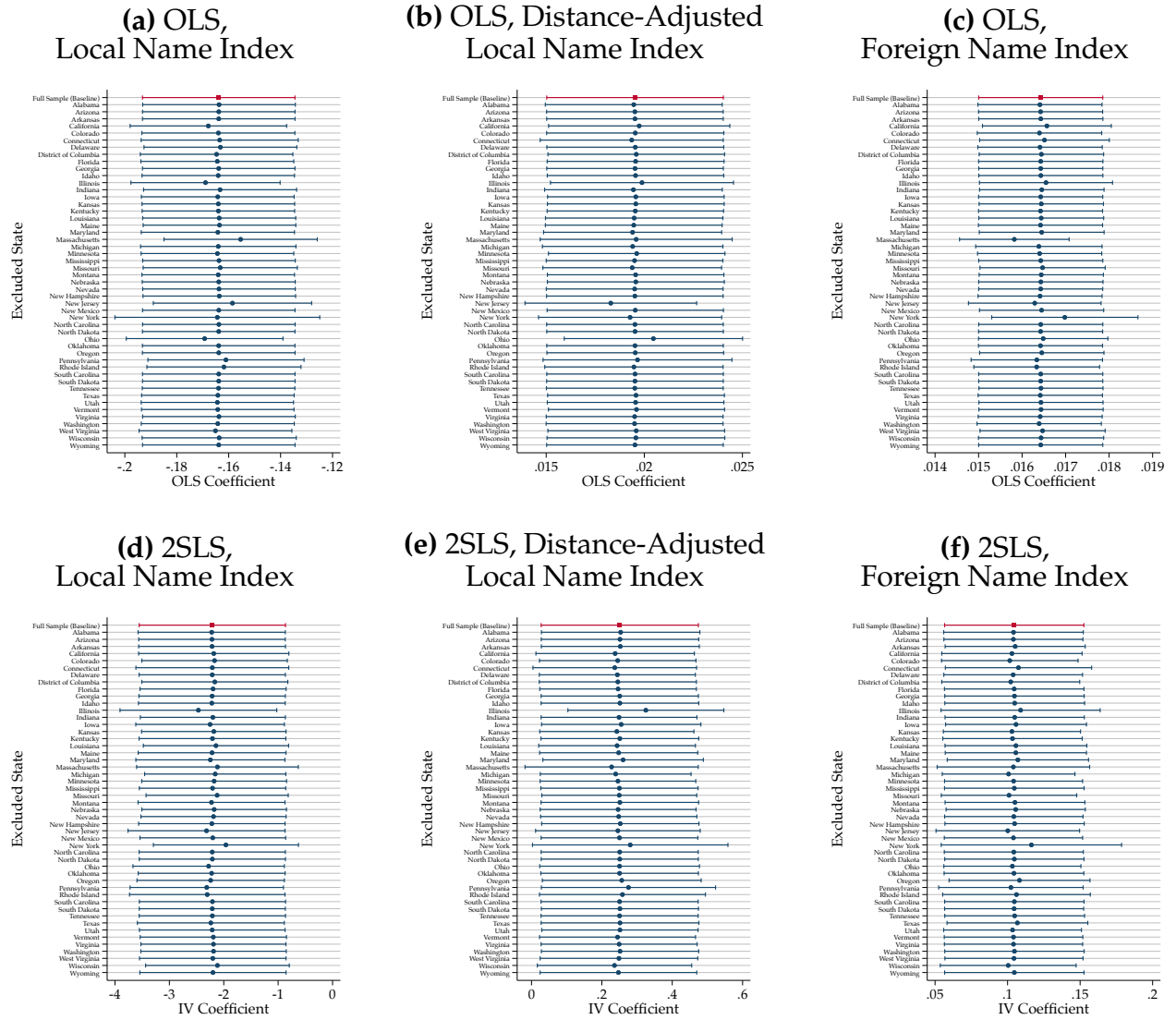
Notes. This figure displays the average distribution of distances between individuals' municipalities of origin within each enumeration district (blue). Gray lines report results from 1,000 simulations in which individuals are randomly allocated across districts; in each simulation, all members of the same household are excluded before reassignment. Excess mass of the blue distribution on the left of the random-matching counterfactual indicates that individuals disproportionately co-locate with immigrants originating from municipalities geographically close to their own. Referenced on page: [19](#).

Figure D.7. Tests of Instrument Balance



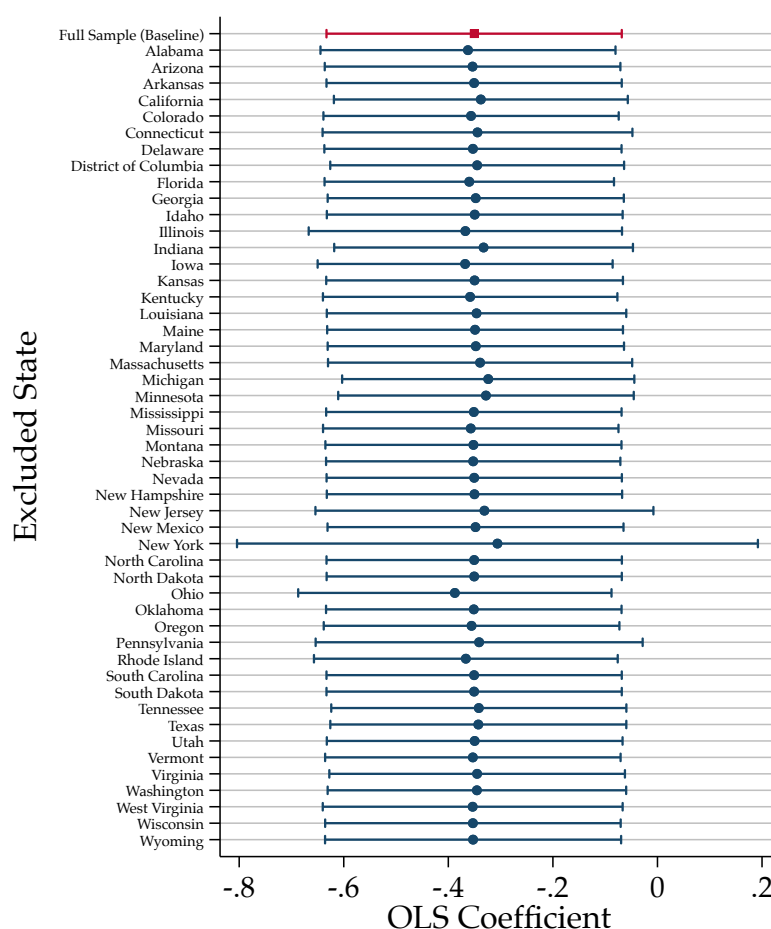
Notes. This figure reports correlations between the area-diversity instrument defined in (5) and area-level (Panel D.7a) and individual-level (Panel D.7b) characteristics. The unit of observation in both panels is an individual observed between 1900 (or the year of immigration, if later) and 1930. In Panel D.7a, each point reports the coefficient from a separate instrumental-variables regression of a predetermined county-level (blue) or hexagon-level (red) characteristic, lagged one census period, on Italian-community diversity instrumented by the ship-based instrument. In Panel D.7b, each point reports the coefficient from a separate instrumental-variables regression of a predetermined individual characteristic, forwarded one census period, on Italian-community diversity instrumented by the ship-based instrument. All specifications include year fixed effects, municipality-of-origin fixed effects, and individual controls (sex, birth year, and immigration year). Panel D.7a additionally includes county or hexagon fixed effects, while Panel D.7b includes hexagon fixed effects. Coefficients are standardized by the standard deviation (Panel D.7a) or the mean (Panel D.7b) of the dependent variable. Standard errors are clustered at the county level for county characteristics and at the hexagon level for hexagon characteristics; bars report 95% confidence intervals. Referenced on page: 22.

Figure D.8. State Leave-Out: OLS and 2SLS Panel Regressions



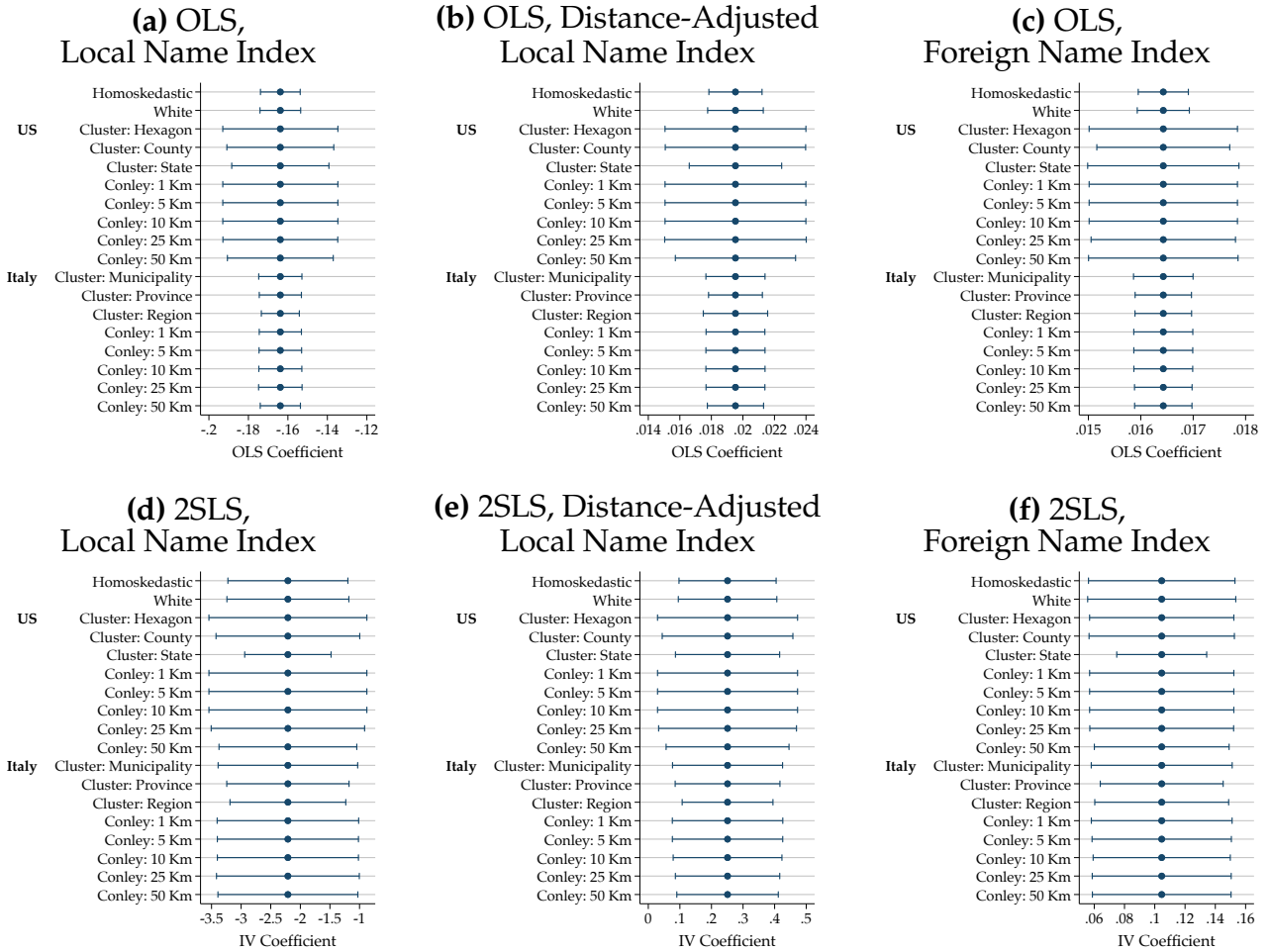
Notes. This figure reports the effect of exposure to diversity within the Italian community on name-based indices. Panels D.8a and D.8d use the local name index defined in (1) as the dependent variable; Panels D.8b and D.8e use the distance-adjusted local name index defined in (2); Panels D.8c and D.8f use the foreign name index of Abramitzky et al. (2020). The unit of observation is an individual observed in the first census in which they appear and, when possible, linked across subsequent censuses between 1900 and 1930. Panels D.8a, D.8b, and D.8c report ordinary least squares estimates; Panels D.8d, D.8e, and D.8f report two-stage least squares estimates. The first marker in each graph (in red) reports the baseline estimate; the subsequent markers report the estimates obtained when dropping one state at a time from the estimation sample. All specifications include municipality, census year, and hexagon fixed effects. Bands report 95% standard errors, clustered at the district level. Referenced on page: 29.

Figure D.9. State Leave-Out: Movers Regressions



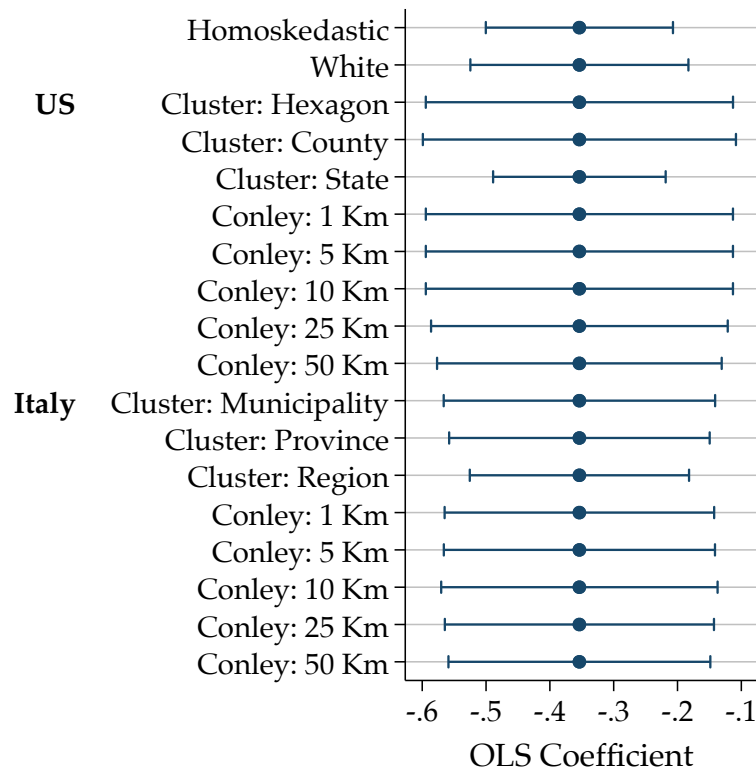
Notes. This figure reports results from the movers design specification in (7). The unit of observation is an individual migrant observed in each year in which they have a child born in the United States between 1900 and 1930. The dependent variable is the local name index defined in (1). The treatment is the interaction between an indicator equal to one after an individual moves to a different hexagon (and zero otherwise) and an indicator equal to one if the difference in diversity between the Italian communities in the destination and origin districts lies in the top 20% of the distribution (and zero otherwise). The first marker (in red) reports the baseline estimate; the subsequent markers report the estimates obtained when dropping one state at a time from the estimation sample. The specification includes individual, event-time (time since move), and hexagon-by-cohort fixed effects. Bands report 95% standard errors, clustered at the district level. Referenced on page: 29.

Figure D.10. Alternative Standard Errors: OLS and 2SLS Panel Regressions



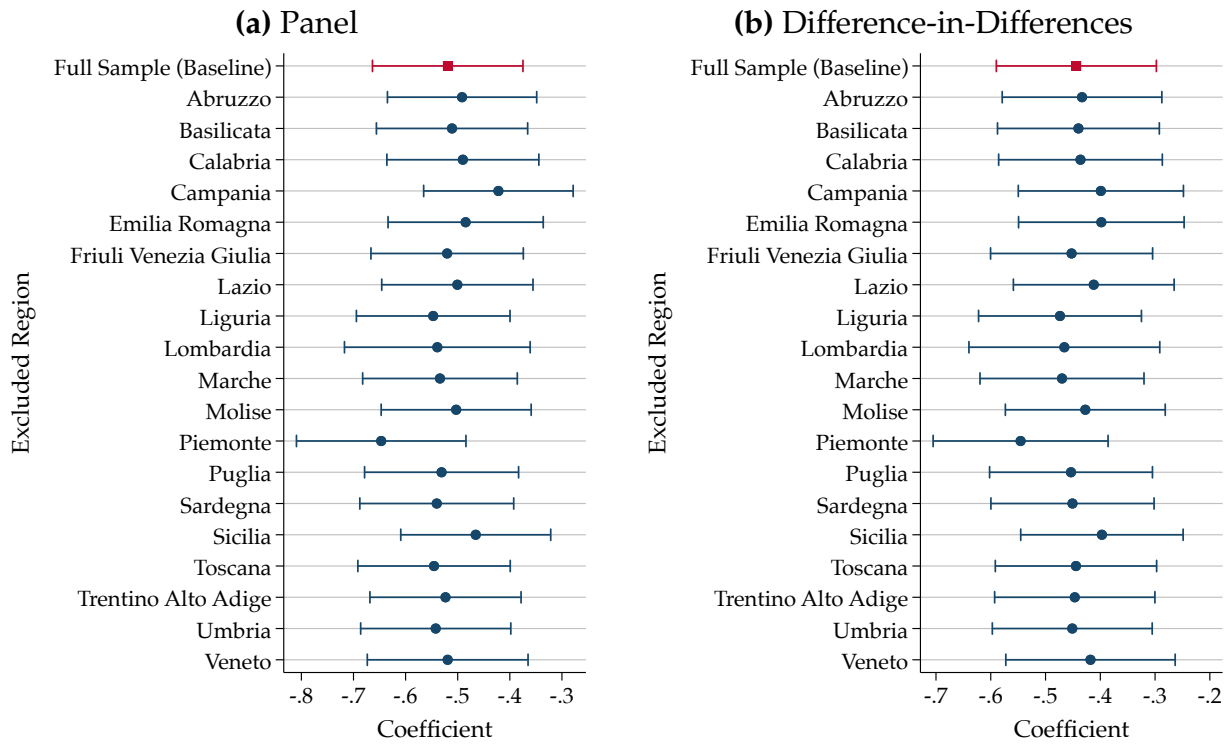
Notes. This figure reports the effect of exposure to diversity within the Italian community on name-based indices. Panels D.10a and D.10d use the local name index defined in (1) as the dependent variable; Panels D.10b and D.10e use the distance-adjusted local name index defined in (2); Panels D.10c and D.10f use the foreign name index of Abramitzky et al. (2020). The unit of observation is an individual observed in the first census in which they appear and, when possible, linked across subsequent censuses between 1900 and 1930. Panels D.10a, D.10b, and D.10c report ordinary least squares estimates; Panels D.10d, D.10e, and D.10f report two-stage least squares estimates. In each regression, we compute the standard errors using different estimators: unadjusted, White, clustered at different US geographical layers, Conley-corrected using various thresholds centered on the hexagon's centroid, clustered at different Italy geographical layers, and Conley-corrected using various thresholds centered on the origin municipality's centroid. All specifications include municipality, census year, and hexagon fixed effects. Bands report 95% standard errors. Referenced on page: 30.

Figure D.11. Alternative Standard Errors: Movers Regressions



Notes. This figure reports results from the movers design specification in (7). The unit of observation is an individual migrant observed in each year in which they have a child born in the United States between 1900 and 1930. The dependent variable is the local name index defined in (1). The treatment is the interaction between an indicator equal to one after an individual moves to a different hexagon (and zero otherwise) and an indicator equal to one if the difference in diversity between the Italian communities in the destination and origin districts lies in the top 20% of the distribution (and zero otherwise). In each regression, we compute the standard errors using different estimators: unadjusted, White, clustered at different US geographical layers, Conley-corrected using various thresholds centered on the hexagon's centroid, clustered at different Italy geographical layers, and Conley-corrected using various thresholds centered on the origin municipality's centroid. The specification includes individual, event-time (time since move), and hexagon-by-cohort fixed effects. Bands report 95% standard errors. Referenced on page: 30.

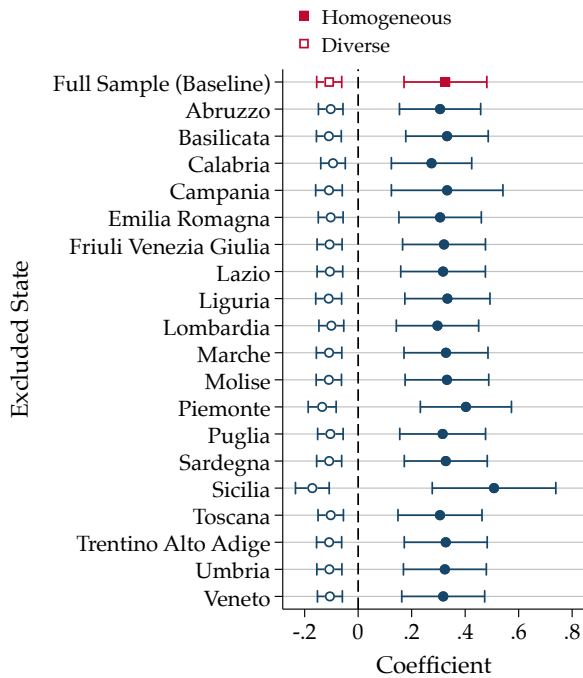
Figure D.12. Region Leave-Out: Name Particularism in Italy



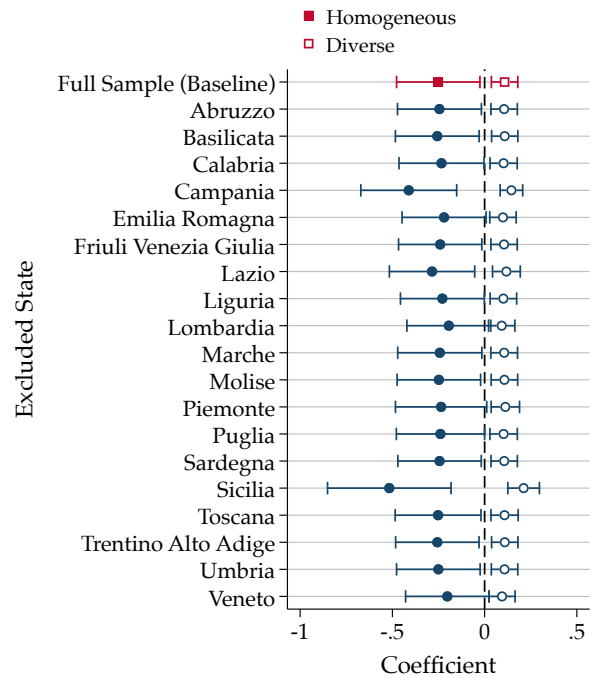
Notes. This figure reports the effect of exposure to diversity within the Italian migrant community on name-based indices in Italian municipalities from equations (8)–(9). The unit of observation is a municipality observed at a decennial frequency between 1880 and 1930. The dependent variable is the local name index defined in (1). In Panel D.12a, each dot reports the coefficient of a treatment term defined as the share of a municipality’s US emigrants who settled in districts in the top quintile of the diversity distribution (3). In Panel D.12b, each dot reports an interaction term between a post indicator equal to one in years at or after the municipality-specific onset of mass migration (defined as the first year with at least one emigrant per 1,000 inhabitants) and the share of the municipality’s US emigrants who settled in districts in the top quintile of the diversity distribution (3). The first marker (in red) reports the baseline estimate; the subsequent markers report the estimates obtained when dropping one region at a time from the estimation sample. All regressions include municipality and year fixed effects. Bands report 95% confidence intervals; Standard errors are clustered at the municipality level. Referenced on page: 35.

Figure D.13. Region Leave-Out: Street Naming in Italy

(a) National Identity

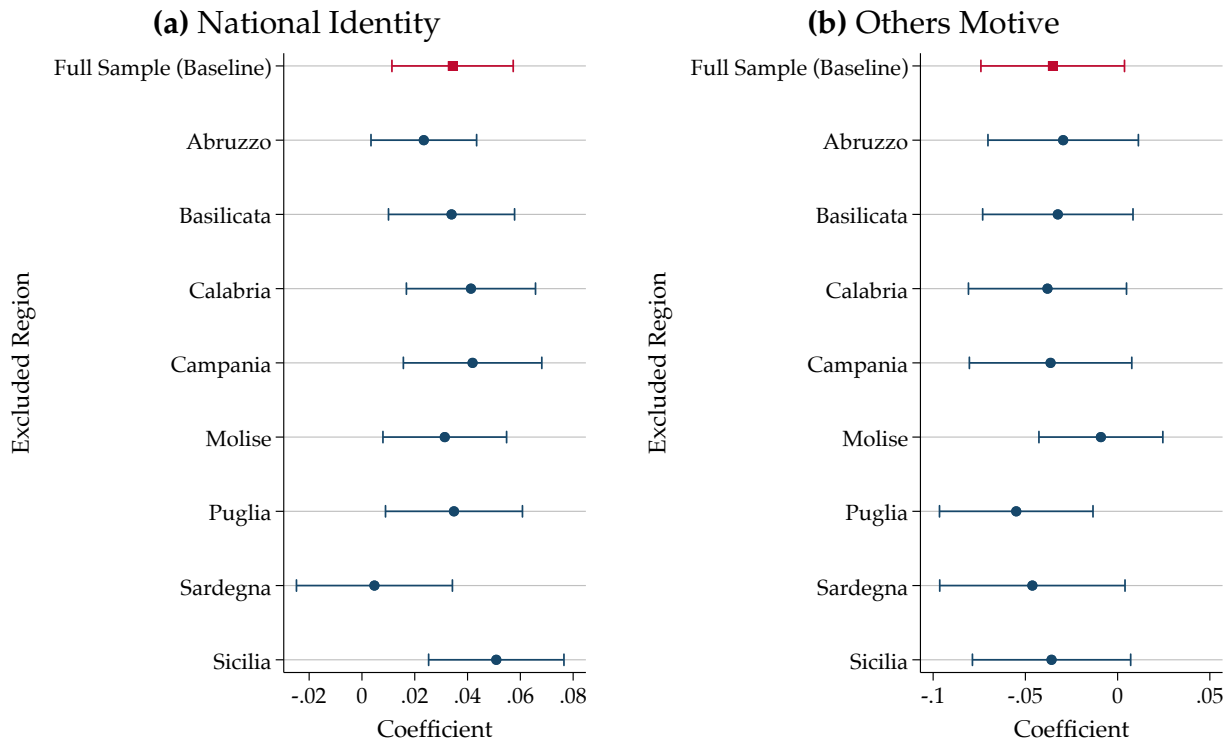


(b) Local Identity



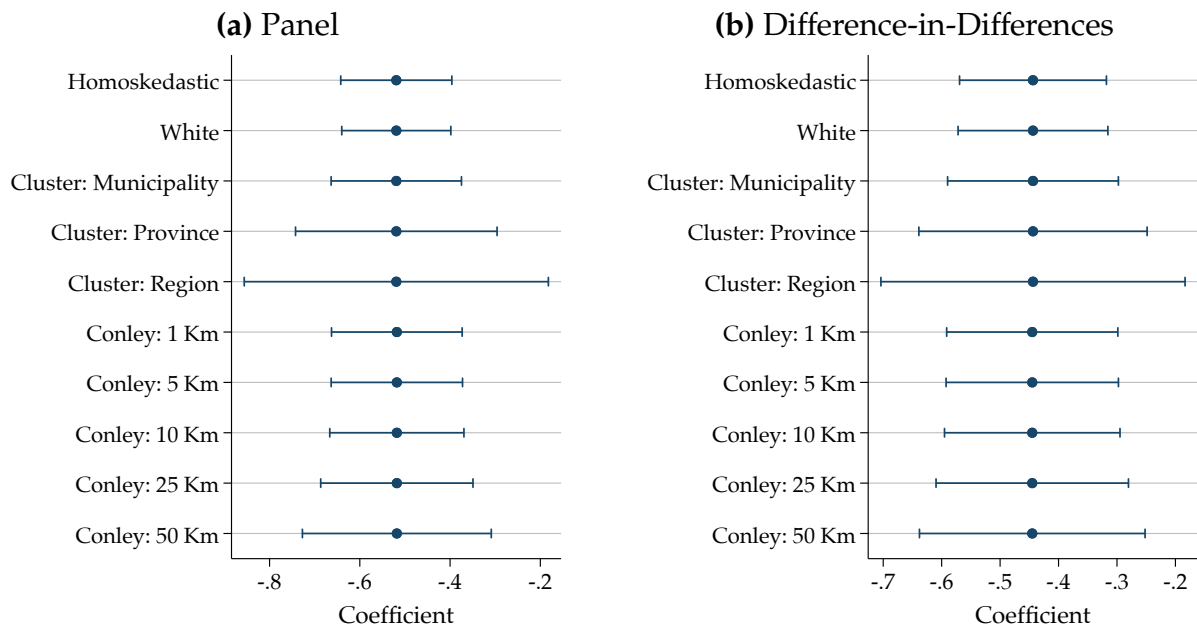
Notes. This figure reports cross-sectional estimates of specification (10) between emigration to diverse Italian communities in the United States and street-naming patterns in Italy. The unit of observation is a municipality. The dependent variables are the shares of streets whose eponyms reflect national identity (Panel D.13a) and local identity (Panel D.13b). The main explanatory variables are the number of emigrants to diverse neighborhoods (displayed by the hollow markers) and the number of emigrants to homogeneous neighborhoods (displayed by the colored markers). The first markers (in red) report the baseline estimates; the subsequent markers report the estimates obtained when dropping one region at a time from the estimation sample. The outcomes are normalized by the total number of streets in the municipality. All specifications include region fixed effects. Bands report 95% confidence intervals; Standard errors are clustered at the municipality level. Referenced on page: 35.

Figure D.14. Region Leave-Out: Monuments in Italy



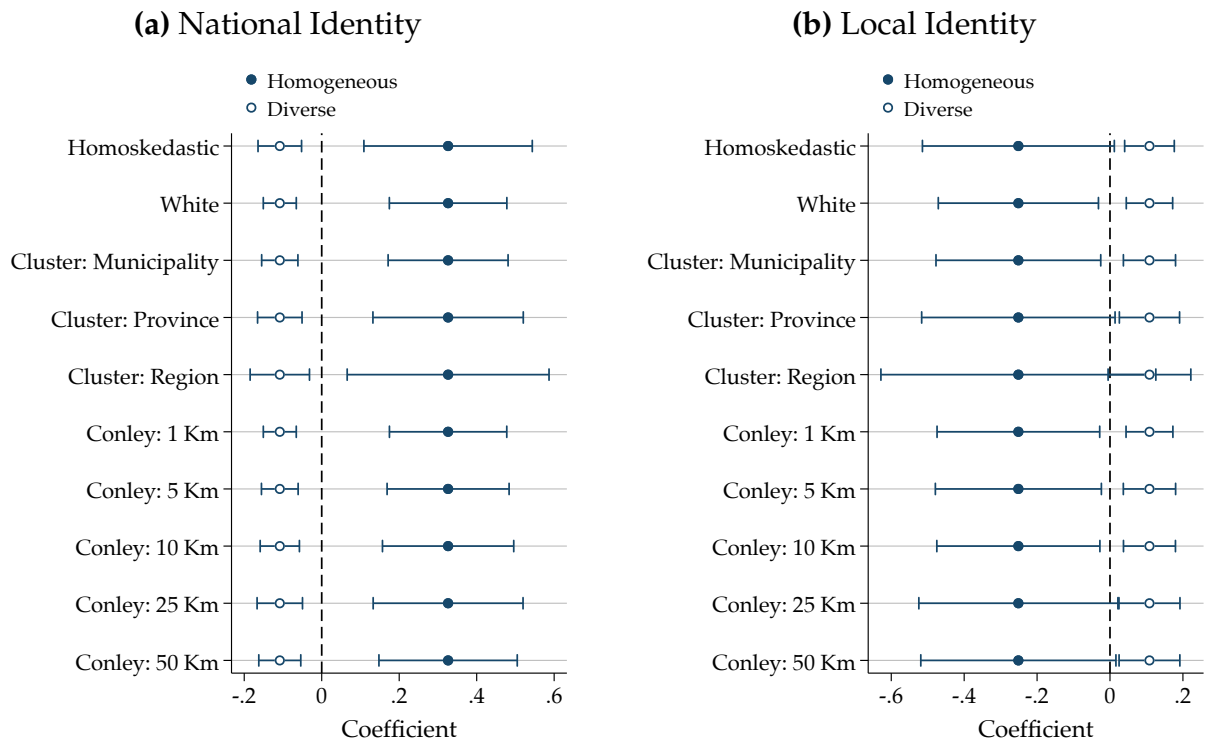
Notes. This figure reports estimates obtained from (11) of the effect of exposure to diversity within the Italian migrant community on monument construction in Southern Italy. The unit of observation is a municipality-decade for municipalities observed between 1861 and 2000. In Panel D.14a, the dependent variable is an indicator equal to one if the municipality has at least one national-identity monument in a given decade. In Panel D.14b, the dependent variable is an indicator equal to one if the municipality has at least one non-national-identity monument in a given decade. The baseline treatment is an interaction term between decade indicators and the share of US emigrants from each municipality who settled in districts in the top quintile of the diversity measure defined in (3). The first markers (in red) report the baseline estimates; the subsequent markers report the estimates obtained when dropping one region at a time from the estimation sample. Each specification includes municipality fixed effects and decade fixed effects. Bands report 95% confidence intervals; Standard errors are clustered at the municipality level. Referenced on page: 35.

Figure D.15. Alternative Standard Errors: Name Particularism in Italy



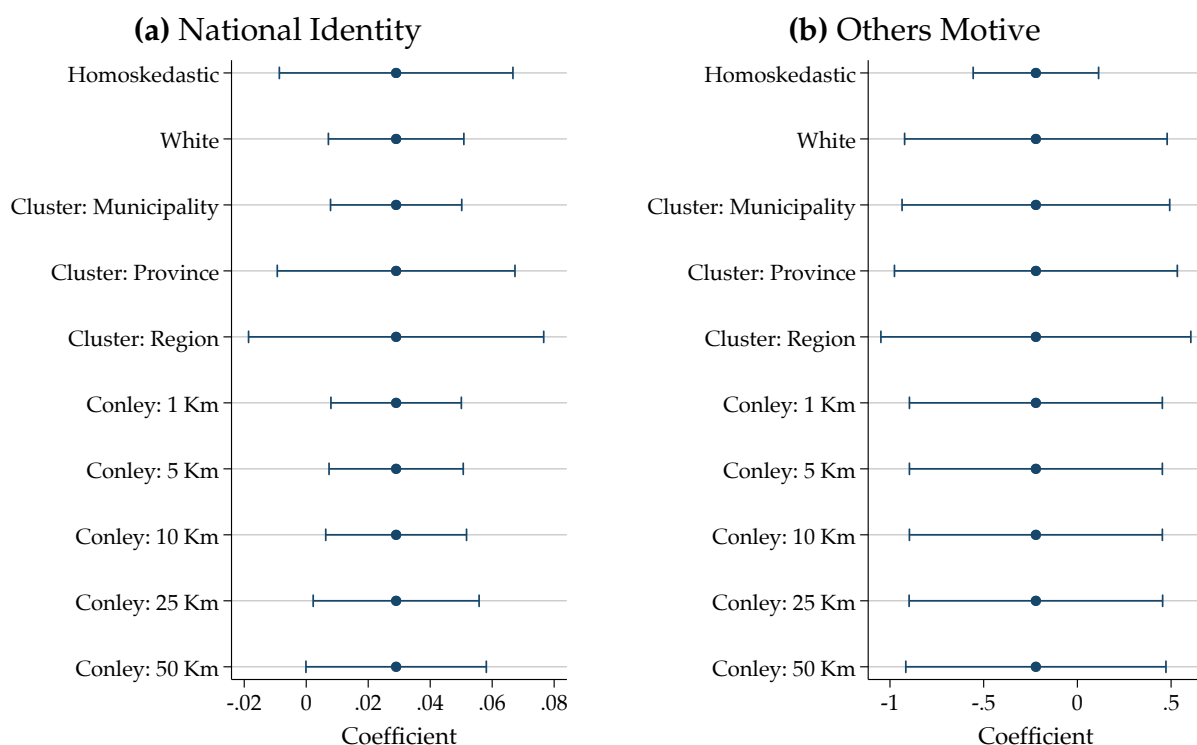
Notes. This figure reports the effect of exposure to diversity within the Italian migrant community on name-based indices in Italian municipalities from equations (8)–(9). The unit of observation is a municipality observed at a decennial frequency between 1880 and 1930. The dependent variable is the local name index defined in (1). In Panel D.15a, each dot reports the coefficient of a treatment term defined as the share of a municipality’s US emigrants who settled in districts in the top quintile of the diversity distribution (3). In Panel D.15b, each dot reports an interaction term between a post indicator equal to one in years at or after the municipality-specific onset of mass migration (defined as the first year with at least one emigrant per 1,000 inhabitants) and the share of the municipality’s US emigrants who settled in districts in the top quintile of the diversity distribution (3). In each regression, we compute the standard errors using different estimators: unadjusted, White, clustered at different geographical layers, and Conley-corrected using various thresholds centered on the municipality’s centroid. All regressions include municipality and year fixed effects. Bands report 95% confidence intervals. Referenced on page: 35.

Figure D.16. Alternative Standard Errors: Street Naming in Italy



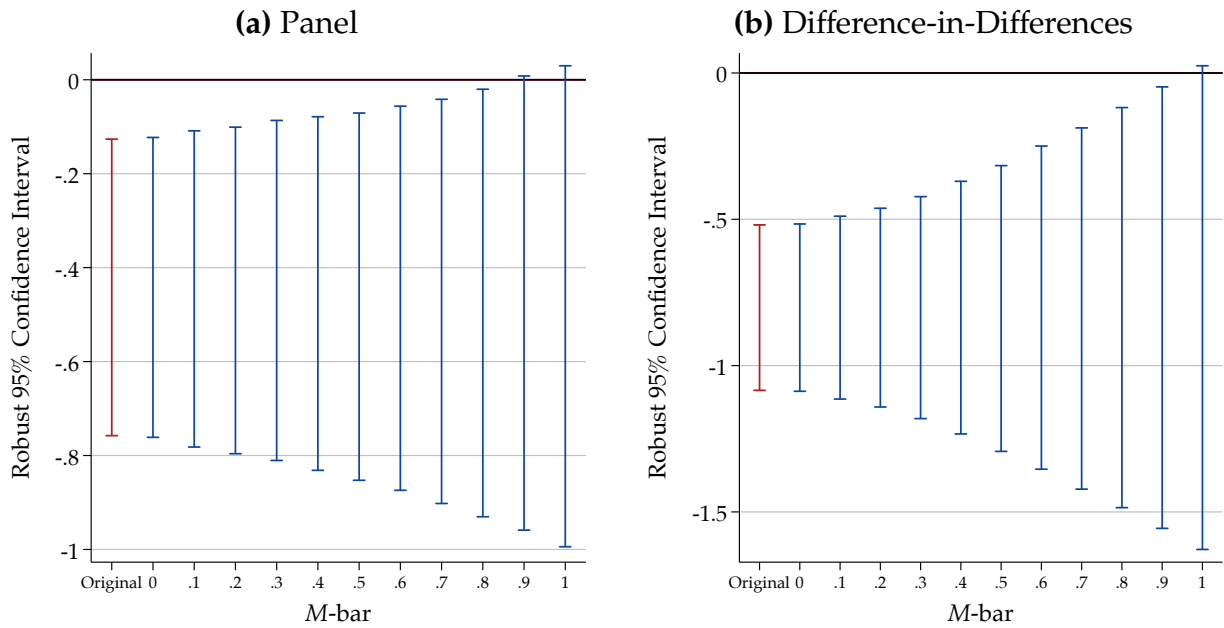
Notes. This figure reports cross-sectional estimates of specification (10) between emigration to diverse Italian communities in the United States and street-naming patterns in Italy. The unit of observation is a municipality. The dependent variables are the shares of streets whose eponyms reflect national identity (Panel D.16a) and local identity (Panel D.16b). The main explanatory variables are the number of emigrants to diverse neighborhoods (displayed by the hollow markers) and the number of emigrants to homogeneous neighborhoods (displayed by the colored markers). In each regression, we compute the standard errors using different estimators: unadjusted, White, clustered at different geographical layers, and Conley-corrected using various thresholds centered on the municipality's centroid. The outcomes are normalized by the total number of streets in the municipality. All specifications include region fixed effects. Bands report 95% confidence intervals. Referenced on page: 35.

Figure D.17. Alternative Standard Errors: Monuments in Italy



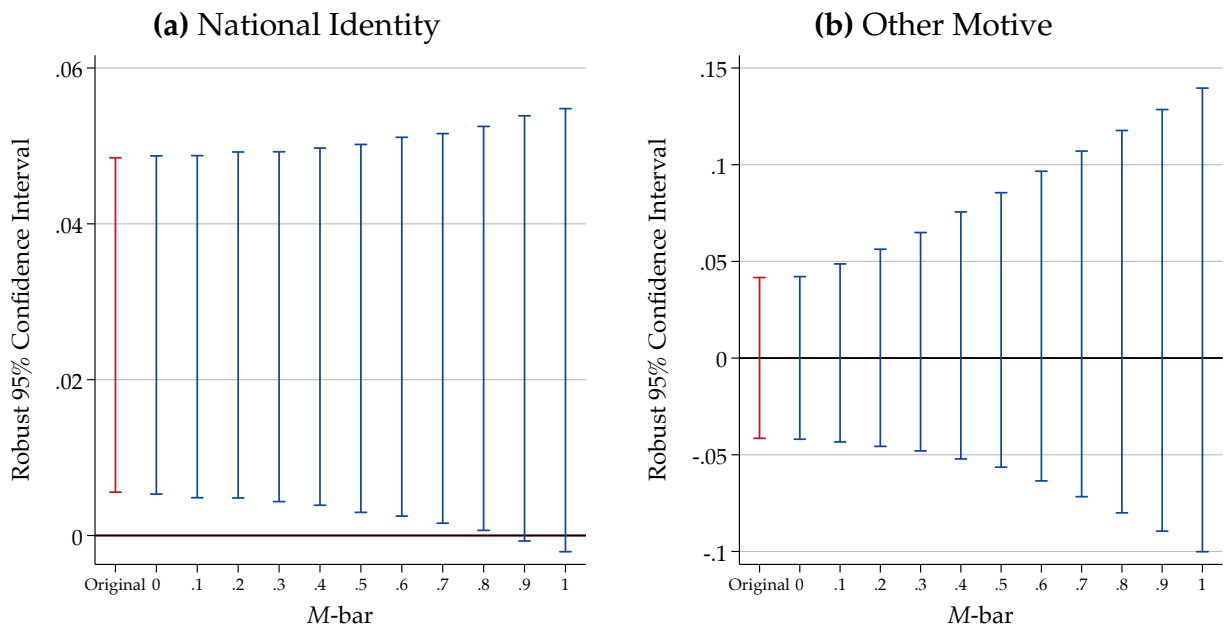
Notes. This figure reports estimates obtained from (11) of the effect of exposure to diversity within the Italian migrant community on monument construction in Southern Italy. The unit of observation is a municipality-decade for municipalities observed between 1861 and 2000. In Panel D.17a, the dependent variable is an indicator equal to one if the municipality has at least one national-identity monument in a given decade. In Panel D.17b, the dependent variable is an indicator equal to one if the municipality has at least one non-national-identity monument in a given decade. The baseline treatment is an interaction term between decade indicators and the share of US emigrants from each municipality who settled in districts in the top quintile of the diversity measure defined in (3). In each regression, we compute the standard errors using different estimators: unadjusted, White, clustered at different geographical layers, and Conley-corrected using various thresholds centered on the municipality's centroid. Each specification includes municipality fixed effects and decade fixed effects. Bands report 95% confidence intervals. Referenced on page: 35.

Figure D.18. Robustness to Violations of Parallel Trends: Naming in Italy



Notes. This figure reports alternative confidence intervals for the effect of exposure to diversity within the Italian migrant community on name-based indices in Italian municipalities from equations (8)–(9). The bands report 95% robust confidence intervals constructed following the method developed by Rambachan and Roth (2023). The unit of observation is a municipality observed at a decennial frequency between 1880 and 1930. The dependent variable is the local name index defined in (1). In Panel D.18a, each dot reports the coefficient of a treatment term defined as the share of a municipality’s US emigrants who settled in districts in the top quintile of the diversity distribution (3). In Panel D.18b, each dot reports an interaction term between a post indicator equal to one in years at or after the municipality-specific onset of mass migration (defined as the first year with at least one emigrant per 1,000 inhabitants) and the share of the municipality’s US emigrants who settled in districts in the top quintile of the diversity distribution (3). All regressions include municipality and year fixed effects. Referenced on page: 35.

Figure D.19. Robustness to Violations of Parallel Trends: Naming in Italy



Notes. This figure reports alternative confidence intervals obtained from (11) on the effect of exposure to diversity within the Italian migrant community on monument construction in Southern Italy. The unit of observation is a municipality-decade for municipalities observed between 1861 and 2000. The bands report 95% robust confidence intervals constructed following the method developed by Rambachan and Roth (2023). In Panel D.19a, the dependent variable is an indicator equal to one if the municipality has at least one national-identity monument in a given decade. In Panel D.19b, the dependent variable is an indicator equal to one if the municipality has at least one non-national-identity monument in a given decade. The baseline treatment is an interaction term between decade indicators and the share of US emigrants from each municipality who settled in districts in the top quintile of the diversity measure defined in (3). Referenced on page: 35.

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